Service

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Warning

Interaction Center

The interaction center provides multichannel collaboration and communication tools to enable efficient and consistent customer service.

Interaction center (IC) agents can handle inbound and outbound service transactions using the phone, e-mail, fax, chat, and letter communication channels. They can process business transactions, such as service orders and enhance their productivity by using alerts and a knowledge search. All relevant account information is available to them in the IC, such as account data, service order status, and product-related information.

The IC provides agents with the tools they need for the processes that comprise their daily work. Generally, the IC supports the following processes:

- Communication process

  This business process involves the communication between an IC agent and a business partner (customer, employee, and so on) in real time, for example using the telephony channel. Since IC agents can also work asynchronously, processing e-mails, service requests, or workflow items from the agent inbox, the communication process is not necessarily performed at all times.

- Interaction process

  The interaction process is the central IC process, which often encompasses the communication process and the business transaction process, and is performed whenever an agent uses the IC to communicate or interact – either directly with a business partner or asynchronously by working on a business transaction belonging to a business partner.

  The interaction process starts in the interaction mode with either the identification and confirmation of the business partner or the selection of a business transaction, e-mail, workflow, or other item belonging to a business partner. The interaction process includes all steps that the agent takes to work on a customer request. The interaction process can optionally result in the creation of one or more business transactions.

- Business transaction process

  Some simple customer or employee requests can certainly be resolved by providing a short answer or other information, not making use of a business transaction. However, frequently an interaction will result in one or more follow-on business transactions, such as a service request. In such cases, each business transaction is linked to the interaction record, providing a connection between the business process and the interaction process.

To start the interaction center, you use the `WUI` transaction and select a corresponding business role, such as **Service IC Agent**.

Related Information

Interaction Center WebClient
E-Mail Response Management System
Integrated Communication Interface

Interaction Center WebClient

Use

The interaction center (IC) WebClient is a thin-client, highly optimized desktop application for interaction center agents. It provides a comprehensive set of interaction center (IC) functions and supports various processes required to answer customer queries and work in corresponding business transactions.

Features

- Account Identification

  A customer search function that agents use to identify customers in the system. You can set up this function to identify customers even before agents accept contacts, based on automatic number identification (ANI) or the address of an incoming e-mail.

- Account Overview

  An overview that agents use to get complete information for an entire account, including individual account details, notes, roles, planned activities, and business partner relationships.

- Interaction Record

  A customer-focused overview of the current interaction and customer history.

- Interaction History

  A comprehensive interaction history that provides one view of a customer. Search and overview of business transactions for a customer. Agents can use various search criteria to find the business transactions they are looking for. Agents can record data both during and immediately after an interaction.

- Knowledge Article Search

  Interaction center agents can search for knowledge articles, which can include documents, links to Web sites, and screenshots, to answer customer queries.

- Agent Inbox

  A central worklist that allows agents to process incoming and outgoing mails, planned activities, and other work items.

- E-Mail

  An interface that agents use for processing incoming and outgoing e-mails, as well as completing unfinished e-mails.

- Chat

  An interface that agents use for processing incoming chat sessions.

- Scratch Pad

  An electronic notepad that agents use to record miscellaneous information during an interaction. Notes from the scratch pad can be imported into fields in the IC WebClient.

- Alerts

  Functions that combine system resources intelligently to guide agents during customer interactions.

- Queue Status

  A feature that agents use to quickly display the channels and queues they are assigned to, based upon configuration in the communication management software.

Specifics for Business Transactions in the Interaction Center

Use

This topic summarizes the specifics that apply to the processing of business transactions in the interaction center (IC).

Features

- The information from a confirmed account is automatically copied to any newly created business transaction.
- The business transactions that you use in the IC have a tiled layout. Alternatively, you can configure an overview layout for these business transactions in Customizing for the UI framework.

Settings

- Business transaction profile in IC

  You can make the following settings in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**:

  - Define which transaction types are available to the agent to create business transactions, and which transaction type is to be the default
  - Define whether dialog boxes are allowed and displayed automatically (automatic display is only supported by the interaction record)
  - Define the transaction type for transaction data and the additional business transactions separately

- Automatic routing

  Set up automatic routing. For information about the necessary settings in the rule modeler, see **Interaction Center WebClient** > **Business Transaction** > **Define Automatic Routing**.

Sales Transactions

Get an overview on how to create and change sales orders and sales quotations, display item proposals, as well as display and print sales transactions.

To use the sales transactions, you use the `WUI` transaction and select a corresponding role, such as **Sales Professional**.

[Entering Sales Documents on the WebClient UI](#)
[Item Proposals in Sales Transactions](#)
[Searching for, Displaying and Printing Sales Transactions on the WebClient UI](#)

Entering Sales Documents on the WebClient UI

Use

You use this function to create and change sales orders and sales quotations.

You carry out your presales activities such as Opportunities and Activities on the WebClient UI, and then, for example, process a rush order without having to exit.

This function is of particular use to sales employees who work mainly with the WebClient UI and therefore do not require the full range of functions offered by Sales Quotation and Order Management. For example, sales assistants can use this function to create simple sales orders.

Prerequisites

You can directly create and change a sales document on the WebClient UI once you have carried out the following tasks:

- You have assigned the relevant sales documents to a profile in Customizing under **Service** > **Transactions** > **Settings for Sales Transactions on WebClient UI** > **Define Profiles for Sales Transactions**.
- You have defined copy control of transaction types in Customizing for **Service** under **Transactions** > **Settings for Sales Transactions on WebClient UI** > **Copy Control for Opportunities and Sales Transactions**.
- If you want to use payment cards in sales orders, you have made settings in Customizing under **Sales and Distribution** > **Billing** > **Payment Cards**.

Features

When you maintain a sales document using the WebClient UI, the system enters data directly in SAP. This means that you can decide yourself which UI you use to process the sales order.

Creating Sales Quotations and Sales Orders as Follow-Up Transactions from an Opportunity

The sales representative or sales assistant has determined a potential sales opportunity, and has therefore created an opportunity. If the customer then requests a quotation, the sales employee can immediately create a sales quotation. To do so, the sales employee opens the opportunity that he or she created earlier, and creates the sales quotation as a follow-up transaction from the opportunity.

When you do so, the following data is copied to the sales quotation:

- Business partners
- Sales organization
- Distribution channel
- Division
- Sales office
- Sales group
- Product configuration
- Items
- Product
- Quantity
- Notes
- Unit

You can only display the following fields:

- Product
- Product ID

- Transfer
- Item number

You can navigate to the sales document using the transaction history in the follow-up activity.

Offering Alternative Items in Sales Quotations

In addition to the item requested by the customer, the sales employee can also offer an alternative item in the sales quotation. In order to offer the alternative item, when the sales employee copies the standard item and the alternative item from the opportunity to the sales quotation, the sales employee must make sure that the alternative item comes directly after (by item number) the standard item in the list of items.

Creating a Follow-Up Activity from a Sales Document

You can create a follow-up activity from a sales document. You can then navigate to the sales document using the transaction history of the follow-up activity. The system shows the follow-up activity in the following:

- Transaction history of sales orders and sales quotations
- Document flow shown in SAP GUI (transactions `VA02`, `VA22`, and `VA42`)

Correcting Errors

The system handles errors according to **Sales and Distribution** rules. This means that if errors occur, you cannot save the sales order. You can only continue processing the sales order after the errors have been corrected.

Note

The assignment of digital payment cards is currently not supported on the WebClient UI.

Activities

Certain restrictions apply to this function. Therefore, it should only be used by sales employees who work with sales documents occasionally.

Example

You create a sales quotation as a follow-up transaction for an opportunity. The system copies data directly from the opportunity to the sales quotation.

Item Proposals in Sales Transactions

Use

- Item proposals are used as input help in sales orders and quotations.
- Item proposals are available on the WebClient UI for Service, including the Interaction Center (IC).

Prerequisites

You have set up the item proposals in sales transactions in Customizing by choosing **Service** > **Transactions** > **Settings for Sales Transactions on WebClient UI** > **Product Proposals**. Under the node **Past Orders**, you make the settings for item proposals from previous orders.

Features

Item proposals are displayed directly in the **Items** assignment block.

You can automatically display item proposals, or manually trigger the display. To manually trigger the display of proposals, set the indicator **Past Orders on Request** in the Customizing of product proposals, in the activity **Assign Method Schema to Transaction Type**.

Item proposals appear at the end of the item list as suggested items, that is without item numbers. You can transfer the suggested items as final order items, by entering an order amount. Only then can the system carry out pricing and an availability check for the items.

- **Item Proposals from Past Orders**

  The system proposes items from the previous sales transactions for this customer, that is products and order quantities.

  - You can define the transaction types and the number of transactions in a certain time frame that the system draws from to determine proposed items, for example the last 10 sales orders from the past three months.
  - You can define how the system calculates the proposed quantities: from the last order, or as an average quantity from the previously defined number of orders.
  - You can select the required transactions using a popup, or the system can immediately (after you have entered the customer data) display all products from the past transactions.

Searching for, Displaying and Printing Sales Transactions on the WebClient UI

Searching for and Displaying Sales Transactions on the WebClient UI

If you want to provide your customer with information on the current status of a sales order that has already been created, you can use the **Search** function to locate transactions quickly, and forward the necessary information to the customer immediately. In addition, this function allows you to enter changes requested by the customer quickly and easily in the sales transaction.

Displaying sales transactions related to the relevant account

You can display sales transactions, that is, sales quotations and sales orders related to the relevant account on the **Account** overview page, under the corresponding assignment blocks. This allows you to gain an overview of all existing sales transactions for a specific customer, for example, in preparation for a customer visit.

Sales representatives can also create new sales transactions, edit or delete existing sales transactions from these assignment blocks. When you create a new sales transaction from here, the **Sold-to Party** field is automatically filled with the account from the **Account** overview page. Once you save and return to the **Account** overview page, the newly created sales transaction appears in the relevant assignment block.

Data retrieval parameters are delivered for each of these assignment blocks, which enable you to filter which data you want to display on the corresponding assignment blocks. For example, you may only want to display sales orders with posting dates for last month.

The following data retrieval parameters are available for the corresponding assignment blocks:

- Quotations: **Posting Date, Transaction Type**, **Valid From/To**
- Sales Orders: **Posting Date, Transaction Type**

For more information on data retrieval parameters, see Configurable Data Retrieval.

Printing Sales Transactions

You can print sales transactions, that is, quotations and sales orders from the WebClient UI. This allows sales representatives to preview, save or print sales transactions in order to check them, share them with their team, or file them to their documents. It is not intended for sales representatives to send transactions directly to their customers; this will continue to be done from the standard **Sales and Distribution** UI.

Once you have saved the sales transaction on the WebClient UI, you can use the **Preview Output** pushbutton for a print preview with a PDF viewer. You can also save locally or print out the preview using a PDF viewer. To print sales transactions, you can choose the output type set up in SAP **Sales and Distribution**, including PDF-based print forms, SAPscript and Smart Forms.

For more information, see PDF-Based Print Forms

Specifics for Interaction Center: Communication Management Software

Use

Communication management software (CMS) refers to software products that manage communication channels and handle individual contacts with customers through different channels, such as phone, e-mail, and chat. The Integrated Communication Interface (ICI) allows the CMS to communicate with the Interaction Center WebClient.

For more information about ICI, see Integrated Communication Interface.

For information about the features that are available in the telephony channel, see Communication Management Software.

For information about the features that are available in the chat and e-mail channels, see Chat and E-Mail in the Interaction Center WebClient documentation.

Prerequisites

- You have made the settings in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Communication Management Software Profiles**.
- Optionally, you have made the settings for checking the continued communication between the communication session, the CMS, and the Web browser in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Heartbeat Profile**.

  In this Customizing activity, you can make settings for the following checking methods:

  - Heartbeat without watchdog
  - Heartbeat and watchdog
  - CMS session availability

Business Process Push

Use

You can use this function to push business processes to external routers or communication management software (CMS) for distribution to interaction center (IC) agents based on their real-time availability. For example, the following business processes can be pushed:

- Business transactions, such as service orders
- Faxes and letters received through SAPconnect or the ArchiveLink scenario

The ID of the business process and a parameter that specifies the type of business process are transferred to the external router or CMS.

In addition, for each business transaction, you can define a set of additional parameters to represent, for example, status and priority. In order for the business process to be pushed and prioritized correctly in the CMS, you define rule policies to set the values for the parameters before these values are transferred to the CMS.

Prerequisites

- You have made the settings for business process push in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Business Process Push** > **Define Settings for Business Process Push**.
- You have assigned parameters to those agent inbox item types that you want to be pushed to external routers or CMS in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Business Process Push** > **Assign Parameters for Business Process Push**.
- You have defined rule policies to set the values for the parameters. For more information, see Rule Policies.
- You have made the settings for pushing faxes in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Settings for Asynchronous Inbound Processing** > **Define Receiving E-Mail/Fax Settings**.
- You have made the settings for pushing letters in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Business Process Push** > **Define Letter Settings**.

Features

Business transactions, such as service orders, are automatically transferred to an external router or CMS if the following conditions are met:

- The business transaction is saved and has ended, or is saved and escalated.
- The values of the parameters that have been defined in Customizing for the business transaction correspond to the values of the parameters that are set in the rule policies.

  For example, the business transaction must have status **In Process** and priority **High** to be pushed to an external router or CMS.

The business transaction is locked and the current employee responsible for the business transaction is removed. No changes to the business transaction and its status are possible until the transaction has been pushed to an IC agent.

When an IC agent is alerted that an inbound business transaction is routed, the business transaction ID and type are shown on the screen. The IC agent can either accept or reject the business transaction. The IC agent who accepts it is assigned as an employee responsible and can process that business transaction and change its status.

- Rule policies

  You define rule policies to set the values for the additional parameters for business transactions and cases before these values are transferred to the CMS. You define rule policies in the **Order Routing** and the **Intent Driven Interaction (IC WebClient)** contexts using attributes, conditions, and actions.

  ### Example

  You can create the following rule policy in the **Order Routing** context:

  Conditions:

  `If Priority Is Not Equal To Very High`

  Actions:

  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA`
  - `Send business process push parameter`

  You can create the following rule policy in the **Intent Driven Interaction (IC WebClient)** context:

  Conditions:

  `If Current Event Equals Order Was Saved`

  Actions:

  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = PRI`
  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA`

  Conditions:

  `If Current Event Equals Interaction Ended`

  Actions:

  `Send business process push parameter`

- Pushing of faxes and letters received through SAPconnect or the ArchiveLink scenario

  If the Customizing settings for pushing faxes and letters have been made, faxes and letters received through SAPconnect or the ArchiveLink scenario are pushed to IC agents based on their real-time availability. The number at which the fax arrives is the indicator for pushing the fax.

  An IC agent is alerted that an inbound fax or letter is pushed. The IC agent can then either accept or reject it. Once the IC agent accepts the fax or letter, the **Identify Account** page and the **Fax Preview** or **Letter Preview** is displayed.

ERMS E-Mail Push

Use

You can use this function to push e-mails to the communication management software (CMS) for distribution to an interaction center (IC) agent.

When e-mail is routed to an agent, the **Accept** and **Reject** context area buttons allow the agent to decide whether or not to accept the inbound message. If the agent accepts the e-mail, the system searches for the business partner based on the e-mail address. The agent can then use standard e-mail functions in the e-mail editor view.

Prerequisites

You have specified Inbox as the e-mail provider for the e-mail profile in Customizing for Service under IC WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles.

Activities

- Create rules in the ERMS context to push e-mail to the CMS for distribution

  Example

  You can create the following rule policy in the ERMS context:

  Condition:

  ```
  If E-Mail Sender Equals jdoe@customer.com
  ```

  Action:

  ```
  Push E-Mail to IC WebClient Agent
  ```

- Configure a CMS system ID that points to a CMS system that supports action routing

  Create an entry for the CMS system ID ERMS_ACTION. You do this in the SAP Easy Access menu under Service > Interaction Center > Interaction Center WebClient > Administration > Communication Management Software > Interface Settings > Maintain Communication Management Software Connections.

More Information

E-Mail Response Management System

Queue Status

Use

The queue status, located in the bottom right hand corner of the interaction center, allows you to quickly display the channels and queues you are assigned to.

Features

The queue status:

- Displays the current date and time (default display) or queue.

  If queue information is available, then an arrow is shown after the date. You can click on the arrow to show which queue the current incoming contact was sent to.

  Clicking on the arrow again returns the current date and time.

- Displays a tool tip with a quick view of multichannel information.

  The queue status tooltip displays the channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and all queues to which to are currently assigned.

  The queue status tool tip is only available if the connection to the relevant system is currently available.

- Displays the Agent Dashboard.

  If you click on the queue status, the agent dashboard is displayed.

Agent Dashboard

Use

You can use the agent dashboard to view information relevant to your current interaction center (IC) session. It is useful when you need more information about what is displayed in the queue status. You can access the agent dashboard by clicking the queue status at the bottom right corner of the browser window.

Features

The agent dashboard displays the following information:

- Your user ID and business role
- Channels and queues

  You see all channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and the queues to which you are assigned.

  This information is only available if:

  - There is a connection to a communication management software system, and
  - The communication management software passes queue and channel information to the interaction center.
- System information

  System information is used primarily by system administrators for troubleshooting. It includes:

  - Information on the domain in which the servers are operating (for example, wdf.sap.corp)
  - Name and description of the communication management software being used

Specifics for Interaction Center: Toolbar

Use

The toolbar provides an interface between your communication management software (CMS) and the Interaction Center (IC) WebClient. The toolbar contains pushbuttons and icons for communication functions that apply to the various communication channels.

You can modify the content of the toolbar to better meet your business requirements. For example, you can specify that different sets of pushbuttons are available in the toolbar depending on the connection of an agent to the CMS and the status of the agent's interaction with a contact.

Integration

The following processes are examples of how you can use the pushbuttons in the Interaction Center toolbar:

- Communication process

  For a phone call, you start the process by choosing the pushbutton **Dial** or **Accept**, and you end the process by choosing the pushbutton **Hang Up**.

- Interaction process

  Includes account maintenance and the interaction record. Additionally, business processes such as sales and service can also be handled in the interaction center. You start the process by confirming the account and you end the process by choosing the pushbutton **End**.

- Communication process and interaction process

  The two processes can take place simultaneously and do not necessarily start and end in the same order, for example in the following cases:

  - Multiple phone calls

    You can dial and end multiple phone calls during the same interaction. This occurs, for example, when a phone line is disconnected during a customer interaction and needs to be dialed again.

  - Multiple interactions

    Multiple interactions can happen during a single phone call. If the agent receives and accepts the next call whilst wrapping up the previous interaction, they can end the first interaction by choosing the **End** pushbutton despite already being in the next active phone call.

Prerequisites

- Your system administrator has set up the CMS and ensured that the user names there match those in the IC. For more information, see Specifics for Interaction Center: Communication Management Software and the documentation for the CMS you are using.
- You have made the settings for the toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

Features

You use the toolbar to work in different channels. The system determines the pushbuttons that appear in the toolbar, depending on the following:

- The communication channel in which you are currently working
- Customizing settings

Example

Pushbuttons in the IC toolbar

Channel	Name	Definition
Universal (available for all channels)	**Accept**	Accepts the contact. In telephony, **Accept** is similar to picking up the receiver.
	Reject	Sends the contact back to its original queue.

Channel	Name	Definition
	Transfer or Blind Transfer (in the telephony toolbar)	In telephony, puts caller on hold while you transfer the call to another agent's extension without first consulting with that agent.
	End	Ends the interaction (indicating that the agent has finished wrap up activities), saves all objects created or changed during the interaction, and clears customer information from the screen and from the business data context.
	Hang Up	Similar to hanging up the receiver. Ends a phone call that was started with Dial or Accept.
	Clear Interaction	Discards the interaction record and resets the account identification. **i Note** Discarding other transactions such as sales or service transactions is not possible in this way. Depending on the process status, this may be done by choosing Cancel if it is part of the transaction toolbar. If you want to clear the entire context of the session including the business data context, you have to choose the Clear Interaction pushbutton followed by the End pushbutton.
	Reset CTI	Resets the context area of the browser session to synchronize it with the communication session state. The different states can for instance be caused by network issues.
	Dial Pad	Displays a dial pad that looks like the numbers on a telephone, which you can use to make a call.
Telephony	Hold	Puts the active call on hold.
	Toggle	Switches between a call on hold and an active call, automatically placing the active call on hold and vice versa.
	Retrieve	Takes a call off hold and makes it active once again.
	Consult	Puts customer call on hold while you call another agent. After you are finished consulting with the other agent, the agent call is dropped and the customer call is active once again.

Channel	Name	Definition
	Warm Transfer	Puts caller on hold while you consult with another agent before transferring the call to the agent's extension.
	Conference	Puts customer on hold while you call another agent. Then you activate both calls at the same time so that all three parties can participate in a discussion.
	Wrap Up	Indicates to the system that you need time to wrap up the call. Wrap up time ends when you choose the **End** pushbutton.
E-Mail	Pushbuttons specific to the e-mail function, such as **Reply**, **New**, or **Forward**, appear on the e-mail screen dynamically, as and when they are required.	Represent standard e-mail functions.
Chat	**Leave**	Finishes the chat.

Work Modes

Name	Use
Ready	When you choose **Ready**, the CMS system sends incoming contacts to you.
Not Ready	When you choose **Not Ready**, the CMS system does not send incoming contacts to you.

Scratch Pad

A temporary workspace in the interaction center that you use as an electronic notepad to store miscellaneous information during an interaction.

Customers usually provide a lot of information during the first few minutes of conversation, before you have a chance to access the relevant transaction. Most agents have separate notepads to take notes and then enter the data into the system later.

The scratch pad replaces the need to write manual notes. It allows you to electronically store information so it can be copied into specific functions that require this data.

The scratch pad allows you to focus on recording and listening to what the customer is relaying. Customers do not have to hold their conversation until you follow through the sequence of the interaction, gather the preliminary information, such as contact information, and then navigate to the appropriate area in the IC WebClient to record the information. This also saves customers from having to provide information more than once.

The scratch pad is located in the top left corner of the IC, under the title bar. It is available throughout the interaction, and can be copied into different IC functions, such as the interaction record or knowledge search by choosing **Import Scratch Pad**. All of the data on the scratch pad is imported, but you can still edit, add, or delete it, if necessary.

Data remains on the scratch pad for the entire interaction until you either select **End** from the toolbar, or select **Clear**. Closing the scratch pad does not delete the contents.

Alerts and Messages

Use

These functions display texts on your page to alert you about items or situations that require your immediate attention. There are three types:

- Alerts

  Alerts are predefined by system administrators or Interaction Center (IC) managers to be triggered by certain events in the system.

- Messages

  There are three types of messages: informational, warning, and error.

Features

- Placement on page

  Alerts appear in the alerts area and system messages appear in the system messages area.

- Alerts

  - If more than two alerts appear, an arrow appears, which allows you to page through alerts in pairs.
  - All alerts disappear when you choose **End**.

Example

Alerts

Marcus is a customer service representative at an interaction center. He accepts a call from Ms. Anna Glenn and confirms her identity in the system. The following alert appears:

Service order 00225

Marcus asks Ms. Glenn if she is calling regarding this order. She says that she would like to know the status of the service order. Marcus simply chooses the link to the service order and can inform Ms. Glenn of its current status.

Creating Alerts

Use

You can create alerts in that use system information to provide agents with information they may need during customer interactions.

i Note

No preconfigured alerts are delivered. Interaction center managers define alerts in the **Alert Editor** and assign them through rule policies. Alerts are created independent of any rule policy context.

Alerts can only be triggered in the Intent-Driven Interaction context. This context allows you to automate specific interaction center functions, for example, triggering and terminating alerts, via business rules. Interaction center alerts are not relevant for other rule policy contexts.

Prerequisites

- You have completed the Customizing for Service under Interaction Center WebClient > Additional Functions > Intent-Driven Interactions > Service Manager > Configure Alerts .
- You have completed the Customizing for Service under Interaction Center WebClient > Additional Functions > Intent-Driven Interaction > Define Intent-Driven Interaction Profiles .
- If you want to user user-triggered alerts, you have selected this alert type in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Communication Processing Profiles .

Features

The following types of alerts are available in the IC:

- Real-time alerts

  With these alerts, the system continually polls the server to check for alerts.

- User-triggered alerts

  With these alerts, the system waits until the user triggers an event to perform a server round-trip to check for alerts.

  This type of alert allows you to show alerts without using server polling and can be used with asynchronous processing.

When creating alerts, you can select a theme that affects the appearance of alerts by changing, for example, the font style, font color, and the alert icon. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Themes must be defined in Customizing before they are available for selection when creating an alert. You can define themes in Customizing for Service under Interaction Center WebClient > Additional Functions > Define Themes for Alerts .

Broadcast Messaging

Use

Interaction Center (IC) supervisors can send messages to IC agents in real time using broadcast messaging. They can also create and maintain distribution lists for broadcast messaging and display sent messages. IC agents can receive messages, but cannot send messages.

IC supervisors can access broadcast messaging under Managing Operations > Broadcast Messaging .

Prerequisites

You have made the settings for broadcast messaging in Customizing for Service under Interaction Center WebClient > Additional Functions > Define Broadcast Messaging Profiles .

Features

- Distribution lists

  To create or maintain distribution lists, click **Maintain Distribution Lists**. Broadcast messaging profiles contain an organizational ID. When you create distribution lists, the names that appear in the input help for **Add User** come from the organizational unit entered in the selected broadcast messaging profile. You can also copy an existing distribution list, delete a user from a distribution list, or delete a distribution list. The default distribution list is the one that appears first alphabetically.

  To share distribution lists, the owner of the list can flag other users within the list as administrators. As an administrator, you can add and remove users from the list, use the list to send broadcast messages, and you will be able to view any broadcast messages that were sent to this list in the **Sent Log**.

- Themes

  You can define themes for broadcast messaging in Customizing for **Service** under **Interaction Center WebClient** > **Additional Functions** > **Define Themes for Broadcast Messages**.

  Once a theme has been defined in Customizing, it is available for selection when creating a broadcast message. The theme affects the appearance of broadcast messages by changing, for example, the font style and color. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Interaction Record

Use

The interaction record is a screen in the interaction center (IC) that contains data for the current interaction with an identified business partner. IC agents can log current interactions in this screen, and postprocess them if necessary.

Interaction with a customer always includes an interaction record, which can then be linked to another business transaction, such as a service process. The interaction record can, however, also be saved without creating another business transaction.

Note

From a technical point of view, the interaction record is created as a business transaction.

Once an agent has identified an account, the system navigates to the interaction record. In the interaction record, the agent logs the interaction, records central data from the call, and postprocesses the interaction when the call is finished.

To increase system performance at high volume call centers, you can use the lean interaction record. The lean interaction record is a modified version of the full interaction record.

Prerequisites

- You have defined a date profile and assigned this to your transaction type to use interaction time recording in Customizing for **Service** under **Basic Functions** > **Date Management** and **Transactions** > **Basic Settings** > **Define Transaction Types**.
- You have activated the rescheduling view and optionally enabled automatic text entry for call rescheduling in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**.
- **Using Activity Reasons:**
  - You have made the settings for activity reasons in Customizing for **Service** under **Transactions** > **Settings for Activities** > **Define Activity Reasons**.

i **Note**

The interaction reason corresponds to the activity reason.

- You have assigned the **Reason for Activity** subject profile in your transaction type to the **Business activity** transaction type in Customizing at the header level.

  You can make settings for transaction types in Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Transaction Types**.

- **Using Multilevel Categorization:**

  You have made the following settings in Customizing for **Service**:

  - Activate categorization for the interaction record under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**.
  - Make the necessary settings under **Transactions** > **Basic Settings** > **Define Transaction Types**.
  - Define the subject profile for the category modeler under **Basic Functions** > **Catalogs, Codes and Profiles**.
  - Make the necessary categorization settings for the interaction record under **Interaction Center WebClient** > **Business Transaction** > **Define Categorization Profiles**.

  See Multi-Level Categorization for more information.

- If you want to use the lean interaction record instead of the full interaction record, you have assigned a business transaction profile that contains transaction type 0100 to the business role. You do this in Customizing for **Service** under **UI Framework** > **Business Roles** > **Define Business Role**.

Activity Clipboard

Use

Interaction Center (IC) agents can use this function to view a list of the business objects involved in a current or past interaction. It is located in the Interaction Record or the navigation area.

i **Note**

The activity clipboard in the interaction record visualizes past and current interactions.

The activity clipboard in the navigation area visualizes current interactions.

The objects that were created or changed during those interactions are listed including navigation links.

Integration

This function is integrated with:

- Interaction Record
- Interaction History
- Business data context

  In the Interaction Center (IC), the activity clipboard is one view of the business data context.

Prerequisites

Depending on where you want the activity clipboard to appear, you have to make different settings in Customizing:

Activity Clipboard in the Interaction Record

- You have made the necessary settings in Customizing to define the display of the activity clipboard so that the business partner appears in the activity clipboard list. Your system administrator can either use the default activity clipboard profile, or define which objects are relevant for linking and should appear in the activity clipboard. You do this in Customizing for Service, by choosing Interaction Center WebClient > Basic Functions > Define Activity Clipboard Profiles.
- Both the programming of the business object itself and your Customizing settings must be in place for a business object to appear in the activity clipboard.

Activity Clipboard in the Navigation Area

You have made additional settings in Customizing for Service:

- Define a direct link group of the type EE under UI Framework > Technical Role Definition > Define Navigation Bar Profile > Define Direct Link Groups.
- Assign the direct link group to a navigation bar profile under UI Framework > Technical Role Definition > Define Navigation Bar Profile > Define Navigation Bar Profiles > Assign Direct Link Groups To Nav. Bar Profile.
- Assign the navigation bar profile to your business role profile under UI Framework > Business Roles > Define Business Role.

Features

If a business object is accessed repeatedly, it only appears once on the activity clipboard, in the order in which it was first accessed.

- By default the data is displayed in three columns, however, in Customizing you can choose the number of columns to be displayed with the maximum being five columns. The first column contains the name of the object or activity. Your system administrator can define what information appears in the other columns in Customizing for the activity mentioned above. By default, the Object ID and the Process Type are displayed.
- You can use the links to navigate to objects that appear in the activity clipboard, such as leads or sales tickets.

Account Identification/Account Fact Sheet

Use

IC agents can use this part of the interaction center (IC) to identify accounts in different ways, such as by name, company, or product, and to obtain an overview of an account and contact. If an account does not yet exist in the system, or if it must be updated, IC agents can also create and change accounts from this screen. In addition, they can quick create an account with just the basic data.

Prerequisites

- You have set up one of the following account identification profiles:
  - The standard account identification profile in Customizing for Service under Interaction Center WebClient > Master Data > Define Account Identification Profiles.

- The profile that enables confirmation of multiple business partners at once in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Account Identification Profiles for Multiple Business Partners**.

- In your account identification profile, you have made settings for the following:
  - Contact attached data (if you want to use this feature)
  - Configurable context area (if you want to use this feature)
  - Embedded object components
  - Search scenario
  - Search results filtered for account roles and contact person roles
  - Default contact type for contact searches using search fields for contact communication details (**Telephone**, **E-Mail**, **Fax**)
  - International address versions (if you want to use this feature)
  - Employee search (if you want to use this feature)

- If you are using contact attached data (CAD), you have mapped the contact attached data from the communication management software to account identification using an XML file.

  You have created a function profile in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Profiles for Contact Attached Data**.

  Additionally, if you want to use CAD attributes when creating alerts or rules, you have enabled the relevant CAD attributes for fact gathering. For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Enable CAD Attributes for Fact Gathering**.

- If you are using international address versions, you must have activated them. You do this in Customizing for **Service** under **SAP NetWeaver** > **Application Server** > **Basis Services** > **Address Management** > **International Settings** > **Activate International Address Versions**.

  Mapping from logon language to address version is available in the default implementation of Business Add-In `ADDR_LANGU_TO_VERSMAP_LANGUAGE_TO_VERSION`, or an implementation exists that maps the logon language to that version.

- If you are using employee search, you have defined employees as business partners.

- If you are using the standard account identification profile, you have defined any object components that you want to be displayed in the upper right area of the screen in Customizing for **Service** under **Interaction Center WebClient** > **Customer-Specific System Modifications** > **Define Object Components**.

  If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you define in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Account Identification Profiles for Multiple Business Partners**.

- You have defined an interaction history profile in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Interaction History Profiles**.

- If you are using the briefcase, you have made the necessary setting in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Briefcase**.

- If you are using the enhanced search, you have made the necessary setting in Customizing for **Service** under **Master Data** > **Business Partner** > **Search** > **Activate Wildcard Search for Accounts and Contacts**.

- If you want to ensure that all rules and events that are triggered when the **Business Partner Confirmed** (`BPConfirmed`) event is raised are also triggered when a call is transferred, you have added the **Business Partner Confirmed for Call**

Transfer (`BPConfirmed_Trans`) event to the relevant rule policy in the rule modeler. Once you have done so, an account that is confirmed by an agent is confirmed automatically if the call is transferred to a second agent, and the system does not create a new interaction record when the call is transferred. For more information about creating rule policies, see Rule Modeler.

Features

- Automatic number identification (ANI)

  If your telephony switch system has ANI, the system can identify accounts even before IC agents accept contacts.

- Business context data (BCD)

  When IC agents confirm an account or product, the system displays all relevant information for the account or product in the activity clipboard. This information is also recorded in the BDC so that it can be reused by other functions.

  For more information, see Activity Clipboard.

- Contact Attached Data

  When IC agents confirm an account, the system can incorporate contact attached data from the communication management software (CMS). This information is displayed on the contact attached data screen.

  You can also enable contact attached data attributes for fact gathering so that these attributes can be used when creating alerts or rules. Using these attributes, you can, for example, retrieve a service request number from the CAD attributes of an incoming call and set up a rule that displays an alert providing this information to the IC agent once the account has been confirmed. The IC agent can then click the service request number in the alert to navigate directly to the service order. When using CAD attributes for this purpose, you must ensure that the data that is passed has the same CAD group name and CAD attribute name that have been defined in Customizing.

- Configurable Context Area

  The context area provides background information for the current interaction. If you would like to display different account information in the context area, ask your system administrator to configure the context area accordingly.

- Embedded Object Components

  If you are using the standard account identification profile, you can replace the upper right area of the **Account Identification** screen with your own views that enable you to search for business partner-related business objects. You can define object components so that the agent can search for:

  - Objects based on the confirmed business partner
  - Business partners based on the confirmed objects

  If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you have defined.

- International Address Versions

  If international address versions are activated in your system, you can use the default character set or the international character set to search for business partners (provided a character set is defined for your logon language).

  The system automatically searches in both the default character set and the international character set. For example, in Japan, agents can search for the caller's name using the furigana version of the name. As a result, all names with the same pronunciation are displayed, irrespective of the character set.

  International address versions are available for all countries for which you have maintained an international address version.

- Search Scenario

You can customize search fields and search views that appear as tabs on the **Account Identification** screen. For each search, you can define the title of the tab and the type of search (for example, business partner or object component).

You can also define the sequence of relationship types that appear in the search criteria.

- Filter search results for account roles and contact person roles

  You can set the system to filter search results for account searches by account roles and by contact person roles.

- Define default contact type for searches using the contact communication data search criteria

  For contact searches using the fields for contact communication details (**Telephone**, **E-Mail**, **Fax**), you can set a default contact type in Customizing that determines whether the search criteria are used to search for a contact of the type account, contact person, or both. IC agents can change this setting in the WebClient UI.

- Use contact name in account search

  When searching for accounts in the IC, entering data in both the contact name fields and the account fields will generate a combined search result and provide more precise results.

  Note

  You cannot search for accounts using the account search fields in combination with the search fields for contact communication details (**Telephone**, **E-Mail**, **Fax**). The contact communication details take priority over search criteria entered in the other search fields. If one of the search fields for contact communication details is filled, the system performs a contact search based on the communication details only.

- Employee Search

  You can configure your interaction center to service internal employees. IC agents can use the employee search to look up employee details such as cost center, personnel area, employee group, and employee subgroup. They can also look up telephone numbers and e-mail addresses, or, when they receive a telephone call or e-mail, the system can identify the employee by ANI or from the incoming e-mail, and display the employee's details.

- Multiple Business Partners

  You can configure your interaction center so that IC agents can confirm more than one business partner at once. You can also define a list of partner functions that can be used to confirm business partners and determine the sequence in which partner functions are displayed. You can then map the partner functions of confirmed business partner to the existing partner functions in business transactions and the interaction record.

- Account Fact Sheet

  The account fact sheet gives an overview of the confirmed account. You can personalize views of the fact sheet. For more information, see Fact Sheet.

- Briefcase

  You can set up a briefcase that contains a collection of briefing cards that give you an overview of relevant information, such as addresses and contacts. Each briefing card can be assigned to more than one briefcase.

- Account Quick Create

  You can quick create an account from every business object such as an opportunity or an activity. When you save the new account, the data you have entered in the fields for the business object is automatically filled in for the new account.

- Enhanced Search

  You can search for an account and contact with the following search criteria:

  - Region by Country

The search criterion refines your search for accounts by filtering the correct region of the respective country.

- E-Mail, Telephone, and Website

  You can enter an asterisk (*) at the beginning or the end of a search string to perform a wildcard search. For e-mail, you can also put the asterisk within a search string.

Chat

Use

Interaction center (IC) agents can use the chat function to process incoming chat messages. The chat function in the IC includes standard chat functions, such as joining chat sessions, sending messages, and transferring chat sessions. Agents can transfer chat sessions to other queues or agents. Agents can also see additional details such as chat subject, location where the customer initiated the chat, and shopping cart items. Further features are described below.

Integration

Chat is integrated with the following IC functions:

- Toolbars

  Agents use the channel-specific toolbar that is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Interactive scripting

  Agents can use scripts to guide them through the chat session. The script is available directly next to the chat text area, so agents can easily and quickly access the information without navigating to a different screen.

- Knowledge search

  Agents can use the knowledge search to find solutions to problems or other issues reported by customers.

- Telephony

  Agents can remain in the chat channel and make a short phone call by using **Dial** in the toolbar.

- Interaction record

  Depending on Customizing settings, the chat transcript can be saved and later accessed in the activity clipboard of the interaction record.

Prerequisites

- The Integrated Communication Interface (ICI) is configured (see Integrated Communication Interface).

- You have defined the chat profiles in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Chat Profiles**.

- You have assigned chat profiles to agents through business roles.

- If you want to use standard responses, you have defined the standard response group in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Standard Response Groups**.

- If you want to add additional knowledge bases for use with chat, you have made the settings for knowledge base implementation in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication**

Channels > **Define Knowledge Base Implementation** >.

- If you want a list of available queues when transferring chat sessions, Communication Management Software settings are defined.

Features

- Account identification

  The customer search is automatically started when an incoming chat session is routed to an available agent. The customer's e-mail address or other business partner details are used to determine if customer data exists. If the account is confirmed, then chat transcripts can be saved to the confirmed business partner's interaction record. For more information, see Account Identification/Account Fact Sheet.

- Standard responses

  IC agents can insert standard responses into the chat text. They can choose from a predefined list of commonly used responses, or can use the standard response screen to search and preview other responses.

- Knowledge base integration

  You can define which knowledge bases IC agents can access while using the chat function. You can, for example, integrate knowledge bases such as interactive scripts, knowledge articles, mail forms (standard responses), solution database (problems and solutions), or chat transcripts.

- Multiple chat sessions

  IC agents can process multiple chat requests simultaneously to maximize efficiency.

  > **i Note**
  >
  > For more information about settings for multiple chat, contact your communication management software (CMS) vendor.
  >
  > When you use multiple chat, several parallel HTTP sessions are created on the server in Service. This affects the sizing on the server side.
  >
  > If you are using multiple chat, the client switch scenario is not supported.

- Web shop integration

  IC also supports chat in Live Web Collaboration in Web Channel. You can integrate IC with your Web shop and with Internet Customer Self-Service (ICSS) so that customer requests for assistance via chat are forwarded directly to the interaction center.

E-Mail

Use

You can use this function to process incoming and outgoing e-mails. This function is integrated with the following functions in the interaction center (IC):

- Toolbar

  IC agents use the channel-specific toolbar, which is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Account identification

  Upon receipt of an e-mail, the customer search is automatically initiated based upon the **From** e-mail address. For more information, see Account Identification/Account Fact Sheet.

- Information search

  Results from the knowledge search can be added to e-mails as attachments or as text in the e-mail message. For more information, see Knowledge Articles.

- Agent inbox

  IC agents can access e-mail through their inbox and process them. For more information, see Agent Inbox.

- Interaction record

  IC agents have access to previously sent or saved e-mails in the confirmed business partner's interaction record. For more information, see Interaction Record.

- Telephony

  IC agents can make a phone call by using the **Dial** function in the application toolbar while remaining in the e-mail channel.

Prerequisites

- You have made settings to display fields (for example, date and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.

- You have made settings for e-mail in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels**. For example, you have made the following settings:

  - You have defined toolbar properties for e-mail in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

  - You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Standard Response Groups**.

  - You have defined outgoing e-mail address groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Outgoing E-Mail Address Groups**.

  - You have defined e-mail profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

Features

- Standard responses

  IC agents can reuse use predefined text. For more information, see Standard Responses.

- Multiple incoming e-mail addresses

  Interaction centers can use more than one incoming e-mail address. Each address is assigned to a group of IC agents. IC agents can be assigned to more than one address.

- Multiple outgoing e-mail addresses

  Interaction centers can also use more than one outgoing e-mail address. IC agents are assigned to one default outgoing e-mail address based upon their e-mail profile as defined in Customizing.

- Unfinished e-mails

IC agents can process unfinished e-mails if they need to deal with another transaction before completing an e-mail response.

Smart Responses

Use

Smart responses are standard responses with the ability to include additional dynamic information not limited to simple business partner attributes or system information.

Text is populated with business values and inserted into the e-mail content at runtime. This reduces the amount of time agents have to spend searching for and compiling information in a response. Agents are unaware if the response they choose is a standard response or a smart response.

Example

A customer sends an e-mail to the interaction center asking what his account balance is. After the agent confirms the business partner and initiates a reply to the e-mail, the agent selects a standard response. Variables such as account balance and account number are populated with the business partner's values in the response.

Note

Interaction center agents cannot insert smart responses manually into an e-mail using the **Standard Response** dropdown.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function in the role of an interaction center manager under **Knowledge Management** > **Mail Forms**.
- You have defined Multilevel Categorization. (Not relevant for Auto Acknowledgements)
- You have defined the rule policy with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.
- You have defined the data source classes in Customizing for **Service** by choosing **Interaction Center WebClient** > **Customer-Specific System Modifications** > **Smart Responses** > **Define Data Source Classes**.

Features

Like standard responses, you can insert placeholders in the mail form that can be filled in by data related to the e-mail or to the business partner of the e-mail. You can define how the values for business and transactional data are retrieved and placed in mail forms. You can use data from either the fact base attributes or from the business data context (BDC).

Smart responses can be sent automatically to customers with or without agent intervention based upon multilevel categorization. They can be inserted into the e-mail text or attached to the e-mail. For more information, see the actions and parameters table in ERMS Rule Policies for the **Auto Respond** and **Auto Prepare** actions. They also can be used in auto acknowledgements and response proposals.

Standard Responses

Use

IC agents can use standard responses to help automate e-mail processing. Standard responses use predefined text configured from mail forms that are available to IC agents in the e-mail editor. Standard responses provide consistent handling and messaging to customers regardless of which IC agent processes the e-mail.

On the e-mail editor screen, two assignment blocks offer standard responses.

1. The Standard Responses assignment block provides a list of standard responses from four different sources:
    - Standard response groups as defined in the Define Standard Response Groups Customizing
    - Standard responses are available in the e-mail editor when you select a relevant category
    - Standard responses from a manual search done by the IC agent
    - Standard responses that the IC agent has saved in their favorites.
2. The Standard Response Tree assignment block displays standard responses based on the assigned multilevel categorization schema. Standard responses are also available in chat text. For more information about e-mail and chat, see E-Mail and Chat. Multilevel categorization and Customizing settings help determine which standard responses are available.

Prerequisites

- You have defined standard response groups in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Standard Response Groups.
- You have assigned the standard response groups to e-mail profiles in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles.
- If you want to use standard responses for chat, you must assign the standard response groups to chat profiles in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Chat Profiles.
- If you want to determine the default language for your standard response texts, you must make settings in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > BAdIs for Communication Channels > BAdI: Language Determination for Standard Response Texts.
- If you want to filter the standard responses to limit their use across channels, you must make settings in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > BAdIs for Communication Channels > BAdI: Filter for Standard Response Texts.
- If you want to use the Standard Response Tree assignment block, you must have your e-mail editor profile to the `EMAIL_PROFILE` parameter for the e-mail application area. You can choose any categorization schema without the restriction of using an email profile when you use the `CRM_IC_STD_RESP_SCHEMA` business add-in (BAdI) definition, within enhancement spot `CRM_IC_STD_RESPONSE`. These standard responses will be in addition to those defined in the Define Standard Response Group Customizing.

Features

IC agents can do the following:

- Perform a search and preview standard response texts before choosing a standard response
- Choose the language of standard responses after searches
- Sort and filter standard responses

- Manage favorite standard responses
- View the source of standard response proposals (for example, from favorites or searches)

You can use variables in the mail form to insert placeholders for business partners. When IC agents choose the standard response or insert the standard response into an e-mail, the system fills these placeholders with values. For example, you can populate the customer's name in the salutation.

More Information

- Auto Suggest
- Smart Responses
- Actions and Parameters
- ERMS Rule Policies

Fax and Letter

Use

Interaction center (IC) agents can create, edit, and receive fax messages and letters in the interaction center. IC agents can also use an internal HTML editor to process faxes and letters.

Prerequisites

- You have made settings to display fields (for example, subject, date, and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- If you would like the **To** field of fax messages to be filled automatically, you have maintained the business partner (BP) data with a fax number that includes the country code.
- If you want to use business process push (BPP) with fax and letter, you have to specify the communication management software (CMS) connection by executing the program on the **SAP Easy Access** screen, under **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**. For more information about BPP, see Business Process Push.

  ⚠ **Caution**

  You must enter the following values:

  - Under **Comm. Mgmt Software System ID** enter **FAX_LETTER**.
  - Under **Connection** enter the RFC connection to your CMS system.

Features

Agents can search for incoming fax messages and letters in the agent inbox and display them. They can also view old fax messages or letters for a confirmed account in the interaction record.

i Note

Once a fax has been sent or a letter has been finished, it can no longer be edited.

Using the HTML editor, IC agents can:

- Create, send, and print fax messages and letters
- Display information for the current interaction, such as:
  - Reason
  - Priority
  - Status
  - Name
- Add and display attachments
- Change the default header data for fax messages and letters

Activities

Creating Letters

Before IC agents can process letters in the interaction center, you must upload them to the interaction center by doing the following:

1. Scan the letters.
2. Save them to your hard drive.
3. Go to transaction `oawd`.
4. Expand the node and double-click the highlighted line that appears.
5. Select **Drag & Drop**.
6. In the dialog box that appears, drag and drop the relevant files from your local drive.

Agent Inbox

Use

You can use the agent inbox (inbox) to call up e-mails, business transactions, and various other business objects for display or processing. The inbox is a central worklist that the entire team can use to work on incoming objects.

Prerequisites

- You have made the basic settings required to use the inbox. For more information, see [Basic Settings for the Agent Inbox].
- If you want to display an alert showing the number of incomplete e-mails linked to a particular account, you have created a rule policy and a rule to set up the alert. The rule policy must include the **Business Partner Confirmed** (`BPConfirmed`) event and the rule must include the **Show Number of Incomplete E-Mails** (`AH_NO_CUST_MAIL`) action.

  For more information about creating rule policies, see [Rule Modeler] and [ERMS Rule Policies].

Features

Business Roles

You can use the inbox in business roles with the following profile types:

- IC WebClient Business Role
- CRM WebClient Business Role

> **i Note**
>
> The following inbox functions are not available for business roles with the profile type CRM WebClient Business Role:
>
> - Interact
> - Link
> - Next item
> - Displaying e-mails, fax messages, and letters

Inbox Items

You can retrieve, access, and process the following items in the inbox:

- Business transactions in Service

  You can call up business transactions, such as service orders.

- Workflow work items

  You can process SAP workflow work items, including those that you have customized for use in the inbox. These work items can originate from the current system (SAP S/4HANA or SAP S/4HANA Cloud Private Edition) or from another SAP system.

- E-mails, fax messages, and letters

  - Outbound correspondence

    The inbox supports outbound e-mails, fax messages, and letters created in the interaction center (IC) and linked to an interaction record. The inbox also supports outbound e-mails created by the E-Mail Response Management System (ERMS).

  - Inbound e-mails, fax messages, and letters

    E-mails and fax messages that were received via SAPconnect appear as workflow work items. Inbound letters are also available in the inbox as work items.

    A workflow forwards inbound e-mails to the appropriate group of agents.

  - Multiple incoming and outgoing e-mail addresses

    Agents can be assigned to various incoming addresses. All items sent to the agent's various addresses appear in the same inbox. Agents can also be assigned to more than one outgoing e-mail address. A default address is proposed, or the agent can select a different address.

  - ERMS routing

    Depending on the rule policies you define, you can use a general task for an ERMS decision to allow e-mail routing to all agents or you can specify individual agents or agent groups to allow e-mail routing only to those agents.

- Number of incomplete e-mails linked to confirmed account

  You can set up the inbox so that once an account is confirmed, an alert is displayed that shows the number of incomplete e-mails linked to the confirmed account. IC agents can click the alert to navigate directly to the inbox, where only the incomplete e-mails from the confirmed contact person of the account are displayed. If no contact person has been confirmed, only e-mails from the e-mail address linked with the account are listed in the inbox.

Inbox Search

For information about the inbox search features, see Inbox Search.

Inbox Result List

For information about the inbox result list features, see Inbox Result List.

Basic Settings for the Agent Inbox

Use

System administrators use this process to set up the agent inbox in the Interaction Center (IC) WebClient.

Prerequisites

- The SAPconnect interface is configured.

  For more information about SAPconnect, search for “SAPconnect (BC-SRV-COM)” on the SAP Help Portal at http://help.sap.com.

- E-mail addresses are created.

- If you want to edit scanned letters in your agent inbox, you have defined an ArchiveLink scenario.

  For more information, see the following Customizing activities under SAP NetWeaver > Application Server > Basis Services > ArchiveLink :

  - Basic Customizing > Edit Document Classes and Edit Document Types
  - Customizing Incoming Documents > Workflow Scenarios > Assign Document Type to a Workflow

Process

Step	Description
Create business partner master data for each employee, with an employee role.	Maintain all inbound e-mail addresses that the agent should be responsible for. Assign a user ID.
Maintain settings for recipient distribution	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Maintain Recipient Distribution .

Step	Description
Maintain automatic workflow Customizing	In Customizing for Service, choose Basic Functions > SAP Business Workflow > Maintain Standard Settings for SAP Business Workflow. For more information, see SAP Business Workflow in the Interaction Center.
Activate event linkage for workflow template `WS 14000164` and assign agents for e-mail handling	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Assign Agents for Fax and Letter Processing. For more information, see Assigning Agents for E-Mail Handling.
Forward to employees/groups	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding. Use this transaction to define the content of the dropdown box Forward To in the result list of the agent inbox.
Define quick searches	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. Use this transaction to define the content of the dropdown box Quick Search in the agent inbox search.
Maintain Customizing activities	In Customizing for Service, choose Interaction Center WebClient > Agent Inbox. For more information, review the Implementation Guide (IMG) documentation for each activity.

More Information

Agent Inbox in the Interaction Center WebClient

Assigning Agents for Fax and Letter Processing

Use

System administrators use this procedure to make the following settings for the agent inbox (inbox) in the Interaction Center (IC) WebClient:

- Activate event linking.

  Workflow can define how incoming fax messages and letters are routed and processed. Activating event linking activates the workflow you defined and allows it to be used for the agent inbox. You only have to activate event linking once.

- Assign agents for fax and letter processing.

i Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see ERMS Rule Policies.

Procedure

Activating Event Linking

1. In the **SAP Easy Access** menu, choose **Service** > **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Agent Inbox** > **Assign Agents for Fax and Letter Processing**.
2. In the **Event Linkage** column, choose **Activate event linking**.
3. Under **IC WebClient: Inbound Process AUI** expand the workflow template `WS 14000164` (**IC: Mail Inbox Handling**).
4. Select the icon in the **Activate/deactivate** column.

   Once selected, the icon should display as **Activated**.

Result

Workflow is activated for the agent inbox.

Define Rules for Fax and Letter Processing

Use

System administrators follow this procedure to define routing rules for incoming fax messages and letters in the agent inbox (inbox) in the Interaction Center (IC) WebClient.

The IC WebClient inbox uses workflow to route incoming fax messages and letters to agents.

Additional routing rules use business partner attributes to route e-mails to agents who specialize in particular areas (see Additional Routing Rules for Fax and Letter Processing).

Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see ERMS Rule Policies.

Process

As a system administrator, you define routing rules in the following steps:

1. Create rules.
2. Assign rules to agents or grouping of agents.

Creating Rules

1. In the **SAP Easy Access** menu, choose **Service** > **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Agent Inbox** > **Define Rules for Fax and Letter Processing**.
2. Choose **Create responsibility**.

3. Enter the following:
   - Object abbreviation
   - Description
   - Start date
   - End date
4. Choose **Continue**.
5. Enter a priority.
6. Enter values for each business partner field that you want to use to further define routing.

   To enter more than one range for the same field, select the field and choose **New row**.

   Enter * (asterisk) for the fields that you do not want to define.
7. Save your data.

Assigning Rules to Agents

1. On the **Responsibilities for Rule** screen, select the rule that you created.
2. Choose **Insert agent assignment**.
3. Select an object type.
4. Search for and select one or more objects of the type that you specified in the previous step.
5. Choose **Create**.

If you want to delete an assignment, select a rule on the **Responsibilities for Rule** screen and choose **Delete agent assignment**.

Example

You want to use the business partner language Spanish as an additional routing field. You make the following settings:

Name	From	To
Address	*	<no entry>
Country	*	<no entry>
Language	ES	<no entry>
Name	*	<no entry>
Region	*	<no entry>
Postal code	*	<no entry>

Additional E-Mail Routing Rules

Use

The Interaction Center (IC) WebClient agent inbox uses workflow to route incoming mails to agents. In the standard workflow, after a customer sends an incoming mail to the interaction center, the system retrieves the address to which the customer sent the mail. It determines which agents are assigned for processing incoming mails for that address. This determination is based on the e-mail addresses entered in the employee's business partner master data.

In addition, you can use certain business partner data to define which agents or groups of agents should process the mails (for example, to route e-mails to agents who specialize in particular areas). The routing rules can be assigned to object types such as users, positions, and organizational units.

Prerequisites

Routing rules for e-mails have been defined and assigned to agents or groups or agents (see Defining Routing Rules for E-Mails).

Features

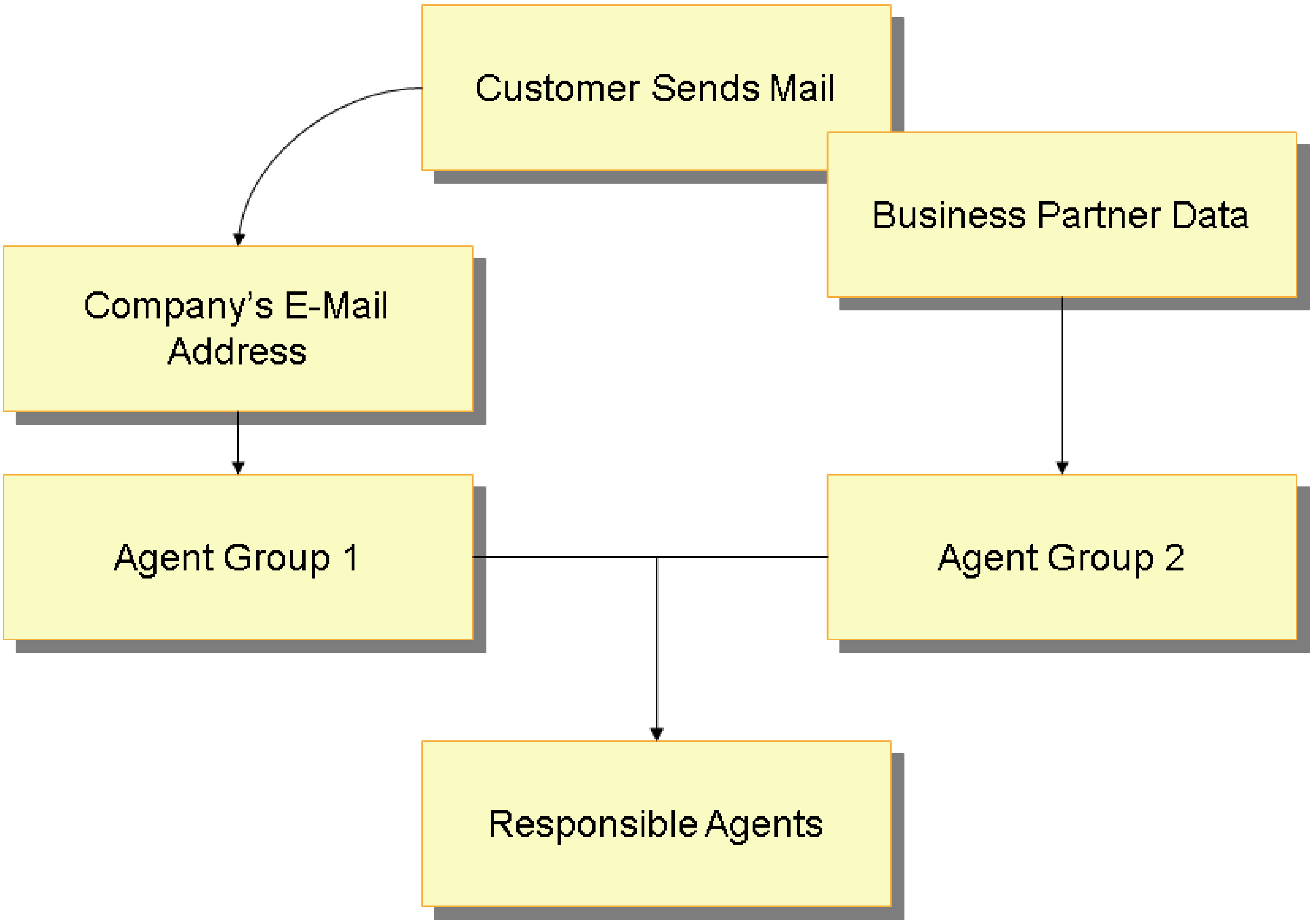
The following business partner data can be used to define how mails are routed:

- Address
- Country
- Language
- Name
- Region
- Postal code

To use other fields, you need to create a new rule. From the **SAP Easy Access** menu, choose **Architecture and Technology** > **ABAP Workbench** > **Development** > **SAP Business Workflow** > **Definition Tools** > **Rules for Agent Assignment.** > **Create**. The **Responsibilities** tab allows the creation of rule definitions and assignments to agents.

Activities

The figure below shows how the system determines which agents process incoming mails and is followed by an explanation:



1. With additional routing rules, the system tries to determine the business partner by the customer's e-mail address. If the business partner is determined, attributes from the business partner master data can be used to further define which agents process the mail.

   Note

   If the business partner cannot be determined, the additional routing rules are not used. In this case, agents who are assigned to process mails for that incoming address or number receive the mail (see [Defining Routing Rules for E-Mails](#)).

2. The system checks which agents are assigned to the incoming mail address (**Agent Group 1**), as specified in the inbound distribution.

3. The system checks which agents are assigned to process the specific business partner attributes (**Agent Group 2**) if the business partner address or number is found. This depends upon the defined rules and which agents are assigned to them.

4. The routing depends on how many agent groups exist:

   - Two agents groups

     The mail is routed only to agents who are in both of the groups (**Agent Group 1** and **Agent Group 2**). If no agents are found in both groups (agents in **Agent Group 1** and **Agent Group 2**, but none belonging to both groups) the mail is routed to all agents in each group.

   - One agent group

     If no agents are in **Agent Group 1**, the mails are routed to agents in **Agent Group 2**. Alternatively, if there are no agents in **Agent Group 2**, the mails are routed to agents in **Agent Group 1**.

Example

Routing works as follows when two agent groups exist:

1. Mr. Smith sends an e-mail to service@company.com.

   Julie and David (Agent Group 1) are assigned to process mails sent to this e-mail address.

2. The system identifies Mr. Smith's business partner data based upon his outgoing e-mail address.

3. The system determines that an additional rule can be used based upon Mr. Smith's country (US).

   Michael and Julie (Agent Group 2) are assigned to an e-mail rule for US customers.

4. The system determines that only Julie should receive the incoming mail because she is the only agent in both of the agent groups.

   If Julie was not in both of the groups, then all of the agents, David, Michael, and Julie, would receive the incoming mail.

Define Recipient Profile for Forwarding

Use

Use this procedure to define to which person(s) you may forward items for further processing from the agent inbox in the Interaction Center (IC) WebClient. Inbox items may be forwarded to an individual employee or a group.

Prerequisites

Business partners specified as recipients must be defined.

Procedure

In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding.

Quick Searches

Use

You can define searches that can be used as options for finding business objects in the agent inbox (inbox).

Prerequisites

Features

You can define quick searches in the SAP Easy Access menu, under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches.

SAP delivers one Quick Search (0001) that is defined to select My Open Items sorted first by priority, then by main category. You can define additional quick searches based on your own selection criteria.

You can use an asterisk (*) as a wildcard to specify the title of a business object that has been derived from the business transaction.

You can also define a default quick search for each business role in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Note that the inbox supports the following time periods for quick searches: creation period, change period, and due period.

Example

You can define quick searches to retrieve item groups such as:

- All overdue items
- All items for which you are responsible
- All high-priority items

Inbox Search

Use

The agent inbox (inbox) search provides a broad range of search attributes and search features that you can use to tailor the inbox search according to your business requirements.

Prerequisites

You have set up and configured the inbox. For more information, see Agent Inbox in the Interaction Center WebClient.

Features

Search Attributes

For information about the available search attributes, see Search Attributes.

Search View: Standard and Advanced

The standard search provides a fixed arrangement of search fields that the user cannot change.

The advanced search provides selectable search attributes, the multiple use of search attributes, and the use of saved searches.

You can specify whether you want to use the standard search or the advanced search. You do this in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Lean Search

During a lean search, the system determines only the required data and does not store any data. It only determines the object attributes that are relevant for the search result list. This enables a faster search and faster processing times when working with the result list. The remaining attributes are determined when the object is used, for example, when the object is opened in the display mode or in the edit mode.

Predefined Searches

Quick Searches

You can create predefined search queries that appear as selection criteria in the **Quick Search** field. For example, you can define a search that agents can use to retrieve all high-priority items. For more information, see Quick Searches.

Quick searches are assigned to business roles through the inbox profile.

Saved Searches

The saved searches function is only available in the advanced search. Saved searches are user-defined.

Search Attribute Validity Check

The following attribute validity check prevents unclear search results:

If a main category cannot be combined with a selected search attribute, the main category is excluded from the search. For example, a search involving the main category **E-Mail** in combination with the **Checklist ID** search attribute is not valid, and the search is not performed.

This system behavior also applies if only one main category has been selected or if no main category has been selected. In the latter situation, the validity check is performed for all main categories that are defined for the corresponding inbox profile in Customizing, and the search is performed for all main categories that have passed the validity check.

Search Traceability

A search processing log consolidates various detailed search messages, such as which search attribute is not supported by which main category, in a structured way. The processing log can be accessed using a corresponding button in the result toolbar.

In addition to the entries in the processing log, the following general messages are displayed on the UI, as appropriate:

- Search has been only partially performed
- Search could not be performed

This enables the user to understand the search behavior and to tailor future searches accordingly, or to contact the system administrator, as required.

Predefined Inbox Search Tables

You can use predefined inbox search tables that contain the inbox search attributes and inbox result attributes that are relevant to specific inbox item types. Predefined inbox search tables are available for the following inbox item types:

- **Inbox Item: OneOrder**
- **Inbox Item: Workitem - Inbound Email, Fax and Letter**
- **Inbox Item: Outbound Correspondence**

For information about the prerequisites and features, see Predefined Inbox Search Tables.

Search Attributes

Use

You can use various search attributes and search attribute combinations to locate the required agent inbox (inbox) items.

Prerequisites

For information about the necessary settings for the inbox search, see Inbox Search.

Features

Available Search Attributes

Area	Search Attribute	Function
Predefined searches	Quick Search	Predefined search criteria (for example, My Open Items) You can configure these searches on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. You must assign the relevant quick searches to the corresponding inbox profile, and you can define a default quick search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Saved Searches	The saved searches function is available in the advanced inbox search.
Inbox item	Main Category	Inbox item type as defined in the following Customizing activities in Customizing for Service under Interaction Center WebClient > Agent Inbox: • Inbox Search Definitions > Define Item Types for Searches • Define Inbox Profiles
	Object ID	ID of the inbox item
	Description	Description of the inbox item
Partner information	Created By	User who created the inbox item
	Account	Account for which the inbox item has been created
	Assigned To	Assigned agent or agent group To use the Assigned To search attribute to search for business transactions and cases, the relevant user or group must be maintained as a business partner in the system. The business transaction and case search is based on the business partner in combination with the inbox partner mapping (inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The search for work items (inbound e-mails, fax messages, and letters), outbound correspondence (outbound e-mails, fax messages, and letters), social media posts, and workflow work items is based on the assigned users. Assigned To: Me The search returns the inbox items that are assigned to the current user. The search returns the following results: • Business transactions and cases to which the current user is assigned as the employee responsible • Work items for which the current user is the employee responsible (assigned agent)

Area	Search Attribute	Function
		• Work items that are available for processing by the user and have not yet been picked by another user • Outbound correspondence that was last changed by the user • Social media posts for which the current user is the employee responsible (assigned agent) Assigned To: My Groups The search returns the inbox items that are assigned to the selected groups. The search returns the following results: • Business transactions and cases to which the selected groups are assigned as the groups responsible • Work items to which the users of the selected groups are assigned as the processor • Work items that are available for processing by the users of the selected groups • Outbound correspondence that was last changed by users of the selected groups • Social media posts to which the users of the selected groups are assigned as the processor Assigned To: <no entry> For work items, the Assigned To: My Groups search is performed. Note that the search for workflow work items always returns the workflow work items assigned to the current user. Therefore, the Assigned To search attribute is not relevant for workflow work items.
	Employee Responsible	Assigned employee responsible The specifics for the Assigned To: Me search attribute also apply to the Employee Responsible search attribute. The Employee Responsible search attribute cannot be combined with the search attribute Assigned To.
Time	Creation Period Change Period Due Period	Predefined periods of time for searching inbox items (for example, Last 3 days) In the standard search, you can use the respective From/To fields to enter a user-defined date or date range. You can define the time periods in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Time Periods for Searches. You can define which time periods are available for the respective search attributes in the inbox search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Created On Changed On Due On	Date or date range These search attributes are only available in the advanced search.
Importance	Status	Inbox status, such as Open, In Process, Completed

Area	Search Attribute	Function
		You can define the inbox statuses in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Inbox Status. You must map the system statuses and user statuses of the inbox item types to the inbox statuses. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Status to Inbox Status.
	Priority	Inbox priority, such as High, Medium, Low You must map the inbox item type priorities to the inbox priorities. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Priorities to Inbox Priorities.
Multilevel categorization	Category ID	ID of category used in multilevel categorization For more information, see Multilevel Categorization.
Checklist	Checklist ID	Not applicable
E-mail addresses	Sender Address Recipient Address	E-mail addresses that can be used to search for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) If you are not using the predefined inbox search table for work items, entries in the Sender Address field are restricted to 20 characters (use an asterisk in case of longer e-mail addresses). • Outbound correspondence (Outbound e-mails, fax messages, and letters)
Sorting	Sort By	Sort attributes by which the search result is sorted A maximum of two sort attributes can be used for each search.
Generic functions	Clear	Clears the search attribute values
	Reset	Sets the inbox search criteria and the inbox result list back to the initial state: • The Quick Search field is cleared. • The search criteria is set back to the predefined combination of search fields (relevant to the advanced search only), and the field values are cleared. • The Maximum Number of Results field is set back to the default. • The result list is cleared, and the sorting and the filtering are set back to the initial state. Note that personalization settings for the result list and the selected display type (table or tree) are not set back.
	Maximum Number of Results	Maximum number of search results You can define the maximum number of search results to improve the search performance, for example. You can define the default values and range of user entries for the Maximum Number of Results field. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.

Search Attributes not Enabled for Multiple Use:

- Category ID
- Assigned To: Me and Assigned To: My Groups cannot be combined with other Assigned To search criteria. You cannot for example perform a search using Assigned To: Me in combination with Assigned To: My Groups.
- Sort By

Predefined Inbox Search Tables

Use

The standard system contains predefined inbox search tables, which contain the inbox search attributes and inbox result attributes relevant for a specific inbox item type. You can improve the search performance by using predefined inbox search tables. For each of the following inbox item types, a separate table is available:

- Inbox Item: Workitem - Inbound Email, Fax and Letter

  Inbox search table for work items (`CRMD_AUI_WI_IDX`)

- Inbox Item: Outbound Correspondence

  Inbox search table for outbound correspondence (`CRMD_AUI_OUTCORR`)

  > **Note**
  >
  > For outbound correspondence, the predefined inbox search table is mandatory for both of the abovementioned inbox item types.

- Inbox Item: Inbox One Order item

  For this inbox items type a predefined search table is not required because the search is optimized already. However, in case of Customizing changes, this inbox item type must be updated. For more information, see the *Features* section.

Prerequisites

General

- You have defined the business transactions and work items that you want to use in the inbox in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches.
- You have assigned partner functions for use in the inbox in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Business Transactions to Responsible Employees and Groups.
- You have defined the inbox status mapping in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Status to Inbox Status.

Inbox Search Table for Work Items

- You have made the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Assign Standard Tasks to Communication Methods.
- You have made the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles > Work Item Table.

Inbox Search Table for Outbound Correspondence

- If you want to adjust the data record of an outbound correspondence item before it is stored in the inbox search table for outbound correspondence, you make the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Business Add-Ins > BAdI: Outbound Correspondence Database Update.

Features

Data Load

To use the predefined inbox search tables, you first need to upload the data for the relevant main categories. You do this in the SAP GUI, on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Process Data for Inbox-Specific Searches (transaction `CRM_IC_INDEX_LOAD`).

Only the main categories that are mapped to the corresponding inbox item types in Customizing activity Define Item Types for Searches are available in the program Process Data for Inbox-Specific Searches.

Note

If you change any of the Customizing settings detailed in the Prerequisites section, you must update the corresponding inbox search table accordingly.

In this case, you must also update the entries for the inbox item type Inbox One Order Item to fill various header fields in the corresponding business transaction header table, for example `CRMS4D_SERV_H` for service orders.

Process Data for Inbox-Specific Searches: "Inbox Item Type"

- Choose the inbox item type `ONEORDER` Inbox One Order Item to update the following header fields in the corresponding business transaction header table, for example `CRMS4D_SERV_H` (only in the case of Customizing changes):
  - INBOX_STATUS
  - INBOX_ASSIGNED_TO
  - INBOX_GROUP_ASSIGNED
  - INBOX_ACCOUNT
  - INBOX_CONTACT_PERSON
- Choose the inbox item type `WORKITEM` Inbox Workitem to process the data for the inbox search table for work items. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.
- Choose the inbox item type `OUTCORRESP` Outbound Correspondence Item to process the data for the inbox search table for outbound correspondence. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.

Process Data for Inbox-Specific Searches: Displayed Figures

- Source Entries (Not available for outbound correspondence)

  The number of entries in the tables, which the system uses to update the corresponding predefined inbox search table

- Target Entries

  The number of entries in the predefined inbox search table that is being updated

Process Data for Inbox-Specific Searches: Processing Options

- Update

  Loads the data for the selected main categories from the source tables to the corresponding inbox search table

  Note that for the selected main categories, all previous entries in the corresponding inbox search table are removed and new entries are provided.

- Remove

  Removes the selected main categories from the corresponding inbox search table

- Calculate

  Updates the displayed figures

- Reset Status

  Resets the status to Not Started

  You can use the Reset Status function if you have canceled a job and the status of the corresponding main category is still In Process.

The functions above are also available if multiple main categories are selected, unless one of the main categories cannot be processed due to the status.

Process Data for Inbox-Specific Searches: Availability of the “Update” and “Remove” Functions

The following table provides an overview of the availability of the Update and Remove functions and the corresponding status:

Context	Status of Corresponding Main Category	Update	Remove
One of the selected main categories is no longer supported according to the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches .	Not Supported	Not enabled	Enabled
One of the selected main categories is already included in another data load.	In Process	Not enabled	Not enabled
You have canceled a job and reset the status to Not Started using the Reset Status function.	Not Started	Enabled	Enabled
The initial data load for one of the selected main categories has not yet been performed.	Not Started	Enabled	Enabled
One of the selected main categories has already been removed from the inbox search table.	Not Started	Enabled	Enabled

Context	Status of Corresponding Main Category	Update	Remove
The data for one of the selected main categories has been uploaded to the inbox search table.	Completed	Enabled	Enabled

Delta Update

The automatic delta update is limited to the main categories that you have uploaded, or are in the process of uploading, to the relevant search tables:

- Inbox search table for work items

  When a work item is saved

- Outbound correspondence

  - Outbound e-mails

    When an e-mail is saved, sent, or deleted

  - Outbound faxes and outbound letters

    When a fax or letter is saved or deleted and the session is ended

- For outbound correspondence, you are not required to upload the data for the relevant main categories to enable the delta update.

- The system status is used to calculate whether a business transaction is overdue.

Specifics for the Inbox Search Table for Outbound Correspondence

The system fills the following fields from the interaction record using your settings in Customizing for **Service** under **Interaction Center WebClient > Business Transaction > Assign Partner Functions to Business Transactions**:

- `ACCOUNT`
- `CONTACT`

If more than one interaction record is assigned to an outbound correspondence item, the interaction record that the fields are supplied from (during data load) is undefined.

During data load, the system loads the outbound correspondence items linked to an interaction record during an interaction center session or during ERMS processing. For more information, see Interaction Record.

⚠ Caution

If you delete outbound correspondence items from the inbox search table, you can use the **Process Data for Inbox-Specific Searches** transaction to restore items that are linked to an interaction record. However, you cannot use this transaction to restore items that are not linked to an interaction record. For example, you cannot restore outbound e-mails sent by ERMS when they were not created with an interaction record.

Inbox Result List

Use

The agent inbox (inbox) result list displays the inbox search results in table or tree format and provides the functions required to process the inbox items.

Prerequisites

- You have set up and configured the inbox. For more information, see Agent Inbox in the Interaction Center WebClient.
- If applicable, you have configured the result toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Toolbar Profiles**.

Features

Result List in Tree Format or Table Format

You can display the result list in tree format or table format:

- The tree format provides context information, such as related business transactions.
- The table format provides better performance, sorting and filtering functions, and the use of table charts.

  You can define whether the result list is displayed in tree format or in table format as a default. You do this in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

  The user can switch between the tree format and table format in the inbox, as required.

Maximum Number of Hits

You can define the maximum number of search results.

For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Advanced Warning for Due Date

You can define an advanced warning for each inbox item type, with reference to the due date.

For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Inbox Search Definitions** > **Define Item Types for Searches**.

Configurable Result Toolbar

You can configure the result toolbar according to your business needs for interaction center (IC) business roles (business role profile type **IC WebClient Business Role**) and for other core business roles in Service:

- Result toolbar buttons

  You can define the following for each button:

  - Description
  - Label
  - Tooltip
  - Icon

    - Availability of button for use in IC business roles, in other business roles for core or both (providing the button function is available in the corresponding business role)

  You can also add user-defined buttons.

- Result toolbar profiles

  You can define toolbar profiles for IC business roles and for other core business roles. You can define the following for each profile:

    - Which buttons are available
    - Button order
    - Button grouping

      You can group buttons using separators.

    - Whether the label, icon, or both are displayed for the button

  If you do not define toolbar profiles for your business roles, the system uses default profiles.

Available Toolbar Buttons and Corresponding Functions:

Button Description	Button Function
Complete	The statuses of the selected inbox items are set to **Completed**. The **Complete** button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) • Workflow work items Whether a workflow work item can be completed depends on the corresponding workflow task. • Outbound text messages
Delete	The selected inbox items are deleted. The **Delete** button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters)
Display	The system navigates to the details page for the inbox item in the display mode: • No interaction record is created. This also applies if the user switches from the display to the edit mode from the object. (For workflow work items and work items (e-mails, fax messages, letters), switching from the display mode to the edit mode is not supported.) • The account is not confirmed and not published in the context area. When switching to the edit mode from the object, the system defines the user who is currently logged on as the employee responsible (according to the partner mapping in the inbox Customizing activity **Map Business Transactions to Responsible Employees and Groups**) and confirms the business partner. The **Display** button is not supported for work items (inbound e-mails, fax messages, and letters) for business roles with the profile type **WebClient Business Role**.
Edit	The following steps are performed:

Button Description	Button Function
	• The system navigates to the details page for the selected inbox item in the edit mode. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity **Map Business Transactions to Responsible Employees and Groups**). • No interaction record is created. • The business partner is confirmed and published in the context area. If other business transactions are opened during the editing process, always the business partner of the business transaction that was most recently opened is confirmed and published in the context area. • The business transaction is linked to the activity clipboard. • All subsequently opened business transactions are also linked to the activity clipboard, unless the user chooses the **End** button. Choosing the **End** button clears the activity clipboard. • If the user chooses **Interact**, all business transactions that were linked to the activity clipboard during the editing process are linked to the interaction record. • Editing during an interaction If the user edits other business transactions during an interaction, the business partners for the other business transactions are not confirmed or published in the context area. The **Edit** button is only supported for business transactions.
Forward	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the **Forward** button, a dropdown menu is displayed that contains the recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile. If the recipient profile contains more than 10 entries, the 10th item of the button dropdown menu is a **More** entry that launches an input help. The input help contains all remaining recipients that you have defined in the recipient profile for forwarding. The forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Forward (Extended)	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the **Forward** button, a dropdown menu is displayed that contains the following entries (maximum of 10 entries): • The recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile • The **More** entry that launches an input help The input help contains the remaining recipients that you have defined in the recipient profile for forwarding. • The **Other Employees** entry that launches the employee search • The **Other Service Teams** entry that launches the business partner search The extended forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Interact	The following steps are performed by the system:

Button Description	Button Function
	- The system navigates to the details page of the inbox item. - The account and contact are confirmed and published in the context area. - An interaction record is created. - The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). - In the case of work items (e-mails, fax messages, letters) and workflow work items, the system navigates to the account identification where a preview of the work item or workflow work item is displayed. The Interact button is not supported for the following inbox items: - Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) - Workflow work items The Interact button is not supported for business roles with the profile type WebClient Business Role.
Link	If the user selects a second inbox item during an interaction and chooses the Link button, the second inbox item is linked to the current interaction record. The Link button is not supported for the following inbox items: - Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) - Workflow work items The Link button is not supported for business roles with the profile type WebClient Business Role.
Next Item	The system navigates to the next inbox item that is available for processing and locks the item for other users. Once the user has chosen the Next Item button, the button is deactivated until the user chooses the End button. If the Next Item button is used for outbound correspondence (outbound e-mails, fax messages, and letters), the system navigates to the next outbound correspondence item that is in process and to which the current user is assigned as the employee responsible. In this case, the outbound correspondence item is not locked for other users. The Next Item button is not supported for business roles with the profile type WebClient Business Role.
Preview (Toggle)	Displays a preview of the currently selected e-mail, fax message, and letter The preview is displayed below the table chart area. The preview does not provide follow-up buttons, such as Reply and Forward.
Reference	Sets a reference to the currently selected inbox item If the user chooses the Reference button, a dropdown menu is displayed that contains the relevant entries from the activity clipboard. If the user chooses an object from the dropdown menu, the currently selected inbox item is referenced from the object (the inbox item is added to the object's business context). Note that the Reference button only works if the `BUSINESS_CONTEXT` function profile is assigned to the business role. In Customizing for the business context, the following is defined: The inbox item types to which a reference can be added Only these inbox item types are available in the dropdown menu of the Reference button.

Button Description	Button Function
	• Which inbox item types can be added as a reference For more information, see Customizing for Service under Transactions > Settings for Service Requests > Define Settings for Business Context. The Reference function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type WebClient Business Role.
Reference (Extended)	Unlike the dropdown menu for the Reference function, the dropdown menu for the Reference (Extended) function contains entries from the Recent Items. The last entry in the dropdown menu is the Other Transactions entry that launches a transaction search. Apart from this, the Reference and Reference (Extended) functions are identical. The Reference (Extended) function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type IC WebClient Business Role.
Refresh	The system performs the most recent search again, based on the selected search criteria or based on a quick search. The Refresh button enables the user to conduct the most recent search again without expanding the Search Criteria view and choosing the Search button, or selecting the quick search again.
Reserve	The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The Reserve button is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation	Resets the reservation of the inbox item and removes the employee responsible Only the employee responsible assigned to the inbox item can reset the reservation of the inbox item. The Reset Reservation function is not supported for the following inbox items: • SAP ERP business transactions • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation (Extended)	Resets the reservation of inbox items on behalf of other users The Reset Reservation (Extended) function is not supported for oubound correspondence (outbound e-mails, fax messages, letters, and text messages).
Separator	The separator is used to group toolbar buttons in the result toolbar profile.
Search Processing Log	Provides access to the search processing log. For more information, see Inbox Search (Search Traceability section).
Show Table View (Toggle)	Displays the result list in table format
Show Tree View (Toggle)	Displays the result list in tree format

Multiple Selection of Inbox Items

The following functions are available when multiple inbox items are selected (unless the function is generally not available for one of the selected inbox items):

- Delete
- Reserve
- Reset Reservation
- Complete
- Link
- Forward
- Reference

The following functions are not available when multiple inbox items are selected:

- Edit
- Display
- Interact
- Preview On

Business Transaction Routing

Use

You can use this function to forward business transactions and cases in the interaction center (IC) manually or automatically.

Prerequisites

You have made the necessary settings for automatic routing in the BSP application rule editor.

For more information, see Customizing for Service under Interaction Center WebClient > Business Transaction > Define Automatic Routing.

Features

You can forward business transactions and cases to agent groups as well as to individual agents. Both manual and automatic routing are available:

- Manual routing

  You can manually enter a responsible employee or a responsible group as the business partner.

  The transaction or case can be selected in the agent inbox of the assigned responsible employee as an open item for processing.

  - Assigning a responsible employee

    You can either enter an employee directly or using a search.

    If you start a search and a responsible group is assigned, the system searches for agents in the responsible group. If there is no responsible group assigned, the system takes all agents into account during the search.

  - Assignment of a responsible group

You can enter the responsible group directly or using a search. If you start a search and a responsible employee is assigned, the system searches for all organizational units to which this employee is assigned in the organizational model. If no responsible employee is assigned, the system searches for all organizational units that are assigned to the business role. The organizational units are displayed in the search results according to the structural authorization of the agent.

- Automatic routing

  You trigger automatic routing by choosing **Escalate**.

  You can include up to five of the following attributes in the routing:

  - Priority
  - Status
  - Confirmed product
  - Confirmed customer
  - Categorization
  - Transaction type
  - Business role

  You can also push business transactions and cases to external routers or communication management software (CMS) for distribution to IC agents based on their real-time availability. For more information, see Business Process Push.

E-Mail Response Management System

Use

E-Mail Response Management System (ERMS) is a tool for managing large amounts of incoming e-mail. Instead of routing all incoming e-mails into one queue, ERMS provides services for automatically processing and organizing incoming e-mail. There are several automated activities that reduce the need for manual intervention by interaction center agents, thus substantially increasing efficiency and processing accuracy, by helping agents to process e-mail in less time and allowing them to provide the same quality response regardless of agent skill level. ERMS provides tools for agents to efficiently and consistently respond to messages, and for managers to administer, monitor, and report on the whole e-mail process.

SAP delivers preconfigured ERMS settings and functions that you can extend using custom coding.

Integration

- Organizational Management

  Defines how e-mails are routed and to what groups or organizations and how they are going to handle and process e-mails. You can route e-mails to an organizational unit or responsible group, a position, or even an individual user. The e-mails are routed to the agents' inboxes where they are processed.

- SAP Business Workflow

  Enables automated processing and routing of e-mails. For more information, see Workflow for ERMS.

- SAPconnect

  Receives inbound and sends outbound e-mails. Once e-mails are received, SAPconnect creates SAP office documents, which are then processed by SAP Business Workflow.

- Multilevel Categorization
- Machine Learning

  You can connect ERMS with **SAP Service Ticket Intelligence** to automatically categorize incoming e-mails.
- E-Mail Response Management System Analytics

  Allows you to generate reports and analyze your ERMS processing on historical data. You can analyze historical ERMS e-mail volume, handling time, response time and service level.

Features

Manager Features and Tools for Productivity

The WebClient UI is available for modeling rules and categories, and for monitoring and viewing analytics.

- Rule Policies
- Category Modeler
- ERMS Simulator
- E-Mail Workbench

Agent Features and Tools for Productivity

- Agent Inbox in the Interaction Center WebClient

  Central location in which agents receive and respond to e-mails. E-mails are integrated with other communication channels.
- E-Mail Threading
- Automatic Business Object Creation

  Interaction records, along with appropriate interaction reasons, and service tickets can be created and associated to incoming e-mails automatically. For more information, see the **Create Interaction Record** and **Create Service Ticket** actions detailed in the actions and parameters table in ERMS Rule Policies.

Automated Features

- ERMS Responses
- E-Mail Filtering

  Certain incoming e-mails that are not relevant for agent processing, such as out-of-office replies, spam, or bounced e-mails, can be filtered or deleted based upon the e-mail subject or content.
- Rules

  These identify and determine the automated processing of e-mails.
- E-Mail Routing

  Incoming e-mails can be routed to particular queues or agent groups that are most capable of responding to the issue in the e-mail. For example, you can route e-mails that contain foreign languages to queues in which agents are fluent in that language.
- Content Analysis and Fact Gathering

The system analyzes attributes and content regarding each incoming e-mail and stores it in a fact base, enabling the use of the data for making decisions in respect to categorizing, routing, and automatic responses.

- E-Mail Escalation

  E-mails can be escalated if they are not processed within a specific period of time. For more information, see Defining E-Mail Escalation.

- Web Forms

  In addition to accepting incoming e-mails, it is also possible to accept information created from a web form on a website.

- Support of Multiple Incoming and Outgoing E-Mail Addresses

  You can define multiple incoming e-mail addresses. You can also define multiple outgoing addresses that agents can change when they are responding to e-mails.

More Information

Setting Up ERMS

ERMS Rule Policies

Use

You can create rule policies in the e-mail response management system (ERMS) context to handle and evaluate incoming e-mails.

Rules are evaluated every time an incoming e-mail is received if the policy in which they belong is invoked during rule processing.

To create ERMS rules and policies, you use objects configured in the ERMS repository.

Prerequisites

- You have defined attributes, actions, and action parameters in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

- You have created authorization groups in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

- You have assigned the authorizations groups to authorization object `CRM_ERMS_P` in **Role Maintenance** (transaction `PFCG`).

- You have assigned the authorization groups to policies in the WebClient UI in Service. You can see (and in some cases select) authorization groups in the WebClient UI under **Rule Policy Details**, after you select a policy (the top node of the policy structure). The possible values are the entries you maintained in the Customizing activity above.

Activities

- Select attributes

  Most attributes are self explanatory. Only those needing further explanations are listed below.

Attribute	Use
ERMS: Content Analysis Accuracy	A value between `0` and `1` describing the accuracy of the content analysis result. Accuracy = 1 indicates the result is highly focused. All matching categories form a single path in the category hierarchy. Accuracy = 0 indicates the result is contradictory. At least two matching categories point to different paths in the category hierarchy.
ERMS: Best Minus Next Best Score	Distance in score between the category with highest score and the category with second highest score. Each matching category has a score which is a value between `0` and `1` describing how good the content-queries of the category match the given textual content.
ERMS: Number of Top Categories	Number of matching categories that all have the highest score.
ERMS: Best Category Score	Highest score among all matching categories.
ERMS: Top Scoring Category	Matching category with the highest score.
Escalation Date	Date when the escalation of an e-mail occurs. The date is calculated based upon either rule evaluation, Customizing, or service level agreements (SLA) at the time the e-mail arrives.
Escalation Time	Time when the escalation of an e-mail occurs. The time is calculated based upon either rule evaluation, Customizing, or SLA at the time the e-mail arrives.
Escalation Type	Depends on the property METHOD of the service UT_ESCALDETERM. If the METHOD is set to `1`, the Escalation Typeis SLA. If the METHOD is set to `2`, the Escalation Type is determined as the shortest time out of one of the following: rule evaluation, Customizing, or SLA.
Response Time Duration (Hours)	Used for escalations. Represents the initial response time in hours.

- Set actions and parameters

  This covers the standard delivered actions and their corresponding parameters.

Action	Use	Parameters
Add Attribute Value	Adds a value to a multi-value attribute	Attribute The attribute that a value is added to. Value The value that is added.

Action	Use	Parameters
Auto Prepare	Same as Auto Respond, except the response is not sent automatically. The agent can review the content first, make any necessary changes, and then send the response. This action is triggered when an agent selects Reply to an incoming e-mail.	Mail Form Name of the mail form that determines the content of the outgoing e-mail. Outgoing E-Mail Address Address used for the outgoing e-mail's sender address. If you do not use this parameter, the incoming e-mail address that the e-mail was originally sent to, will be used. Category Schema Schema to determine which Top Scoring Category is used. Inline Determines whether standard responses and solutions tied to the category are inserted directly into the e-mail content instead of as an attachment. If both are inserted into the e-mail content, then the standard responses are shown first in the order they are attached to the category. The solutions are then added, also in the order they are attached to the category.
Auto Respond	Sends an automated response with proposed answers and/or solutions, to an incoming e-mail, without agent intervention. The solutions are inserted into the e-mail as attachments and are determined from the Top Scoring Category of the incoming e-mail. Not to be confused with the Send Auto Acknowledgement action.	Category Schema Mail Form Outgoing E-Mail Address See descriptions above from Auto Prepare. Create Interaction Record Creates an interaction record, after the response is sent, which includes both the incoming and the outgoing e-mails. Route To (On Exception) Organizational unit that the incoming e-mail is sent to if no solutions are attached to the Top Scoring Category. Inline See description above from Auto Prepare.
Create Interaction Record	Creates an interaction record in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to interaction record, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the interaction record.

Action	Use	Parameters
Create Service Ticket	Creates a service ticket in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to service ticket, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the service ticket. You can configure the transaction type (for example, TSRV or TSVO) for this service ticket. You do this in Indirectly Called Services and Properties in Customizing for Service by choosing E-Mail Response Management System > Service Manager > Define Service Manager Profiles .
Delete E-Mail	Deletes an incoming e-mail. Can be combined with the Create Interaction Record action, so the interaction record shows an e-mail was received and was also deleted.	(No parameters)
Forward E-Mail		Forward To Forward From
Invoke Policy	Temporarily suspends the processing of the current policy and continues with the processing of the specified policy. After the specified policy is finished processing, the action stops processing the original policy, unless the processing was stopped previously by the Stop Further Rule Processing action. This action can be nested, that is a policy that is invoked may invoke other policies. If a policy has multiple variants, then the active variant is invoked.	Policy Name of the rule policy that is invoked
Push E-Mail	Allows e-mail routing through the communication management software instead of directly going to an inbox. E-mails that are pushed to agents using this action are also routed to the Agent Inbox by the ERMS default routing, unless another routing destination is specifically maintained. Therefore, we recommend directly specifying a Route E-Mail action after the Push E-Mail action, allowing all pushed e-mails to be monitored by a supervisor or special organizational unit, and avoiding incorrect routing of push e-mails to the default routing that are already completed.	Organizational Object
Route E-Mail	Directs the incoming e-mail to an agent or group of agents. The e-mail is then accessible from the agent inbox.	Organizational Object You can enter an agent's system user name or an organizational unit. If you enter an organizational

Action	Use	Parameters
		unit, then the e-mail can be processed by any agent who belongs to that organizational unit.
Route to Case Processor	Directs the incoming e-mail to the agent who is assigned as the case Processor in the Case Overview, only if the incoming e-mail is related to an existing case and the case number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no case associated to the incoming e-mail or if there is no processor assigned to the case.
Route to Service Ticket Responsible	Directs the incoming e-mail to the agent who is assigned as the Employee Responsible in the Service Header, only if the incoming e-mail is related to an service ticket and the service ticket number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no service ticket number associated to the incoming e-mail or if there is no employee responsible assigned to the service ticket.
Send Auto Acknowledgement	Sends an automatic notification that the e-mail was received.	Mail Form Outgoing E-Mail Address See descriptions from Auto Prepare. Create Interaction Record Creates an interaction record to record that an acknowledgement was sent after receiving the e-mail. (The outgoing e-mail acknowledgement text is not tied to the interaction record.)
Set Attribute Value	Allows you to set an attribute value in the fact base if certain conditions are met. This attribute can then be used in rule evaluation or passed on to actions.	Attribute Value
Show Number of Incomplete E-Mails	Searches for incomplete e-mails linked to an identified account.	(No parameters)
Stop Further Rule Processing		(No parameters)

- Set rule policy authorizations

  You can define your own groupings for authorization, for example, by department or by organization.

Authorization Field	Description	Values
ACTVT	Activity	The following values affect authorizations for policies: 01: Create or Generate 02: Change 03: Display 06: Delete 43: Release(For releasing draft rules)

Authorization Field	Description	Values
ERMS_AUGR	Authorization Group	The values you defined for the authorization groups in Customizing activity **Define Repository**.
ERMS_CTXT	Context	The context in the WebClient UI in Service as defined in Customizing activity **Define Repository**.

More Information

Rule Modeler

Rule Policies

Rules

Conditions

Actions and Parameters

Conflict Resolution

Rule Policy Authorizations

ERMS Simulator

Use

The **ERMS Simulator** allows an interaction center (IC) manager to simulate the results of new or changed rules contained in rule policies that are used for e-mail handling. IC managers also see a detailed view of the resulting actions and categorizations for those e-mails. An IC manager can run the simulation in the background and have the result sent to an e-mail address.

The functions in the **ERMS Simulator** are available in the WebClient UI.

Prerequisites

- You have created rule policies.
- You have set up ERMS and already processed incoming e-mails, using ERMS.

More Information

Setting Up ERMS

E-Mail Workbench

Use

The E-Mail Workbench is an overview of e-mail processing in an interaction center, which you can also use as a means to mass process e-mails.

You can access the E-Mail Workbench in the WebClient UI by choosing IC Manager > E-Mail Workbench.

You can perform manual operations for e-mails, such as searching, routing, forwarding, re-categorizing, and deleting e-mails. You can also view the specific rules that were invoked for a particular e-mail and process escalated e-mails.

You can view and in some cases edit the following elements of an e-mail:

- E-mail text
- Attachments
- Rules for processing the e-mail
- Associated categories, that you can add or remove. Automated processing can assign categories to e-mails as determined from Multilevel Categorization. You can change these categories if the assignment is not accurate or complete.
- The Fact Base
- Related e-mails

Prerequisites

- You have set up and actively used ERMS to process incoming e-mails.
- You have set UT_ERMS_REPLICAT as your ERMS service manager profile for Directly Called Services. You do this in Customizing for Service by choosing E-Mail Response Management System > Service Manager > Define Service Manager Profiles.

ERMS Responses

Use

There are several types of E-Mail Response Management System (ERMS) responses you can use to automate e-mail processing. This automated processing saves agents time in compiling and searching for information and ensures consistent handling and messaging to customers. You can use mail forms to configure the e-mail content for the responses.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function under Knowledge Management > Mail Forms.
- You have defined multilevel categorization. (Not relevant for Auto Acknowledgements)
- You have defined a rule policy with the Send Auto Acknowledgement, Auto Respond, and/or Auto Prepare actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.

Features

The following is a list of responses you can determine for ERMS:

- Auto Acknowledgement

When an e-mail is received, the system can send an automatic acknowledgment informing the sender that the e-mail was received. If the business partner is identified, you can use business partner attributes, such as first and last name, to personalize the response.

- Auto Response

  The system can send a solution or a response automatically, without intervention from agents, based upon the e-mail content.

- Auto Prepare

  Response proposals are similar to auto responses. However, instead of automatically sending the response, the response is reviewed by agents for accuracy. Agents can send, modify, or delete the proposal.

More Information

ERMS Rule Policies

E-Mail Threading

Use

You can use e-mail threading for service requests and incidents to link e-mails to the associated business transaction, keeping all communication and related e-mails together for easy access to the information, regardless who handles the business transaction. E-mail threading also enables the e-mails to be routed to the person responsible for that business transaction if the customer sends the e-mail back to the interaction center.

Mail forms are used to create a tracking text.

Prerequisites

- You have created tracking text in WebClient UI under **Knowledge Management** > **Mail Forms**. You can add the **Email Response Management** context attribute to the **Text Element**.

  The **Service Request Tracking Text** mail form field is available as a variable for service requests and incident numbers.

  If the variable is not available in the mail form, you have to manually add the `CRMS_ERMS_SCENARIO_FIELDS` structure to the **ERMS** context in Customizing. You do this in Customizing for **Service** by choosing **Basic Functions** > **Communication Channels** > > **Maintain Attribute Contexts for Mail Forms**.

- You have enabled tracking text insertion by maintaining the **Tracking Text Form** field and selecting the **Track Text on Reply** indicator in Customizing for **Service**. You do this by choosing **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

- You have defined rules using the **Route to Service Request Responsible** action. For more information, see the actions and parameters table in ERMS Rule Policies.

Features

- **E-Mail Tracking Thread**

  The threading is enabled by the use of tracking text you define. You use a variable that is filled at runtime with the service request number to insert in the tracking text. This variable allows e-mails to be linked to business transactions.

- **Tracking Text Insertion**

The system inserts the tracking text in the e-mail content when an e-mail is created, either manually or automatically through automated processing, or when the e-mail is replied to. If a service request or incident is automatically created for an incoming e-mail from the auto acknowledgement action, you can link both the incoming e-mail and the outgoing reply to the service request or incident.

- **Thread Detection and Routing**

  When customers reply to e-mails, the system determines whether tracking text is inserted. The incoming e-mail can be associated automatically to the related business transaction. Attributes from these transactions, such as the service request number, can be used during rule evaluation. When you create rule policies, you can create rules that route the e-mail to the agent who is currently assigned to the business transaction.

Example

Example

You create a mail form with the following as the tracking text.

************** DO NOT DELETE **************

{SrvReqNo:[8000006814]}

************** DO NOT DELETE **************

You define a rule, “If Service Request Responsible Is Not Equal To "", Then Route to Service Request Responsible to forward the e-mail”.

More Information

E-Mail Threading Process

E-Mail Threading Process

Use

This process describes e-mail threading using service tickets as the relevant business object.

Prerequisites

You have defined e-mail threading.

Process

1. The customer sends an e-mail.
2. Either an agent sends a manual response or the system sends an automatic response.

   Manual Response

   a. The agent processes the e-mail and creates a service ticket.

   b. The agent replies to the e-mail.

c. The system uses the mail form you entered in the agent's e-mail profile. It inserts the mail form text and tracking text into the outgoing e-mail. The system populates the variable with the actual service ticket number that was just created.

Automatic Response

a. The e-mail response management system processes the e-mail automatically and creates a service ticket, as defined through an action.

b. The system sends a reply to the customer as defined in an action, such as **Send Auto Acknowledgement**.

c. The system uses the mail form that is defined as a parameter in the **Send Auto Acknowledgement** action. The mail form includes the service ticket tracking variable. When the system sends the reply, the variable is populated with the service ticket number that was just created.

3. The customer replies to the e-mail and sends it back to the interaction center.
4. The fact gathering services scan the e-mail and pick up the tracking text and store this in the fact base during automated processing.
5. The e-mail is associated to the service ticket.
6. The e-mail is routed to the agent who is processing the service ticket through a rule that uses the **Route to Service Ticket Responsible** action.

E-Mail Escalation

Use

E-mails can be escalated if they are not processed within a specific period of time, for example, e-mails that are not processed within 24 hours after the initial receipt are escalated. You can define the time periods that determine the escalation and you can be notified and view escalated e-mails as soon as the escalation occurs. From reviewing escalated e-mails, you can reroute the e-mail or decide on another action to speed up processing.

Integration

The SAP Business Workflow is used for the escalation process. In Workflow for ERMS, the escalation times you define are used as the **Latest End** date and time.

Prerequisites

You have defined e-mail escalations.

Features

Escalation Definition

Escalated e-mails are determined when defined initial response times are not met. The initial response time is the time between an e-mail is received in the system until the time that an agent processes the e-mail.

Initial response time is calculated by one of the following:

- Service Level Agreements (SLA)

You can set a deadline for escalation by using a service contract. The system uses the service contract that is assigned to the business partner and reads the duration time from the response profile associated with the service contract.

- Rules and actions

  In the WebClient user interface, you create a rule using the **Set Attribute Value** action with the value **Response Time Duration** to determine at what point an e-mail is considered escalated. You can also use some fact base attributes as additional criteria, for example, if a customer is a high priority and valued customer with a specific system status, and sends an e-mail with a request, but the e-mail is not processed within a specific number of hours as determined in the rule, the e-mail is escalated.

- E-Mail Addresses

  You can determine an escalation time based upon the receiving e-mail address.

Escalation Determination

You can choose which method is used to determine the initial response time. You can either use the:

- Service Level Agreement

- Combination of SLA, rules and actions, and e-mail addresses

  The system uses the definition with the shortest initial response time out of all three methods.

Escalation Process

See E-Mail Escalation Process.

Escalation Notification

You are notified of a new escalated e-mail by receiving an e-mail. Mail forms are used to create the notification text.

Escalation Monitoring

After receiving the escalation notification, you use the overview in the E-Mail Workbench to monitor the number of escalated e-mails per organizational unit and navigate directly to the individual e-mail for further action.

E-Mail Escalation Process

Use

This process describes the e-mail escalation process.

Prerequisites

You have defined e-mail escalations.

Process

1. An e-mail is received in the interaction center (IC).

2. Automated processing begins.

3. The initial response time (IRT) is determined by the escalation definition.

4. If the IRT limit is reached, the e-mail becomes escalated and the workflow escalation process starts.

5. The notification e-mail is sent to the IC manager.

6. The manager accesses the E-Mail Workbench, views the situation and takes further actions with the escalated e-mail.

More Information

E-Mail Escalation

Defining E-Mail Escalation

Prerequisites

You have authorization to read mail forms.

Context

You use this procedure to define and setup E-Mail Escalation.

Procedure

1. Define the method used for determining initial response times.
   - Service Level Agreements
     a. Run transaction `CRMD_SERV_SLA`.
     b. Maintain a response profile and assign a response time. For **Name of Duration**, select **Duration Until First Reaction**. The response profile is associated to a service contract. For more information, see Service Level Agreements (SLA).
   - Rules and Actions

     Define a rule in the WebClient user interface and use the **Set Attribute Value** action with the **Response Time Duration** attribute.
   - E-Mail Addresses

     Define the escalation time period for each receiving e-mail address in Customizing for **Service** by choosing **Interaction Center WebClient** > **Agent Inbox** > **Settings for Asynchronous Inbound Processing** > **Define Receiving E-Mail/Fax Settings**.
2. Assign the method used for determining initial response times and assign the fact gathering service.
   a. In Customizing for **Service** choose **E-Mail Response Management System** > **Service Manager** > **Define Service Manager Profiles**.
   b. Select your service manager profile.
   c. Under **Directly Called Services**, find the Service ID `RE_RULE_EXEC` and add the following services underneath to ensure that they are executed in the correct order with the attribute **Invocation Order**. If you do not use Rules and Actions to determine your initial response times, the order of the services could be invoked before `RE_RULE_EXEC`:
      i. Under **Directly Called Services**, add the Service ID `FG_ESCALTIME` if it is missing. You can maintain the property values using the F1 field help.

ii. Under Directly Called Services, add or select Service ID UT_ESCALDETERM and choose Properties.

iii. In the Property ID field add or select METHOD, and in the Property Value field enter the value. Refer to the F1 field help for e-mail escalation values.

i Note

Do not include the service UT_SEND_ESCL as part of your DEFAULT service manager profile.

3. Customizing of the notification mail.

a. In the WebClient UI, define the mail form used for notifying managers of the e-mail escalation. This is done in the interaction center (IC) manager role under Knowledge Management > Mail Forms. Insert escalation specific placeholders by clicking Attributes in the mail form and adding those specific escalation fields (for example, escalation date, time and type, sender's name and e-mail address and the e-mail subject).

If these escalation objects are not available, ensure that object CRMS_ERMS_ESC_FIELDS has been added and configured to the correct context in the mail form. In Customizing for Service choose Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms.

b. Assign the mail form and the receiver's and sender's e-mail information to the service manager profile.

i. In Customizing choose Service > E-Mail Response Management System > Service Manager > Define Service Manager Profiles.

ii. Select service manager profile ERMS_ESCALMSG and maintain the property values of the service UT_SEND_ESCL under Directly Called Services. Refer to the F1 field help for e-mail escalation values.

c. In Customizing choose Service > E-Mail Response Management System > Service Manager > Assign Service Manager Profiles. Assign the service manager profile ERMS_ESCALMSG to the object ObjectID 15107965 without entering the Address/Number.

d. In transaction SWU3 under Maintain Runtime Environment, maintain the Schedule Background Job for Workflow Deadline Monitoring and specify the interval until next deadline check.

Automated Processes and Architecture

Rule Processing Services

Use

Rule processing, or rule invocation, services is one of the four types of services found in the E-Mail Response Management (ERMS) service manager. There are standard delivered rule processing services available. For more information, see Customizing documentation for Service > E-Mail Response Management System > Service Manager.

Features

Rule processing services evaluate a set of predefined rules and determine a set of actions that need to be executed during rule processing.

Rules consist of conditions and actions that are triggered if the conditions evaluate to true. The rule processing services support rules and nested rules. Nested rules are only processed if the condition in the higher-level rule is true.

Rule processing starts with the first rule defined within a given policy and continues sequentially with all subsequent rules in exactly the order defined in the rule modeler. All actions that are triggered during rule evaluation are collected and passed back to the service manager. For each of these actions, the service manager determines the appropriate action handler and calls these handlers accordingly.

Special rule processing

The **Stop Further Rule Processing** action can immediately interrupt rule processing. If this action is triggered, only that actions that have executed to that point are passed to the service manager.

The **Invoke Policy** action is executed by the service manager after the current policy is completely processed. It is not executed immediately.

Content Analysis

Use

This function identifies the categories that best match specific content and makes automatic category suggestions.

Content analysis is used in the E-Mail Response Management System (ERMS) to analyze the content of inbound e-mails. It uses predefined search queries that are assigned to categorization schemas, to determine suitable categories.

Integration

Function	Description
Category Modeler	You define content queries for content analysis in the category modeler. If necessary, you can reset the content analysis in the page area **Application Areas** for each application area. You do this by choosing **Reset Content Analysis**.
E-Mail Response Management System (ERMS) - automated processing	ERMS manages a high volume of inbound e-mails. This system partially automates routing and responding to e-mails. The automated processing of e-mails in the ERMS uses content analysis to evaluate conditions assigned to categories. The following ERMS components relate to content analysis: • Fact gathering • Rule operators **Matches Category** and **Rule Attributes**
Interaction Center	Content analysis provides suggested categories based on the content of e-mails received in the interaction center and selected for processing in the agent inbox. These categories are then displayed in the relevant fields of the following IC-specific business transactions: • Service order • Service request • Interaction record

Prerequisites

You have made content analysis settings in Customizing for **Service** by choosing **Cross-Application Components** > **Multilevel Categorization** > **Define Application Areas for Categorization**.

Features

- Automated categorization

  Category suggestions are made automatically using content queries.

- Language detection

  Content analysis also supports automatic language detection for textual content.

More Information

E-Mail Response Management System

Fact Base

Use

The fact base is an E-Mail Response Management System (ERMS) component that serves as a data container for any information that is related to an inbound e-mail and can be relevant during rule evaluation.

Integration

Fact gathering services gather data relevant to the inbound e-mail and add the data to the fact base.

The rule engine runtime retrieves data from the fact base and uses it for rule evaluation and processing.

Features

The fact base stores information that is relevant for exactly one instance of an inbound e-mail. Such information may include e-mail content, business partner information, and the incoming e-mail address to which the e-mail was originally sent.

The fact base is organized as an XML document. Its structure is flexible to include any tree structure and content, depending upon your business requirements.

Updates and read access to the fact base occurs through XSL transformations (XSLT) and XPath queries. These standardized methods are platform- and programming-language independent. This provides maximum flexibility if you need to extend the fact base, for example, for data residing in legacy systems.

Overview Report

This report provides a real-time snapshot of the activity and current status of the E-Mail Response Management System (ERMS). It gives you an overview of the volume of incoming e-mails, which you can display by organizational unit, e-mail address, or Web form ID.

To use this monitoring function, you must be actively using ERMS. To use this function, you must be running Service, and you must activate the necessary BSP service.

Performance Report

You can use this report to monitor the overall effectiveness of the E-Mail Response Management System (ERMS) in your interaction center. This report provides a real time overview of key performance indicators such as volume, average handling time, average response time, and service level. You can sort this information by organizational unit or by e-mail address.

To use this monitoring function, you must be actively using ERMS.

Workflow for ERMS

Use

This workflow is used for the E-Mail Response Management System (ERMS) in the Interaction Center.

ERMS allows automated processing of inbound e-mails received via SAPconnect. E-mails can be automatically responded to or routed to an agent group for manual processing. An inbound communication triggers the workflow template `WS00200001`, which controls the work item processing and fulfills several tasks.

The following system tasks are the most important:

- Determining the responsible agent group and routing the work item to that group
- Deleting and stopping further manual processing

For a complete list of tasks, see the workflow template `WS00200001` in the workflow definition tools.

Prerequisites

ERMS configuration activities are completed.

Process

1. The incoming e-mail is received.
2. Task TS00207915 **ERMS Rule Execution** invokes the service manager.

   The service manager provides capabilities (necessary to process e-mails) such as fact gathering, content analysis, and rule processing.
3. Depending on the outcome of the service manager invocation, the system continues processing the work item by one of the following steps:
   - Delete

     You can determine when the system deletes an item by using a rule, for example, in cases of junk e-mail (spam). Task TS66107918 **ERMS Delete E-Mail** removes the office document. The workflow is ended.
   - Do not process further

     You define when processing is stopped in either a rule or in one of the services in the service manager.

- Route

  Task TS00207914 **ERMS Decision** forwards the work item to an agent or agent group as selected by the processor determination.

The processor determination decides which agents are responsible for a work item. Task TS00207914 determines which agents or agent groups are eligible for processing e-mails. The actual determination of which agent or agent group receives the work item is defined in rules.

Technical Execution

The following technical information is useful if you wish to customize your own workflow:

- Workflow template

  The actual procedure is implemented in the workflow template WS00200001 **(ERMS1).**

- Triggering events for the workflow template

  The event **MailReceived** of Business Object Repository (BOR) object `ERMSSUPRT2` is maintained as a triggering event in the template WS00200001.

- Processing methods

  The following table explains how methods of BOR object `ERMSSUPRT2` are used in the corresponding tasks of the workflow template WS00200001:

Method	Use
`RECEIVE`	Receives the mail from SAPconnect
`RULEEXECUTION`	Raises the event **MailReceived** and triggers workflow

Setting Up ERMS

Use

This procedure describes the basic settings needed for E-Mail Response Management System.

Prerequisites

Your IT administrator has created e-mail addresses for receiving incoming e-mails.

Procedure

Mandatory and Recommended Steps

Step	Description
Create an organizational model.	In Customizing for **Service** under **Master Data** > **Organizational Management** > **Organizational Model** > **Create Organizational Model**.
Maintain workflow standard settings.	In Customizing for **Service** under **Basic Functions** > > **SAP Business Workflow** > **Maintain Standard Settings for SAP Business Workflow**.

Step	Description
Define receiving e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Define Receiving E-Mail Addresses/Fax Numbers.
Assign receiving e-mail addresses to ERMS workflow.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Recipient Distribution.
Assign agents for e-mail handling.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling. Choose Assign Agents. Under workflow ERMS 1 either: • Use a general task for ERMS decision to allow e-mail routing to all agents, depending on the rules, or • Specify individual agents or agent groups to allow e-mail routing only to those agents, depending on the rules.
Activate event linking. Incoming e-mails can use workflow to define how mails are routed and how they are handled. Activating event linking activates the workflow you defined and allows it to be used for ERMS.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling. Choose Activate Event Linking. For New Recipient, choose possible entries and select SAP Object Instance. Use business object repository (BOR) Object Type ERMSSUPRT2 (ERMS Support 2). When this BOR object type is invoked, it triggers workflow WS00200001 (ERMS1). For more information, see Workflow for ERMS.
Define outgoing e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Sender E-Mail Addresses.
Define service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Service Manager Profiles. Choose Default and double click Directly Called Services to set the DEF_ROUTING for Invocation Order 50 to an organizational unit.
Assign service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Assign Service Manager Profiles.
Create ERMS policies and rules	In the Interaction Center, choose Process Modeling > Rule Policies. Choose E-Mail Response Management System as the Context.
Assign standard tasks to communication methods	In Customizing for Service under Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Assign Standard Tasks to Communication Methods. Choose INT as the communication method for Object ID 207914.
Define e-mail templates and e-mail forms.	In the Interaction Center, choose Knowledge Management > Mail Forms.

Optional Steps/Extensions of Default Customizing

Step	Description
Define additional services. You can define services that are available in the service manager. This activity determines how all necessary services for evaluating policies are invoked. You should only	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Services.

Step	Description
execute this activity if you want to change the default settings or create new services.	
Extend the ERMS repository.	In Customizing for Service under E-Mail Response Management System > Define Repository.
Define additional e-mail profiles.	In Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles.
Define catalogs, codes, and subject profiles. Catalogs are resources which help ensure the uniform use of terms	In Customizing for Service under Basic Functions > Catalogs, Codes and Profiles.
Define additional categorization profiles. This influences which categorization schema(s) are displayed in the corresponding business transactions, and the behavior of the auto suggest function.	In Customizing for Service under Interaction Center WebClient > Business Transaction > Define Categorization Profiles.
Define reporting for e-mail statuses and events.	In Customizing for Service under E-Mail Response Management System > Reporting.
Delete reporting data.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Utilities > Delete Reporting Data.

Integrated Communication Interface

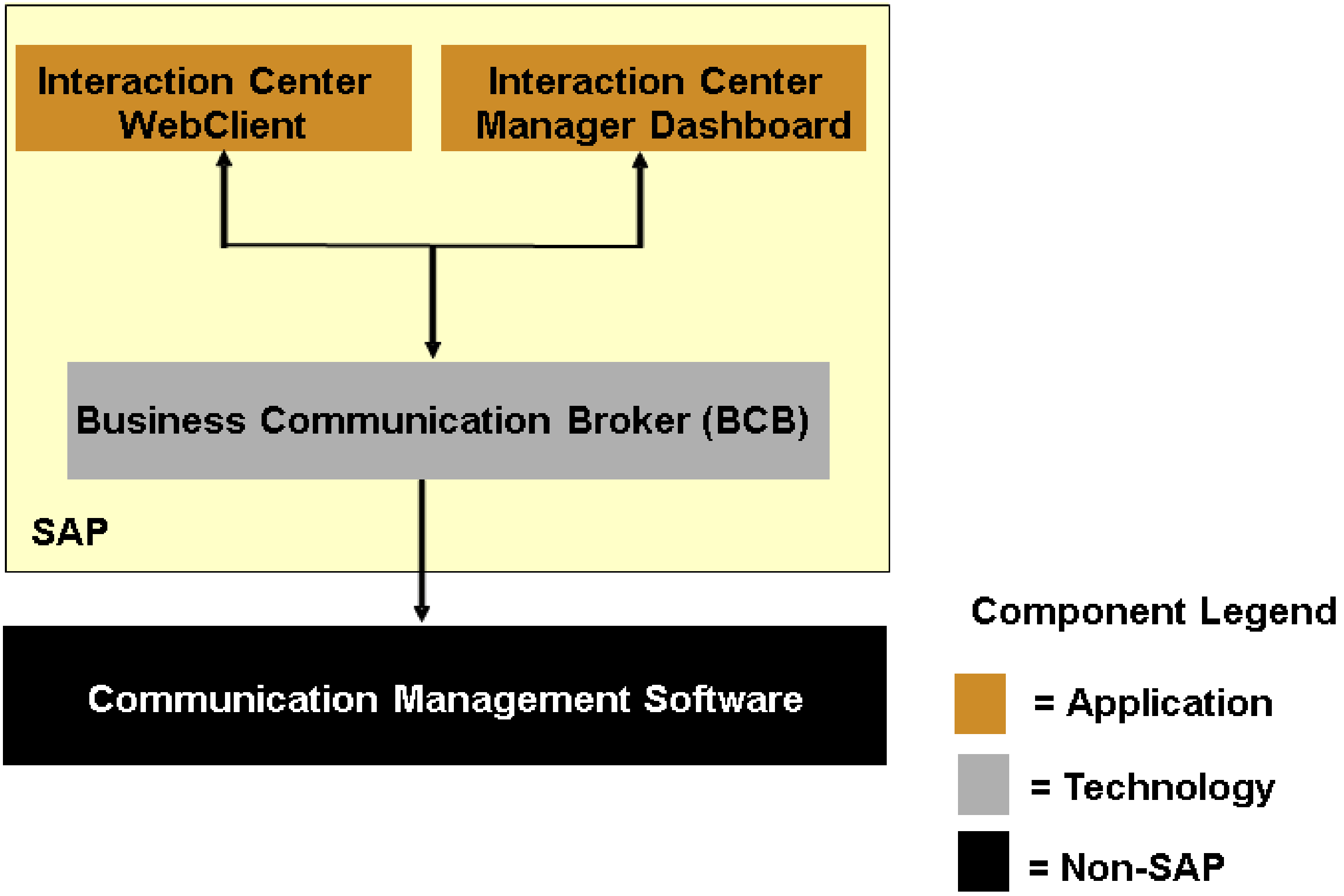
Use

The Integrated Communication Interface (ICI) supports the integration of non-SAP communication products with SAP components that use communication services. The ICI is intended in particular for scenarios involving multiple communication channels such as telephony, e-mail, and chat, but it can also be used in single-channel scenarios. It enables activities such as an agent accepting an inbound phone call or deflecting an inbound chat request, or a supervisor monitoring a queue. The ICI is only one of several components required to implement communication scenarios of this kind.

i Note

An example scenario is the integration of communication management software systems with the interaction center (IC), specifically the IC WebClient and IC manager dashboard. For illustration purposes, this example is used throughout this documentation.

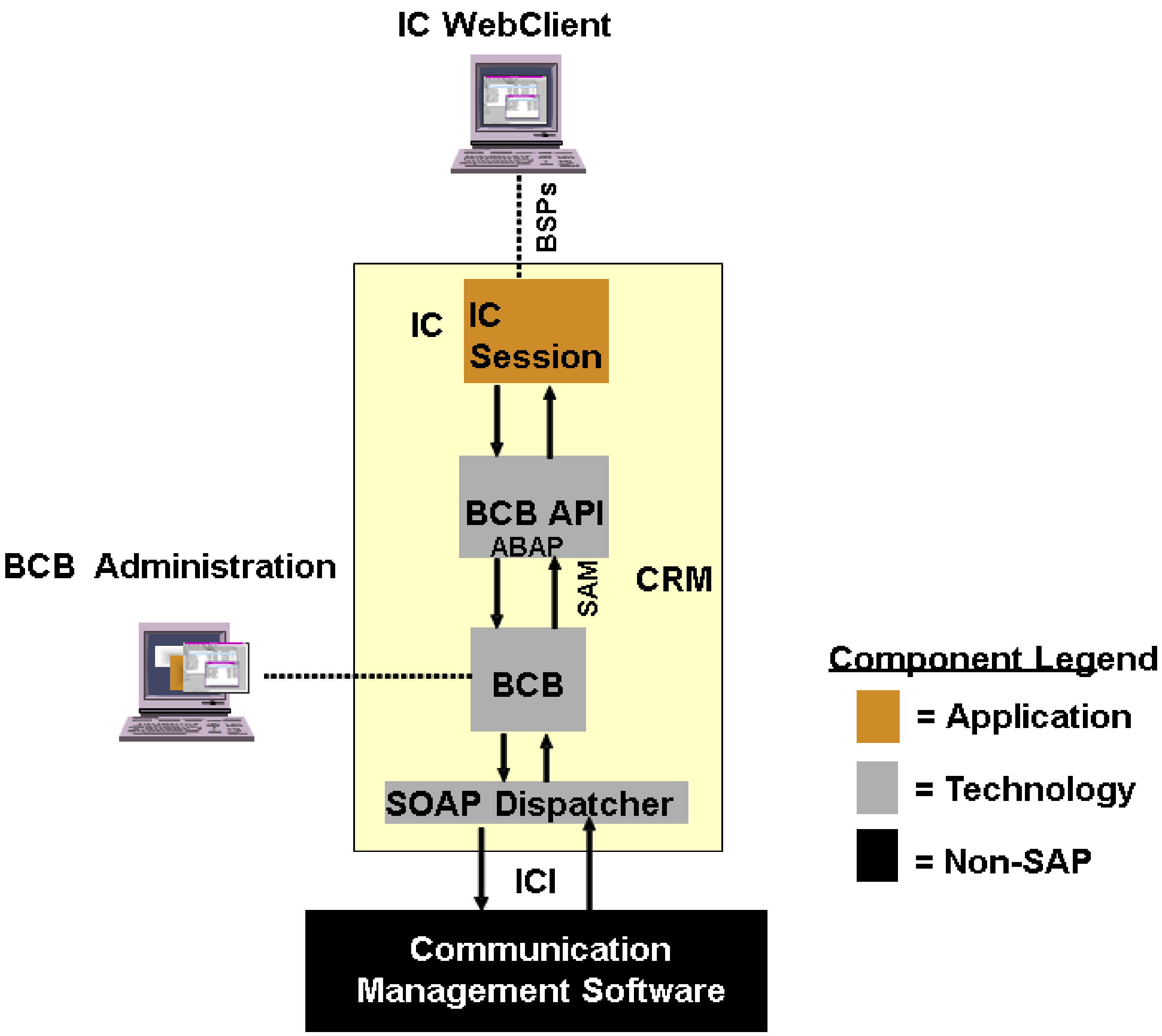
The graphic below represents a simplified view an example scenario without detailing the interfaces. The scenario is made up of functionality from three distinct sources – an application component, a technology component, and an external (non-SAP) component.



Integration

The ICI is based on Simple Object Access Protocol (SOAP) and Extensible Markup Language (XML); they are non-proprietary, Web-oriented, open interface technologies. HTTP is used at the transport level.

The system architecture required to connect the interaction center with external communication management software using the ICI is made up of several components. The system architecture of the interaction center example scenario in a live environment is shown below:



Live System Architecture

BCB Components

The BCB, BCB API, and SOAP dispatcher components enable the CMS. The messages passed between the application component and the external component are passed via the ICI.

ℹ Note

CMS Customizing for IC WebClient is done using transaction `CRMM_BCB_ADM`.

Application-Specific Components

The application-specific components are linked to the ICI-specific components via the Business Communication Broker Application Programming Interface (BCB API). This is the API for application customizing.

In the interaction center example scenario, the application consists of the IC WebClient, which provides an ABAP-based user interface for interaction center agents.

The IC server sends actions to the communication management software via the ICI-specific components and receives events back from the communication management software also via the ICI-specific components. The actions might include agent logon, initiate outbound phone call, or remove chat line from chat session. The events might include new inbound phone call, or new posting in chat session.

External Components

These are non-SAP communication systems. In this scenario, the external product is the communication management software, which is a multichannel communication product. This is responsible for merging requests from different media, routing requests to specific agents or agent groups, queuing requests, signaling inbound requests to the IC, and so on. The external products communicate with SAP via the ICI using SOAP and XML.

Certification

A certification process is implemented to verify the connectivity of external communication products with SAP using the ICI. An external product does not have to support the ICI in full. Partial certification is possible to enable the connection of single-channel communication products, such as telephony products.

Features

Many application components use external communication services. Integrating an external communication product with an application component using the ICI can provide:

- Integration of multiple communication channels such as phone, e-mail, and chat
- Active reporting of new, changed, or completed phone calls, e-mails, or chat sessions (known generically as items)
- Monitoring and statistics

Note

Multichannel management (that is, the synchronization and serialization of incoming communications) is not part of this SAP functionality. The ICI is a server-to-server interface. It does not support direct communication between the SAP client and the external communication product.

Integration

You can integrate Service with other systems to optimize and automate your service processes. The integration solutions cover both the integration with other solution capabilities within SAP S/4HANA or SAP S/4HANA Cloud Private Edition, and the integration with external products, such as SAP Field Service Management.

Related Information

Logistics Integration
Finance Integration
Integration with SAP Global Trade Services
Integration with SAP Field Service Management
APIs for Service
Data Exchange in Service

Logistics Integration

[Logistics Integration in Procurement Process](#)

You can use logistics integration to trigger the procurement of services and service parts planned in service orders.

[Logistics Integration in the Material Withdrawal Process](#)

You can use this function if you want to integrate the material withdrawal process.

[Manual Goods Issue for Service Parts](#)

With this logistics integration, you can post a goods issue for reserved service parts manually before the completion of a service confirmation. This way, the required service parts can be handed over to the service employee before the repair. You can also adapt stock levels once the service parts have been handed over to a service employee. The integration of Extended Warehouse Management (EWM) supports the picking and packing and the return of unused parts.

[Advance Shipment](#)

Advance Shipment enables the shipment of service parts to the service technician or the customer before the repair.

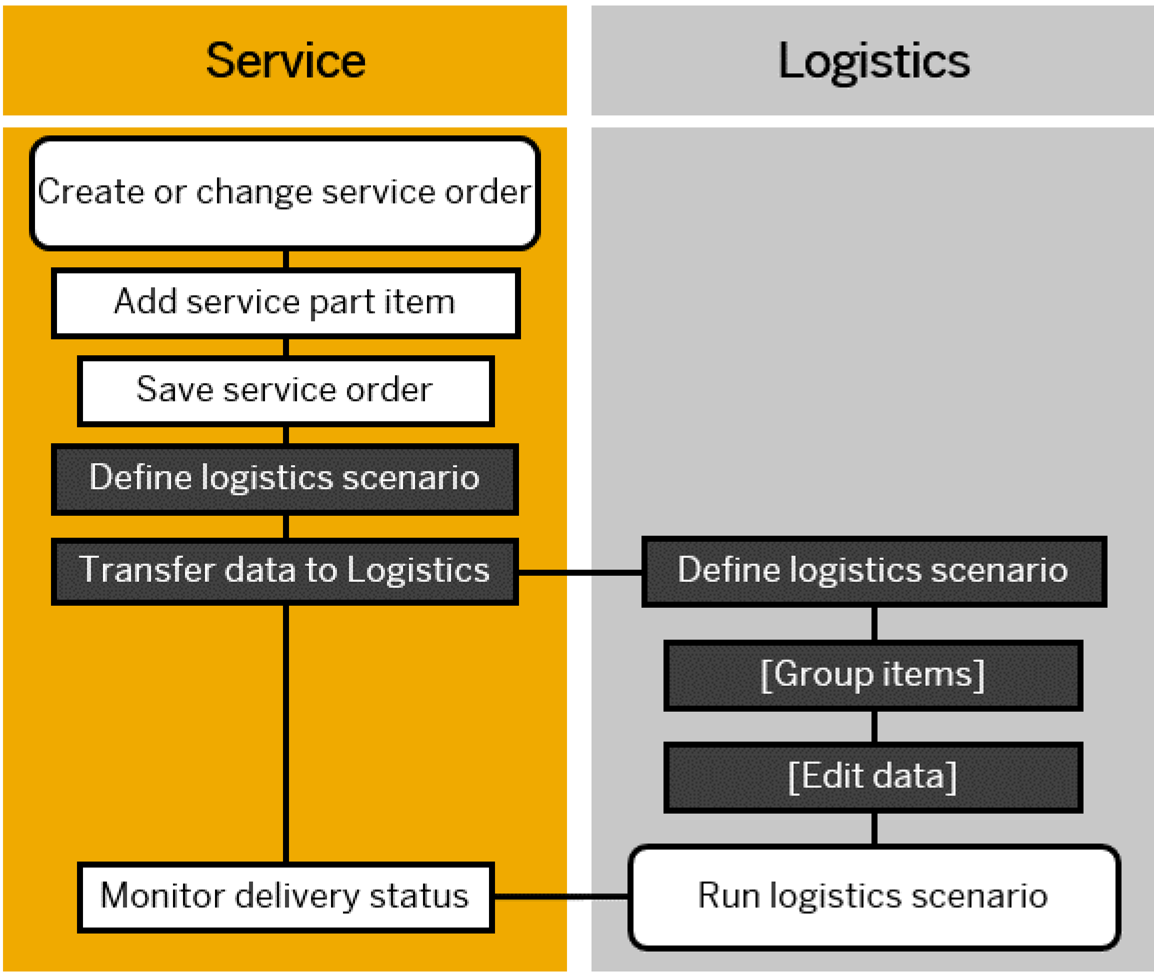
Logistics Integration in Procurement Process

You can use logistics integration to trigger the procurement of services and service parts planned in service orders.

The following diagram provides an overview of the steps that are carried out as part of logistics integration in procurement.

i Note

To provide a streamlined overview of the process, only the service order and service parts appear in this diagram.



Depending on whether the service part is already in stock or whether it first needs to be procured or even manufactured, you carry out different processes (logistics scenarios).

You can use the following logistics scenarios for procurement of materials and services:

- Reservation of materials
- Purchase requisitions for services and materials
- Purchase orders for services and materials

As a key user, you can configure the process in Customizing under **Service** > **Transactions** > **Settings for Service Transactions** > **Integration** > **Logistics Integration** > **Enterprise Organizational Management** or **Organizational Management (Legacy)** depending on the organizational model used. To use the enterprise organizational model, you first have to enable it under **Service** > **Basic Functions** > **Enable Enterprise Organizational Model**.

For more information, see Organizational Management in Service.

Prerequisites

The Purchasing data in the material master data must be maintained if the service part needs to be procured.

Related Information

Reservation of Stock Service Parts
Service Orders

Reservation of Stock Service Parts

After a successful Available-to-Promise (ATP) check for stock service parts in a service order, you can reserve these service parts if needed.

Creating and Completing Reservations

The system creates a reservation for a service part when you change the status of the service part item to **Released**. Once the reservation has been created, a reservation document is shown in the transaction history of a service order. If you change the quantity of the reserved service parts, the system updates the reserved quantity in the reservation document.

When a service confirmation is set to **Completed**, the system updates the reservation document with the number of the service parts confirmed as used in the service confirmation and creates a goods movement. If all parts have been consumed, the reservation is set to **Completed**. When the service order is set to **Completed**, the reservation is set to **Completed** even if not all parts have been consumed. Non-consumed stock is released for general availability.

If you use Advance Shipment, the system updates the reservation when a goods issue for the withdrawn quantity of the reserved parts is posted. The reservation is set to **Completed** after the final goods issue. For more information, see Advance Shipment.

Configuration Activities

To create a reservation for a service part, the system uses the **RSRV (Reservation)** logistics scenario. As a key user, you can configure this logistics scenario using one of the following Customizing activities depending on the organizational model used:

- **Service** > **Transactions** > **Settings for Service Transactions** > **Integration** > **Logistics Integration** > **Organizational Management (Legacy)** > **Define Logistics Scenario for the Procurement of Materials and Services**
- **Service** > **Transactions** > **Settings for Service Transactions** > **Integration** > **Logistics Integration** > **Enterprise Organizational Management** > **Define Logistics Scenario for Procurement of Materials and Services**.

To use the enterprise organizational model, you first have to enable it under Service > Basic Functions > Enable Enterprise Organizational Model.

This logistics scenario supports item-based accounting.

Related Information

[Availability Check](#)
[Advance Shipment](#)

Stock Transfer of Service Parts

You can initiate a stock transfer from a service part item of a service order, for example from a central storage location to a van stock storage location of a service employee.

Use

When the required service parts are not available in the executing service employee's storage location, a stock transfer process can be triggered to transfer the required stock to the storage location of the service employee. If the source and target plant are different they must belong to the same company code.

Prerequisites

You have made the relevant mandatory and optional settings in Customizing for Service:

- You have defined the criteria used to trigger the stock transfer process in the logistics scenario for the procurement of materials and services. In this activity, you map a purchase requisition or a purchase order to the logistics subscenario **03 Stock Transfer Item**.
    - If you use the Enterprise Organizational Model, you set this up in Customizing under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Define Logistics Scenario for Procurement of Materials and Services.
    - If you use Organizational Management (Legacy), you set this up under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Define Logistics Scenario for Procurement of Materials and Services.
- You have specified which purchasing document type is to be used for a combination of service transaction type and item category for the relevant logistics subscenario for procuring materials and services under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Set Purchasing Document Types for Service Transaction Type and Item Category.
- If you use the Enterprise Organizational Model,
    - Specify the relevant **employee-specific plant and storage location** under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Assign Plant and Storage Location to Service Employee and Country/Region
    - Specify the relevant **supplying plant and storage location** under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Assign Plant and Storage Location to Country/Region.

  For more information, see [Enterprise Organizational Management](#).

- If you use Organizational Management (Legacy), specify the **supplying plant and storage location** and the **employee-specific plant and storage location** under Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Assign Plant and Storage Location to Service Organizational Units.

  For more information, see [Organizational Management (Legacy)](#).

Note

The system uses a predefined sequence when searching for the plant and storage location. For details of the search logic, see [Determination Sequence for Plant and Storage Location](#).

Features

A stock transfer can be triggered from a service order by means of a purchase requisition or directly by means of a purchase order.

Related Information

[Organizational Management in Service](#)

External Workforce Procurement

In service order management, you can procure third-party service performers to provide services.

Depending on your Customizing settings for logistics integration you can initiate the automatic creation of a purchase requisition or purchase order from a service order item. If the Customizing has been set to automatically create a purchase requisition, then the purchase order is manually created using the **Create Purchase Order - Advanced** app, or you can plan a job to create the purchase order automatically using **Schedule Purchasing Jobs - Advanced** app.

You can create a purchase requisition or a purchase order with one of two different procurement methods: One is through the use of Lean Services (recommended) and the other is through the older External Services Management (**MM-SRV**) method.

Lean Services

Lean services procurement uses a material master of type **SERV Service** for service procurement. The purchase requisitions and purchase orders use the procurement item category **Standard**.

Prerequisites

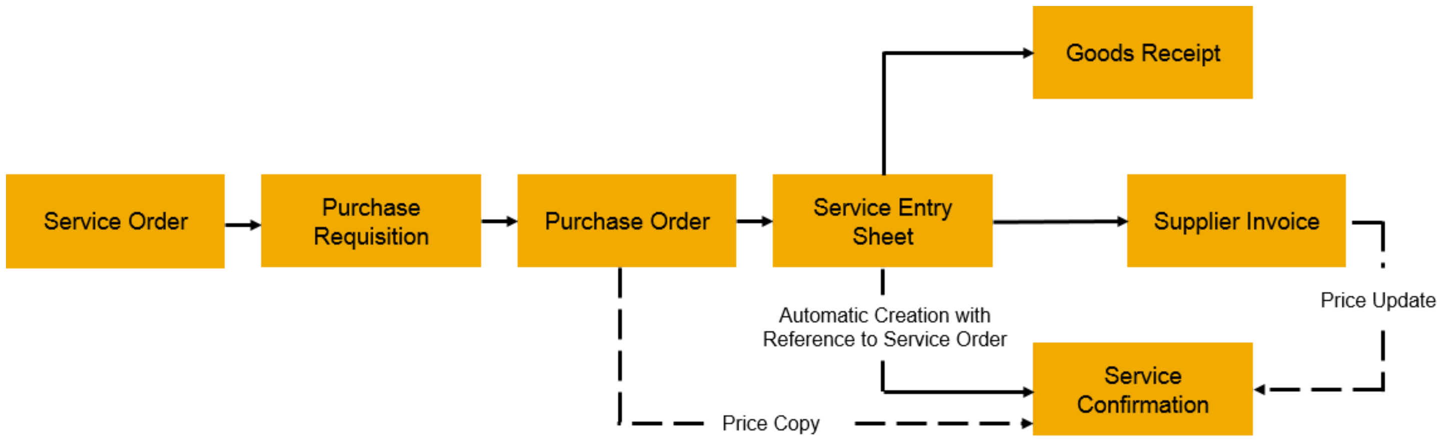
Depending on the organization model used, you have made the following settings in Customizing for Service:

- Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Define Logistics Scenario for Procurement of Materials and Services
- Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Define Logistics Scenario for the Procurement of Materials and Services

To activate lean service procurement you have to assign the logistics scenario **Purchase Requisition (PREQ)** or **Purchase Order (ORDR)** and the logistics subscenario **02 Lean Procurement Service** to the service order transaction type and item category.

Process Flow

The figure shows the process flow used in lean services:



Service Performer

When you add a service provided by an external service performer to a service order, the system automatically creates a purchase requisition or purchase order depending on the Customizing settings. You can also assign a preferred service performer to the service order by choosing the partner function **Service Performer** in the **Parties Involved** assignment block of the respective item. The service performer will be transferred to the purchase requisition and purchase order.

The external service performer uses a service entry sheet to record and confirm the services that have been provided. For more information on service entry sheets, see [Manage Service Entry Sheets - Lean Services](#).

Creation and Update of the Service Confirmation

- **When Item-Based Accounting Is Activated**

  If item-based accounting is active, a service confirmation is created automatically when the service entry sheet has been approved. The service confirmation transaction type used for the confirmation is defined in the Customizing details of the service order type.

  The system transfers the gross price of the purchase order or supplier invoice to the service confirmation item. Any changes to the price in the supplier invoice are updated to the service confirmation as long as the service confirmation has not been set to status **Completed**. Once the confirmation has the status **Completed**, the supplier invoice does not update the price anymore.

  The condition type maintained in the item category of the service confirmation item is used as the target condition type. For example, the condition type **PMP0** or **PCO4** can be used. You can also create your own condition types in Customizing for pricing and assign them to your item category.

- **When Item-Based Accounting Is Not Activated**

  If item-based accounting is **not active**, the service confirmation is not created automatically. The price from the purchase order is transferred to the service confirmation. If the supplier invoice is available before the creation of the confirmation, this price will be copied instead of the purchase order price. Any changes to the supplier invoice after the service confirmation has been created will not be updated in the confirmation.

Pricing

The system requires a price to be able to automatically create purchase requisitions. To determine the price of an externally procured service or service part from the purchasing information record or the purchasing contract, a pricing routine has been introduced. This pricing routine can be found using transaction **VOFM** under **Formulas** > **Condition base value** with the routine number **31**.

The pricing procedure for service orders is **S17J01** and it can be found under **SPRO** > **Sales and Distribution** > **Basic Functions** > **Pricing** > **Pricing Control** > **Define and Assign Pricing Procedures** > **Set pricing Procedures**.

Pricing procedure **S17J01** includes condition type **PCO4**. Pricing routine **31** is assigned to the condition type **PCO4** in the column **Alt. Cndn Base Value**.

During the creation of a service order, when an externally procured service or service part is added to the item, the price will be determined as follows:

1. Condition type **PCO4** will be automatically added to the assignment block **Price Details** of the item.
2. Pricing routine **31** will be executed and the price is determined from the purchasing contract or the purchasing info record.
3. The price of the externally procured service or service part will be assigned to the condition type **PCO4** since the pricing routine is assigned to **PCO4**.
4. The determined price can be overwritten by entering a manual price for **PCO4** under **Price**.

To create a purchase requisition for the service order item, the price is determined from the condition type **PCO4**. To enable this determination **PCO4** is assigned to spare part or service part item category in the Customizing for item categories.

If the price is not determined from **PCO4**, the material valuation price will be considered. If the material valuation price is not maintained an error will be raised during the creation of the purchase requisition.

If the system doesn't find a price automatically from the purchasing info record, you must enter a proposed price manually. You can define which condition type can be used for the price transfer in the customizing for service order item category, for example, the condition type **PMPO Manual Price**. You can create your own condition types in the Customizing for pricing and assign them to your item category.

For more information, see **Lean Services** in Service Purchasing and Recording.

External Services

In External Services Management the purchase requisitions and purchase orders will be created using the procurement item category **D - Service**. The procurement item will take over the description of the service order item or the service master, if there is a service master defined with the same ID as the service material in the service order item.

This service procurement integration scenario cannot be used with item-based accounting.

Prerequisites

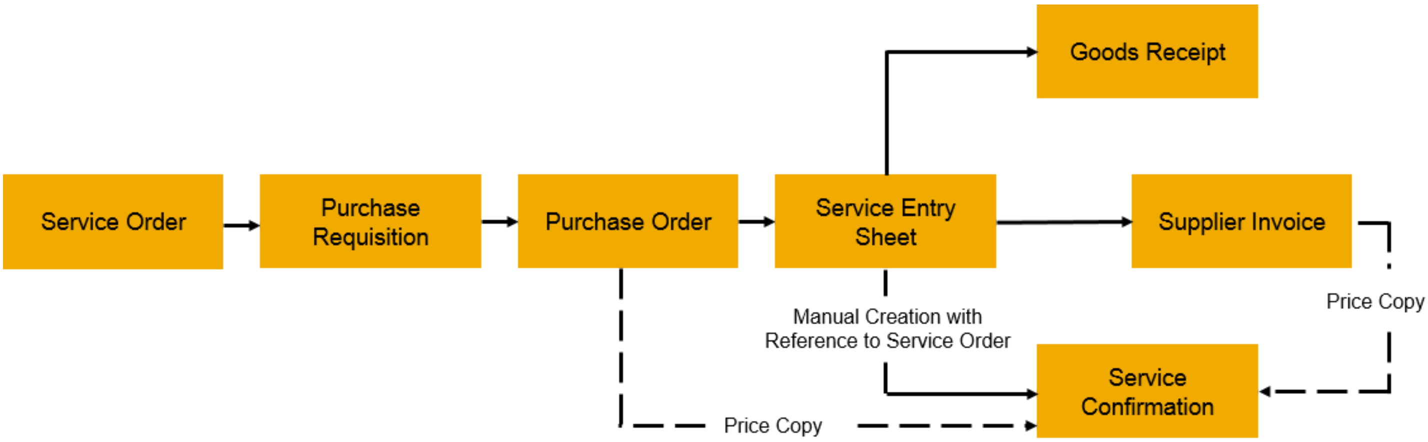
Depending on the organization model used, you have made the following settings in Customizing for Service

- **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Define Logistics Scenario for Procurement of Materials and Services**
- **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Define Logistics Scenario for the Procurement of Materials and Services**

To activate the service procurement scenario without lean service, set the logistics scenario to **Purchase Requisition (PREQ)** or **Purchase Order (ORDR)** and don't select a logistics subscenario for the service order transaction type and item category.

Process Flow

The figure shows the process flow used in external services:



The use of the service performer partner function is not supported in this scenario. To record the service entry sheet you use the **Maintain** function in **Service Entry Sheet**. For more information, see Entering Services Performed. The app **Manage Service Entry Sheets - Lean Services** is not supported.

Since item-based accounting is not supported in this process, automatic creation of service confirmations is not possible. The price from the purchase order will be copied to the confirmation. If the supplier invoice is posted before the creation of the service confirmation, this price will be copied instead of the purchase order price. If the supplier invoice is posted after the service confirmation has been created, the price from the supplier invoice is not updated in the service confirmation.

The system does not require a price for the creation of the purchase requisition. There is no price transfer from the service order to the purchase requisition.

For specific issues during the creation of the purchase requisition or purchase order with external services management check the following SAP Note:3085016

Information for Key Users

You can assign custom condition types in your configuration environment by using the **Assign Condition Type to Item Categories** activity.

Logistics Integration in the Material Withdrawal Process

You can use this function if you want to integrate the material withdrawal process.

Prerequisites

- You are using Inventory Management (MM-IM).

  For more information, see Inventory Management and Physical Inventory (MM-IM).

- You have made the necessary settings in the Customizing activities listed in the table. Depending on whether you are using **Organizational Management (Legacy)** or **Enterprise Organizational Management**, the navigation path to the Customizing activities differs slightly, as follows: **Service** > **Transactions** > **Settings for Service Transactions** > **Integration** > **Logistics Integration** > **Organizational Management (Legacy) or Enterprise Organizational Management**.

Organizational Management (Legacy)	Enterprise Organizational Management
Define Logistics Scenario for Material Withdrawal	

Organizational Management (Legacy)	Enterprise Organizational Management
Assign Plant and Storage Location to Service Organizational Units	◦ Assign Plant and Storage Location to Service Employee and Country/Region ◦ Assign Plant and Storage Location to Country/Region
Business Add-Ins for Logistics Integration ▸ Business Add-In: Determination of Plant and Storage Location In the standard system, the Business Add-In has one implementation:Plant/Storage Loc. Determ. from Service Org. Using Orgfinder	

Features

Logistics Scenarios

Logistics integration supports the following three logistics scenarios for the withdrawal of materials:

- Service parts at the plant and storage location

  In this scenario, materials are managed at the plant and storage location specified in the one of the following Customizing activities: Assign Plant and Storage Location to Service Organizational Units (Organizational Management (Legacy)) or Assign Plant and Storage Location to Service Employee and Country/Region (Enterprise Organizational Management).

- Service parts in the service employee's consignment stock

  > **Note**
  > In Customizing, the service employee is referred to as **technician**.

  For this purpose, the personnel number must be maintained in the company code data of the business partner role **Customer (FI) - FLCU00** on the Customer: Account Management tab under Reference Data. The system uses the personnel number and company code to find the customer number. The customer number determined in this way is used to book from the consignment stock.

- Service parts in the customer consignment stock

  This is a stock of service parts that does not belong to the customer and that is managed as customer consignment stock.

Service Parts Planning and Availability

You can call up availability information for service part items. The availability information checks the planned quantity against the stock available in the assigned service plant. The service plant is assigned in Customizing.

The check results show the available quantity and the next available date. This information is only available until the service order is saved, because it is not possible to create temporary reservations during order processing.

When the document is saved, a reservation for the service part item is created for the service plant if Customizing settings have been made for the logistics scenario for reservations.

Service Parts Fill-Up

There are different scenarios possible for filling the different storage locations:

- Scenario: Service parts at the plant and storage location

The storage location can be filled with service parts either by means of a regular procurement or production process without a direct reference to the service order, for example, whenever stocks fall below a certain level, or using a stock transfer process that is triggered from the service part item of the service order. For more information, see Stock Transfer of Service Parts.

- Scenario: Service parts in the consignment stock of the customer or of the service employee (referred to as **technician** in Customizing)

  i Note

  The type of storage location used depends on your Customizing settings in the activity **Assign Plant and Storage Location to Service Organizational Units**.

  In the case of the customer's consignment stock, a customer consignment fill-up order can be used to do the fill-up manually. This process is independent of the service order. For more information, see Creating a Consignment Fill-Up

  In the case of the technician's consignment stock, if you use advance shipment, the advance shipment sales order can be used to do the fill-up. For more information, see Advance Shipment Scenarios and Customizing for Advance Shipment.

Service Confirmations

You enter the service parts that were used in the service confirmation. The relevant stocks are updated in the inventory after you set the service confirmation to **Completed**.

The system doesn't cancel a service confirmation automatically if you cancel a goods movement created from this service confirmation. Therefore, if you decide to cancel a goods movement for service parts created from a service confirmation, you have to cancel the service confirmation first, which will subsequently automatically cancel the goods movement.

If you cancel the service confirmation manually after canceling a related goods movement, the system will create a double cancellation.

Manual Goods Issue for Service Parts

With this logistics integration, you can post a goods issue for reserved service parts manually before the completion of a service confirmation. This way, the required service parts can be handed over to the service employee before the repair. You can also adapt stock levels once the service parts have been handed over to a service employee. The integration of Extended Warehouse Management (EWM) supports the picking and packing and the return of unused parts.

i Note

If you use this integration option, you cannot post a goods issue for these items later from the confirmation since the goods issue has already been posted manually from the reservation.

Prerequisites

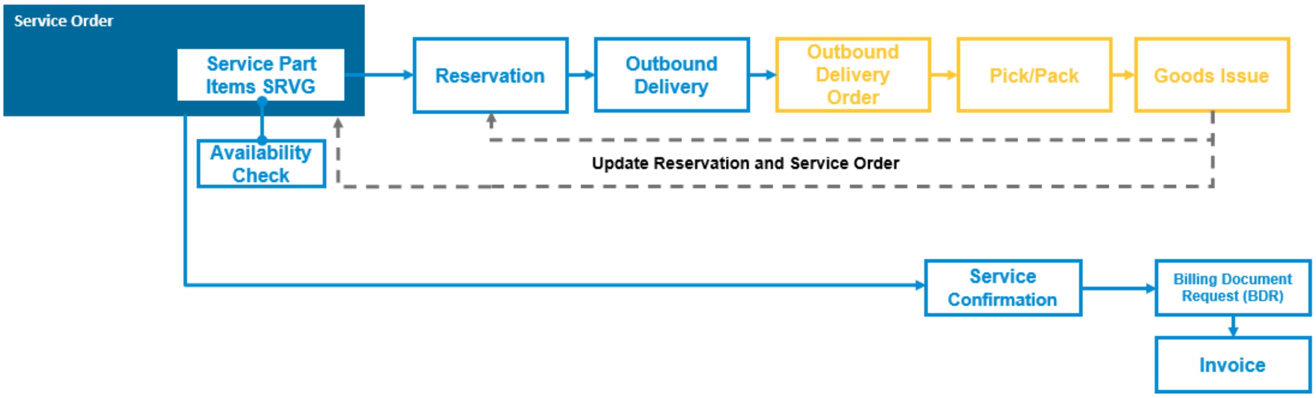
In order to create a reservation for a service part and use the manual goods issue process, the system uses the logistics scenario **RSRV - Reservation** and the logistics subscenario **04 - Manual Goods Issue (for Reservation)**. As a key user, you can configure the logistics scenario for the transaction type and item category using one of the following Customizing activities depending on your organizational model:

- **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Define Logistics Scenario for the Procurement of Materials and Services**

- Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Define Logistics Scenario for Procurement of Materials and Services

If you want to use the enterprise organizational model, you have to enable it in Customizing under Service > Basic Functions > Enable Enterprise Organizational Model. In the standard delivery, the item category SRVG (Service Part Man.GI) is configured for this process for the legacy organizational model. This can serve as a template for your own item categories. The corresponding service confirmation item category is SVCG.

End-to-End Process Flows



Process Flows for a Manual Goods Issue

Manual Goods Issue Without Extended Warehouse Management

1. If the service parts are already in stock in a warehouse and the service part item is released, a reservation is created from the service part item of the service order. You can control this process by configuring the relevant item categories in Customizing. You can check whether the required parts are actually available by triggering an ATP (Available-to-Promise) simulation from the service parts item list of the service order.

2. To post the goods issue **manually**, you can use transaction MIGO (Goods Movement, Goods Issue With Reference to a Reservation) or MB26 (Pick List) to select the relevant reservations. In both transactions, you can either search for the reservation using the reservation number or by entering the service order ID.

3. After you have posted the goods issue, the system adds the goods movement document to the transaction history of the service order on header and item level, and updates the goods issue quantity and goods issue status.

4. In the next step, you can create a service confirmation for the service part item. A quantity check is performed so that you can only confirm as many of the service parts as have been posted goods issue previously. When you complete the service confirmation, no goods issue document is created since the goods issue has already been done manually.

Manual Goods Issue With Extended Warehouse Management

Note

The process with Extended Warehouse Management (EWM) works the same way as without EWM with one difference in the third step:

1. If the service parts are already in stock in a warehouse and the service part item is released, a reservation is created from the service part item of the service order. You can control this process by configuring the relevant item categories in Customizing. You can check whether the required parts are actually available by triggering an ATP (Available-to-Promise) simulation from the service parts item list of the service order.

2. To post the goods issue **manually**, you can use transaction MIGO (**Goods Movement**, Goods Issue With Reference to a Reservation) or MB26 (**Pick List**) to select the relevant reservations. In both transactions, you can either search for the reservation using the reservation number or by entering the service order ID.

3. If you use the EWM integration and post the goods issue with transaction MIGO or MB26, the goods issue is not posted directly. Instead, an outbound delivery is created. This outbound delivery is sent to EWM and an EWM outbound delivery order is created. Picking, packing, staging, and the goods issue is done in EWM and the goods issue in EWM is transferred back- The goods issue is created in inventory management and the outbound delivery is updated. The service order is updated with the goods issue document.

4. After you have posted the goods issue, the system adds the goods movement document to the transaction history of the service order on header and item level, including an update of the goods issue quantity and goods issue status.

5. In the next step, you can create a service confirmation for the service part item. A quantity check is performed so that you can only confirm as many of the service parts as have been posted goods issue previously. When you complete the service confirmation, no goods issue document is created since the goods issue has already been done manually.

Returning Unused Parts Without EWM

To return unused parts into the storage location, you can use transaction MIGO with the action A02 (**Return Delivery**). Select the reservation via the reservation number or via the service order and enter the quantity you want to return. Make sure the flag **Final Issue** is set on the item tab **Reservation** if you want to keep the reservation closed, even after you have returned some parts.

After you posted the goods movement, the system will add the goods movement document to the transaction history of the service order on header and item level, including an update of the goods issue quantity and goods issue status.

Returning Unused Parts With EWM Integration

If you post the return delivery with the transaction MIGO, the goods movement is not posted directly. Instead, an inbound delivery is created which is then sent to EWM where an inbound delivery is created. The goods receipt is also done in EWM. This goods receipt in EWM will be transferred back to create a goods receipt in inventory management and to update the inbound delivery. The service order is updated with the goods movement document.

Additional Information

- If the storage location you are using is EWM-enabled, you require additional settings for EWM. For details, see Service Supply.
- If you are using transaction MB26, see 1919931.
- If you want to maintain the recipient location, you can use transaction IRL01.
- You can disable the above mentioned quantity check in the service confirmation in the following Customizing table depending on the organizational model you use:
  - **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Define Logistics Scenario for Procurement of Materials and Service**
  - **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Define Logistics Scenario for Procurement of Materials and Service**

Here, you have to set the flag **Disable Qty Check** (Disable Quantity Check) for the service confirmation item category you have used.

- If you are using the manual goods issue process and the standard goods issue process for reserved material which is triggered from the confirmation, you must make sure that you are not using EWM-enabled storage locations for the standard goods issue process as this is not supported. You can implement your own plant/storage location determination with the provided BAdIs. These are available in Customizing under
  - **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Business Add-Ins for Logistics Integration > Business Add-In: Determination of Plant and Storage Location**
  - **Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Business Add-Ins for Logistics Integration > Business Add-In: Determination of Plant and Storage Location**

Advance Shipment

Advance Shipment enables the shipment of service parts to the service technician or the customer before the repair.

Features

Advance Shipment is typically used when service parts need to be shipped in advance to the customer location because the part is too bulky to be carried in a technician's van. Based on the goods receipt or the planned delivery date at the customer location, the technician can then perform a repair using the service part. Shipment of service parts is done by using a service order with a sales order. The use of sales orders allows customer consignment stock to be created. As a result, parts that are actually required can be invoiced after the consumption has been posted.

Until the technician confirms that the service parts have been used in the repair process, the parts still belong to the company and not the customer.

Related Information

[Advance Shipment Scenarios](#)
[Customizing for Advance Shipment](#)

Advance Shipment Scenarios

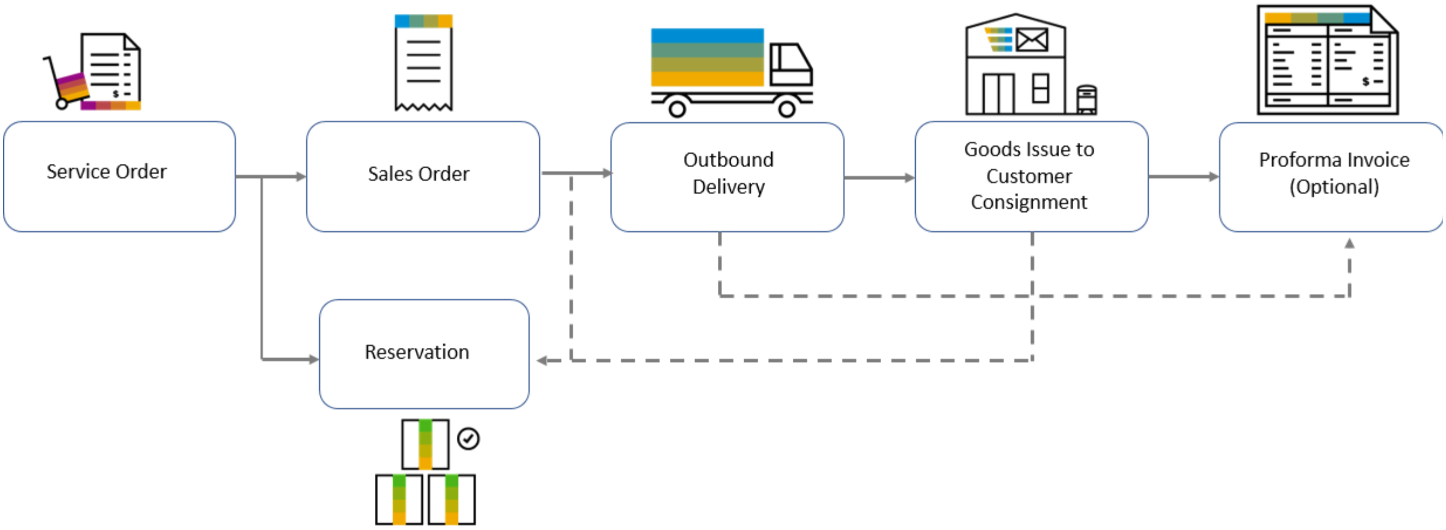
Use

Advance Shipment supports the following scenarios:

- The service parts are already in stock in a warehouse.
- The service parts aren’t in stock and must be procured from a vendor.

Process Flow for Advance Shipment Scenarios

Scenario 1: Service Parts Are Already in Stock in a Warehouse



If the service parts are already in stock in a warehouse and the service part item is released, a reservation and a sales order is created from the service part item of the service order. This process is controlled by configuring the relevant item categories in Customizing for Logistics Integration.

An ATP (Available-to-Promise) check or simulation can be started from the service parts item list in the service order to verify if the parts are available. You can enable this check by specifying the relevant ATP type for the service transaction type in the **Define Transaction Types** Customizing activity.

An outbound delivery can be created from the sales order. After picking and posting of the goods issue, the service part is transferred to the customer consignment stock of the plant and the reservation is updated during the goods issue process. The reservation is set to **Completed** after the final goods issue. The outbound delivery and the goods movement posting is added to the transaction history of the sales order and the service order, along with the delivery status and quantity update.

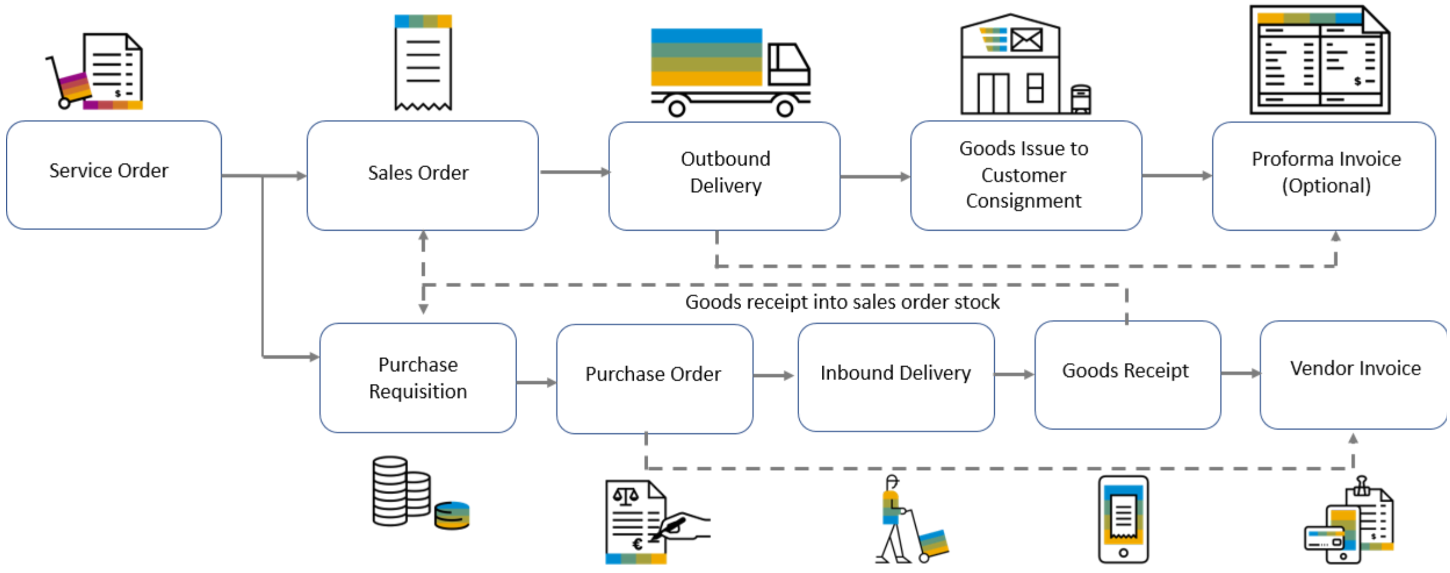
After the service technician confirms the use of the service part with a service confirmation, the final goods issue from the customer consignment stock to the customer is posted including the material consumption.

i Note

Details of the required service part are taken from the reservation, not from the sales order or delivery. The sales order is only used as the basis for creating the delivery. It is not ATP-relevant and does not transfer requirements. So, when the delivery is created, it is not actually known whether the parts are in stock. The final check of whether the service parts are in stock is only done when the goods issue is posted.

If this process doesn't fit your business process, it's possible to change the logistics integration customizing so that a sales order is created without a reservation. In this case, a sales order item category which transfers requirements and is ATP-relevant can be set up and used. The result of the ATP check of the sales order isn't transferred back to the service order.

Scenario 2: Service Parts Are Not in Stock in a Warehouse



If the service parts aren't in stock, they can be procured by means of a purchase requisition or a purchase order created from the service order item. Additionally, a sales order is created and used as the basis for creating the outbound delivery. This process is controlled by configuring the relevant item category in Customizing for logistics integration.

When the goods receipt for the purchase order is posted, the service parts are recorded in the sales order stock (special stock “E”), so that no other process can use these parts. The confirmation control key of the purchase order item determines whether an inbound delivery needs to be created.

The rest of the process, delivery of service parts to the customer, is identical to the processing of service parts from stock. Additionally, the procurement process transactions are added to the transaction history of the service order, along with the delivery status and quantity update.

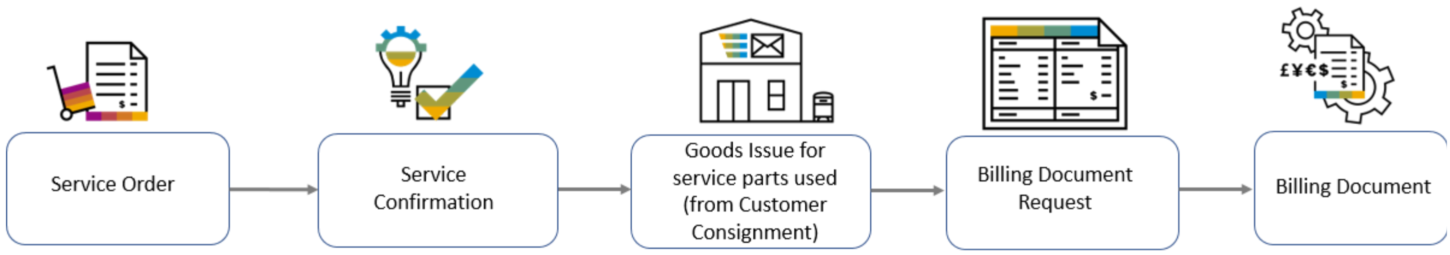
i Note

If you need to create a purchase requisition from the created sales order and not directly from the service order, you can do this by configuring Customizing for Logistics Integration so that only a sales order is created, and not a purchase requisition or purchase order.

In this case you have to set up the sales order item categories that trigger the procurement process. With this configuration, the transactions of the procurement process aren't added to the transaction history of the service order, and also no quantity or status updates are done. To track the progress of the procurement process, the sales order apps or transactions must be used.

Additional Information for Both Scenarios

The Confirmation Process



- The confirmation process is the same for both scenarios. Once the repair has been completed, the service technician enters the confirmation details. This is followed by the goods issue for the service parts used.

- The service confirmation can only confirm the service parts that have already been delivered via the outbound delivery of the sales order. If the quantity in the confirmation exceeds the delivery quantity, a warning message is triggered when the confirmation is created or changed. If the confirmation is set to **Completed**, an error message is issued and the status automatically resets to open.
- The sales order must use the same number range as the service order. For this, the external number range of the sales order type must be identical to the number range of the service order.
- More details to set up these scenarios can be found in the SAP Note https://launchpad.support.sap.com/#/notes/3281697.

Monitoring Advance Shipment Stock

You can use the Monitor Advance Shipment Stock app to track consignment stock at customer locations. This provides real-time visibility on stock consumed and stock remaining.

Related Information

Service Orders
Service Confirmations

Customizing for Advance Shipment

Use

In Advance Shipment, there are multiple Customizing activities. The settings that you must make in Customizing are described in the SAP Note https://launchpad.support.sap.com/#/notes/3281697.

Finance Integration

Introduction to the integration of Service with Financial Accounting and Controlling, and to the functions that are provided by Financial Operations

Integration with Financial Accounting and Controlling

Integrating with Financial Accounting (FI) ensures an accurate recording of cost and revenue that occur in service transactions. Controlling (CO) enables you to analyze cost and revenue that have been recorded in Financial Accounting based on different points of view and dimensions, and to calculate work in process (WIP) for long-running transactions.

Service provides the following concepts to integrate with Financial Accounting and Controlling:

- Item-Based Accounting

  This concept is based on dedicated account assignment objects (also known as **controlling object** or **CO object**) that are created for service order items and service contract items. In Financial Accounting, the items are recognized as the account assignment object to which actual costs and revenues are posted. In Controlling, the planned and actual costs and revenues are analyzed based on the accounting assignment objects. You use Event-Based Revenue Recognition to recognize costs and revenues when processing service orders and service contracts.

  Note

You can activate item-based accounting in the **Enable Item-Based Accounting for Service** Customizing activity. For more information, see Setting Up Item-Based Accounting.

- Integrating with Account Assignment Manager

  This concept provides single-object controlling or a mass controlling. Depending on your system configuration, an internal order, a WBS element, or a profitability segment is used as an account assignment object (also known as **controlling object** or **CO object**). In Financial Accounting, the account assignment object is used to post cost and revenue that have occurred in Service.

Note the following:

- The selected concept impacts the financial value flow and the supported capabilities such as the used apps and tools in SAP S/4HANA or SAP S/4HANA Cloud Private Edition.
- You can't run both concepts in parallel and you can't revert your decision in a productive system once you've decided for one concept.
- Service with Advanced Execution: You can only use Service with Advanced Execution if you've activated item-based accounting in the **Enable Item-Based Accounting for Service** Customizing activity.

  For more information, see Setting Up Item-Based Accounting and Prerequisites (settings for Service with Advanced Execution).

Financial Operations

Service is integrated with the following functions that are provided by Financial Operations:

- Service Transactions with SEPA Mandate
- Contract Accounts in Service Transactions
- Credit management

  For more information, see:

  - Credit Checks in Service Orders
  - Credit Management for Service Contracts

Item-Based Accounting

Use item-based accounting to record and analyze cost and revenue that have been incurred by service transactions.

Item-based accounting enables you to record and analyze cost and revenue in real-time that have incurred in the service order items and service contract items, and to report the profitability of services performed. For this purpose, the system creates an account assignment object (also known as **controlling object** or **CO object**) for each item that is added to a service order or a service contract.

In the context of item-based accounting, account assignment objects that are related to service transactions are defined by the following attributes:

- Service document type (transaction type that is used in Service)
- Service document (ID of the service transaction)
- Service document item (number of the service transaction item)

For more information about account assignment objects, see Account Assignment of Controlling Objects.

Item-Based Accounting in Service

Item-based accounting is provided for the following in Service:

- Item-Based Accounting for Service Orders
- Item-Based Accounting for Service with Advanced Execution
- Item-Based Accounting for Service Contracts

Item-Based Accounting in Finance

Using Account Assignment Objects in Business Transactions

Account assignment objects that are related to service transactions and that are processed in item-based accounting, are retrieved in Financial Accounting and Controlling as follows:

- Universal Journal: In the universal journal, the account assignment object that is related to the line item of the journal entry is displayed.
- Event-Based Revenue Recognition for Service Documents: The account assignment objects for the service transactions items are used as a reference to post revenue.
- Contract Accounting: The account assignment objects for the service transactions items are used as a reference for revenue and cost posting in Contract Accounting.

  For more information, see Integration with Service.

Account assignment objects also allow you to report and analyze cost and revenue in billing, procurement, or in supplier invoices.

Apps in Financial Accounting and Controlling

Financial Accounting and Controlling provide the following apps that are used to manage cost and revenue that incur in service orders and service contracts, and the related items:

> **i Note**
>
> The mentioned listed use the account assignment object as a reference.

App	Used To	For More Information, See
Service - Plan/Actuals **Service - Actuals**	View planned and/or actual cost and revenue You can use the account assignment objects to get an overview of planned and/or actual cost and revenue that have been incurred by service orders and service contracts (service documents). Additional information that is provided by the universal journal, such as the market segments, is also displayed.	Financial Reporting and Analytics

App	Used To	For More Information, See
Product and Service Margin	View recognized revenue and cost after period-end closing	Product and Service Margins
Manage Settlement Rules - Service Documents	Provide distribution rules for settlement for service documents	Manage Settlement Rules - Service Documents
Run Settlement - Actual	Settle actual costs that have been incurred by service orders	Run Settlement - Actual
Schedule Overhead Accounting Jobs	Run background jobs for overhead accounting applications, such as overhead allocation or calculation, actual cost rate calculation, or settlement	Schedule Overhead Accounting Jobs
Run Revenue Recognition - Service Documents	Run revenue recognition	Run Revenue Recognition - Service Documents
Event-Based Revenue Recognition for Service Documents	Perform manual adjustment of costs and accruals	Event-Based Revenue Recognition – Service Documents

Setting Up the System

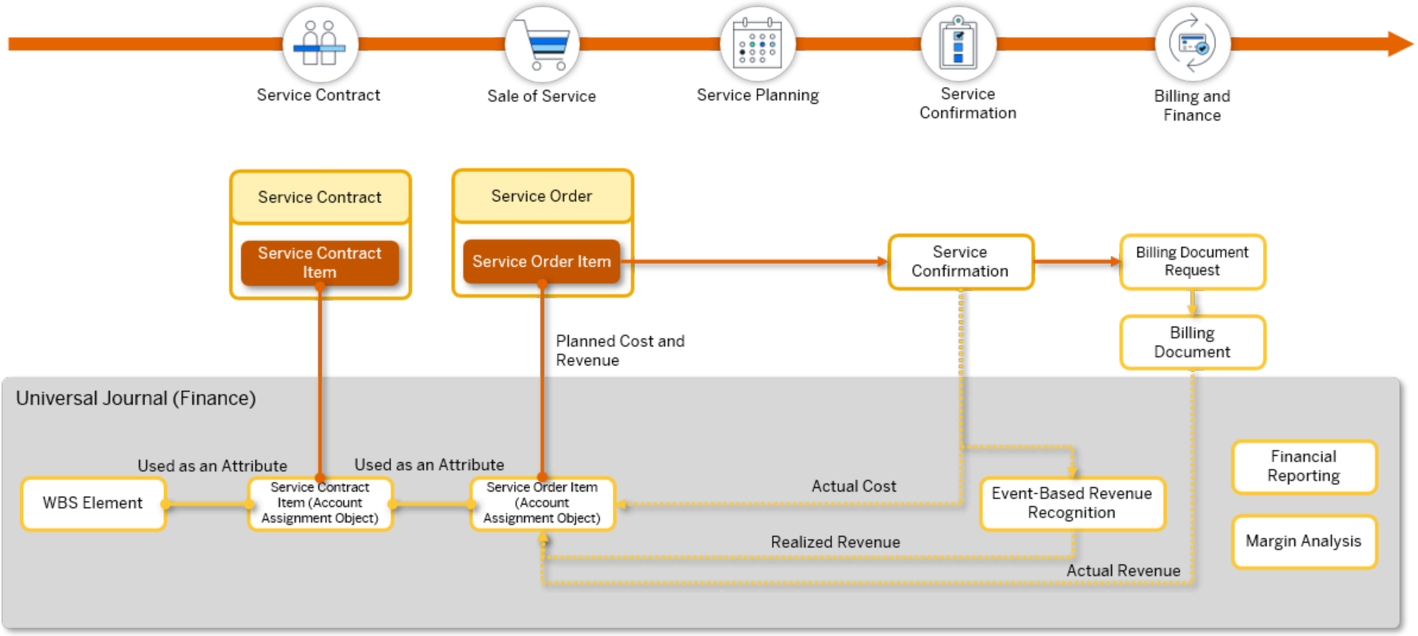
In order to manage cost and revenue that incur in service transactions, key users need to set up the system accordingly. For more information about the required steps, see Setting Up Item-Based Accounting.

Item-Based Accounting for Service Orders

Information about the process of item-based accounting that is related to service orders

In item-based accounting, cost and revenue are processed along with the life cycle of the service order and the related items:

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

1. Initiate the service: The system creates the account assignment object (controlling object) for the items that are added to the service order.

   For more information, see [Triggering Item-Based Accounting]

2. Plan the service: The planned cost and revenue that incurred for the service order items is recorded.

   - Planned cost is recorded in the service order items.
   - Planned revenue derives from the price that includes, for example, any discounts or rebates that have been defined for the service order item.

   For more information, see [Service Planning: Managing Planned Cost and Revenue].

3. Execute the service: The actual cost that has been incurred for the service order item and that has been confirmed by means of a service confirmation is recorded.

   For more information, see [Service Confirmation: Managing Actual Costs].

4. Trigger billing for the service: The actual revenue that derives from the price of the performed service and that is charged to the customer in billing is recorded.

   - For service order items, actual revenue derives from the service confirmation items that are billed.
   - For fixed price service order items, revenue derives from the fixed price service order items that are billed.

   For more information, see [Billing: Managing Actual Revenue].

5. Complete the service: Finalize the service processing in the service order.

   For more information, see [Finalizing the Service Order].

6. Event-Based Revenue Recognition for the service order is used to recognize cost and revenue in real time that have occurred for the service order items.

   For more information, see [Event-Based Revenue Recognition for Service Orders].

Triggering Item-Based Accounting

Initiate item-based accounting for service orders

Procedure

Starting the Process

For service orders, the processing of cost and revenue starts when you add items to the service order. When you save your entries, the system automatically creates an account assignment object (also known as **Controlling object** or **CO object**) for each service order item.

i Note

An account assignment object is created only if the service order item doesn't contain any errors.

The account assignment object is used to record cost and revenue that occur in service order processing. The item category that is assigned to the service order item controls the business process and therefore affects the recorded cost and revenue.

For more information about account assignment objects in Service, see [Item-Based Accounting].

For more information about how the assigned item category controls cost and revenue posting, see Service Confirmation: Managing Actual Costs.

Profit Center Determination

Having added the item to the service order, the system determines a profit center for the item. During profit center determination, the system first determines the cost center of the employee responsible that is assigned to the item and then determines the profit center that is defined for the cost center in the cost center master data. For more information about the processing, see Profit Center Determination.

Subitems such as price items inherit the profit center that has been assigned to the higher-level item.

> **Note**
>
> You must not change the assigned profit center that has been determined by the system after the first actual posting.
>
> If no profit center has been defined for the cost center, the system uses the standard profit center that has been defined for the controlling area.

You can use the **Manage Substitution/Validation Rules - Service Documents** app (see Substitution/Validation for Service Documents) to tailor the profit center determination to your business requirements.

Assigning WBS Elements

WBS elements that are assigned to items can be used for profitability reporting and analytics.

Subitems such as price items inherit the WBS element that has been assigned to the higher-level item.

For more information, see WBS Elements in Service.

Service Contracts as Predecessor Transaction

If you've created the service order as a follow-up to a service contract, the system creates an account assignment for each item that is taken over from the service contract. In the universal journal, the service contract is assigned as an attribute to the account assignment object of the service order item.

For more information, see Item-Based Accounting for Service Contracts.

How to Avoid Inconsistent Data

Note the following to avoid inconsistent data:

- If you change the employee responsible for an item, the system redetermines the profit center. However, if you've already posted costs to the item, you can't change the profit center.
- Don't change the service contract that has already been assigned to the item if you've already posted costs to the item.
- You can manually change the profit center that has been assigned to an item. Once you've made such a manual change, the system no longer redetermines the profit center if you change the employee responsible that is assigned to the item.
- If you change the employee responsible that has been assigned to the item, the system redetermines the profit center only if you've removed the manually assigned profit center. Alternatively, you can define rules in the **Manage Substitution/Validation Rules - Service Documents** app to tailor the profit center determination to your business requirements.

Service Planning: Managing Planned Cost and Revenue

This section explains the features and functions of planned cost and revenue in service orders.

Use

You can check the expected cost and revenue, see the margin, and update the relevant data in a service order in your service planning process.

You can view two types of planned cost and revenue data:

Baseline

When the status of a service order has been set to **Released** for the first time, the planning process of a service is considered as completed. The cost and revenue that have been planned up to this point are considered as the baseline of planned cost and revenue. The figures in the baseline section are calculated when the service planning process is complete. So, the baseline represents the original data at the initial stage of the planning process.

Ongoing

While the baseline of planned cost and revenue serves as a basis for comparison, the ongoing planned cost and revenue reflect the continuously updated data when changes occur in the service planning process. The changes can be a change of item quantity or an added item in a service order.

> **Note**
> You can disable the function of planned cost and revenue in the Customizing activity **Define Transaction Types** under **Service > Transactions > Basic Settings**.

Features

Planned Cost and Revenue

The tab **Planned Cost and Revenue** is available at both service order header and item levels for you to view the relevant data.

Service Order

In the service order details, you can view the following baseline and ongoing data of a service order on the tab **Planned Cost and Revenue**:

- Planned Cost
- Planned Revenue
- Calculation Status

Status	Description
Not calculated	No baseline or ongoing planned cost and revenue of the service order has been calculated.
Not successful	The last attempt to calculate the baseline or ongoing planned cost and revenue for all items of the service order wasn't successful.

Status	Description
Partially successful	At least one item of the service order's baseline or ongoing planned cost and revenue has been calculated.
Successful	The last attempt to calculate the baseline or ongoing planned cost and revenue for all items of the service order was successful.

- Planned Margin

  You can view the difference in percentage between planned cost and planned revenue of a service order. The fields **Baseline Planned Margin** and **Ongoing Planned Margin** allow you to compare the margins of baseline and ongoing figures in your planning process.

Service Order Item

If you want to know the baseline and ongoing planned cost and revenue of a specific item, you can find the relevant data in two ways:

1. At service order header level, you can adjust the view settings of the **Items** tab and add the columns that display the baseline and ongoing data of each service order item in a service order.
2. At service order item level, you can also find the relevant data by navigating to the **Planned Cost and Revenue** tab in the service order item details.

You can view the following baseline and ongoing data in either way:

- Planned Cost
- Planned Revenue

  If you've assigned a price item to an item as a subitem, the planned revenue of the price item is added to the planned revenue of the higher-level item.
- Calculation Status

Status	Description
Not calculated	No baseline or ongoing planned cost and revenue of the service order item has been calculated.
Not successful	The last attempt to calculate the baseline or ongoing planned cost and revenue for the service order item wasn't successful.
Successful	The last attempt to calculate the baseline or ongoing planned cost and revenue for the service order item was successful.

Update of Planned Cost and Revenue

Update of Baseline Data

When the status of a service order has been set to **Released** for the first time, the baseline planned cost and revenue are calculated automatically and displayed. You can update the baseline of planned cost and revenue by choosing **Update Baseline** under the **Planned Cost and Revenue** tab. This function enables you to manually trigger a recalculation of the baseline of planned cost and revenue of a service order after the initial release of the service order.

You can decide to trigger the recalculation in the following situations:

- If a service order item has been forgotten during the planning process, and you've added the item in the service order and released the service order again.
- If you've changed the quantity of an item and released the service order again.
- If you've canceled a service order item.

i **Note**

To use this update function, the service order must be error-free and set to the status **Released**.

Update of Ongoing Data

The ongoing planned cost and revenue are calculated before a service order is released. In addition, the ongoing data is updated automatically whenever changes are made to a service order. For example, when you add a service order item or edit an item, a recalculation of the ongoing figures is triggered. For the ongoing data to be calculated and updated, the items in a service order or service bundle must be error-free.

i **Note**

The ongoing planned cost and revenue are recalculated when you choose **Update Baseline** since the recalculation of the baseline figures also affects the ongoing figures.

Search Function

You can adapt the search filters **Baseline Calculation Status** or **Ongoing Calculation Status** to search for a service order. The search is based on the calculation status of the baseline or ongoing planned cost and revenue of a service order.

Calculation of Planned Cost and Revenue for Service Order Items

Both baseline and ongoing planned cost and revenue are calculated using the same method.

The planned cost and revenue of a service order are based on the planned cost and revenue of items in the service order. The items that you can use for a service order are as follows:

- Service item

  If you've assigned a price item to a service item as a subitem, the higher-level service item inherits the planned revenue that has been calculated to the price item. For more information about price items, see Price Items in Service Orders and Service Quotations.
- Expense item
- External service
- Service part
- Stock service part
- Execution order item

These items can be used in the following cases:

- As standard items, which are items that are billed based on the consumption of time and materials or fixed price items

- As standard items in a hierarchy

Both baseline and ongoing planned cost and revenue can also be calculated for unplanned items. This is especially useful if you use the event-based revenue recognition method **Recognize Revenue on Cost-Based Percentage of Completion (POC)**, and the unplanned items are created as subitems of the main item in a service hierarchy.

The following table provides an overview of how planned cost and revenue are calculated by items:

> **Note**
>
> Each item in the table can be an item billed based on the consumption of time or a fixed price item.

Item	Description	Planned Revenue	Planned Cost
Service Item	Time required to provide the service, which is defined by the reported time of work multiplied by work duration	Item net price multiplied by the planned or agreed working hours	Cost rates multiplied by the working hours **Note** Cost rates are determined based on a variety of criteria. Several apps are available for defining cost rates. For more information, see Cost Rates.
Expense Item	Expenditure spent on the service (such as travel cost)	Item net price multiplied by the quantity	• Based on the pricing condition type assigned and the amount for the corresponding expense item • Or if no pricing condition type is assigned, the item net price multiplied by the quantity
External Service	External workforce procured to provide the service	Item net price multiplied by the quantity	• Determined by the relevant price conditions from the purchasing information records • Or if no records are found, define manually.
Service Part	Parts procured as required for the service	Item net price multiplied by the quantity	• Determined by the relevant price conditions from the purchasing information records • Or if no records are found, define manually.

Item	Description	Planned Revenue	Planned Cost
Stock Service Part	Parts available in stock and picked by technicians	Item net price multiplied by the quantity	Net price from material master record multiplied by the quantity

Note

Planned cost and revenue aren't supported for intercompany items in the customer-facing service order if they are part of a service hierarchy.

Service items that have a price item assigned inherit the planned revenue that has been calculated to the price item. For more information about planned revenue that has occurred for price items, see Price Items in Service Orders and Service Quotations.

Fixed Price Items with Ad Hoc Billing

For fixed price service order items using ad hoc billing plans, the planned revenue is determined by billing dates. The planned revenue of these items is calculated according to the dates on which billing is to occur and the value that is to be billed instead of the net value of the items. For more information, see Ad Hoc Billing for Service Orders.

Event-Based Revenue Recognition

You can conduct Event-Based Revenue Recognition for service orders based on planned cost and revenue using the method **Recognize Revenue on Cost-Based Percentage of Completion (POC) (Method 3)**. This revenue recognition method allows you to recognize revenue on a cost-based percentage of completion (POC) using planned costs from estimation at completion and planned revenue from the billing plan. In addition, it allows for multiple revenue accounts to be used, corresponding to each price condition or discount/surcharge contributing to the planned revenue. For more information, see Revenue Recognition Methods for Service Order Items.

Note

You can only use the revenue recognition method 3 if you've activated planned cost and revenue in Customizing for **Service** under **Define Transaction Types**.

Related Information

Analytical Apps for Management Accounting

Service Confirmation: Managing Actual Costs

Record actual costs in service order processing

Actual costs are costs that have actually been incurred during service processing. They're recorded once the executing service employee creates a service confirmation as a follow-up to a service order item to confirm the performed services. When the status of the service confirmation changes to **Completed**, actual costs are posted according to the posting rules that apply to the item category assigned to the service order items.

Note

You can only create service confirmations as a follow-up to service order items and thereby trigger the recording of actual cost if the service order item has the status **Released**.

Service confirmation items and unplanned service confirmation items don't create their own account assignment objects (also known as **Controlling objects** or **CO objects**) but use the account assignment objects of the related service order items.

Service Product Item (Service Item)

Service items are used to record and confirm the time that is required to provide the service.

When you create a service confirmation for a service item, actual costs are processed as follows:

1. The confirmed activity is automatically transferred to the time sheet of the executing service employee. The sender cost center is determined from the personnel number of the executing service employee.
    - The following data is used to post actual costs:
        - The activity type that is related to the service item classifies the labor performed.
        - The account assignment object is used as a reference to the service order item.
    - The duration (for example, in hours) of the labor performed is transferred to the time sheet of the executing service employee who has provided the service. The quantity that has been entered in the service order item for the labor performed is used for billing and thereby is used to create the actual revenue.

    For more information about managing time sheets, see Cross-Application Time Sheet (CATS).
2. Once the status of the service confirmation is set to **Completed**, the actual duration of the service (for example, in hours) is posted to the universal journal as costs. Costs that occur for service items have the reference document **CATS** (cross-application time sheet) assigned.

    The confirmation of the activities results in the following postings:
    - The sender cost center that is assigned to the executing service employee is credited.
    - The profit center that is assigned to the service order item and that is related to the account assignment object (service document item) is debited.

i Note

You can use the **Manage Substitution/Validation Rules - Service Documents** app (see Substitution/Validation for Service Documents) to derive the profit center, for example, by the product that is used as a service item.

If the financial period during which the labor was reported has already been closed, the system can’t automatically post the data recorded in an approved time sheet to Financial Accounting and Controlling. In this case, you must post the data manually using the **Process Unposted Time Confirmations** app. For more information, see Process Unposted Time Confirmations.

Expense Item

Expense items are used to record and confirm the expenditures incurred for the service, for example, travel costs.

When you create a service confirmation for an expense item, a cost reposting is performed in Financial Accounting and Controlling by distributing actual costs from the cost center that is assigned to the service technicians to the service order:

- The sender cost center that is related to the executing service employee who has confirmed the expense item is credited.
- The service order item that is referenced in the account assignment object is debited.
- The service rendered date that is entered in the service confirmation defines the posting date in Financial Accounting and Controlling.

For more information, see Service Confirmations.

Note

You can use the **Manage Substitution/Validation Rules - Service Documents** app (see Substitution/Validation for Service Documents) to derive the profit center, for example, by the product that is used as an expense item.

Stock Service Part Item (Stock Service Part)

Stock service parts are used to record and confirm the service parts that are in stock and picked by the executing service employee to perform the service.

When you set the status of the service confirmation for a stock service part to **Completed**, the system posts a goods movement for the stock service part using the goods movement type 291 (**Consumption for all account assignments from warehouse**).

This results in the following postings:

- The inventory for the stock service part is credited.
- The service order item that is referenced in the account assignment object is debited.
- The service rendered date that is entered in the service confirmation defines the posting date in Financial Accounting and Controlling.

Procured Service Part (Service Part)

Procured service parts are used to record and confirm service parts that are procured in order to fulfill the service.

The costs that are posted for the service parts are initially obtained from the goods receipt. Once you receive the supplier invoice for the service part, the price listed in the invoice is posted to the account assignment object of the related service order item. The service order item that is referenced in the account assignment object is debited.

For more information, see Reservation of Stock Service Parts and Stock Transfer of Service Parts.

Procured Service Item (External Service Item)

External service items are used to record and confirm the external workforce that is procured to provide the service.

When the service entry sheet that is used to record and confirm the provided services, is approved, the system creates a service confirmation with status **Open**, and posts the price that is stated in the purchase order or the supplier invoice to the account assignment object that has been defined in the service entry sheet. Costs are posted to the account assignment object of the related service order item. The sender cost center that is related to the external service performer who has confirmed the labor performed is credited.

For more information, see External Workforce Procurement. For more information about cost postings that are triggered by service confirmations, see the section Lean Services.

Intercompany Service Order Item

Intercompany service order items are used to record and confirm services that have been delegated to another company code (that is the **receiving sales organization**) in the same system.

The receiving sales organization bills its revenues that have been incurred for the performed services as actual costs to the distributing company that owns the customer-facing service order. So, the costs are posted to the account assignment object that is related to the intercompany service order item as accounts payable.

For more information about the intercompany service order items and the integration with Financial Accounting and Controlling, see Intercompany Service Orders in the section Process under **Service Confirmation**.

Execution Order Item

Costs that occur in Service with Advanced Execution are recorded in the maintenance order that has been created as a follow-up to an execution order item. For more information about managing costs processed in the advanced execution process, see Item-Based Accounting for Service with Advanced Execution in the section Processing Planned and Actual Cost.

Billing: Managing Actual Revenue

Posting actual revenue in the billing process

When the invoice is created for the billing-relevant items, actual revenue is posted to Financial Accounting and Controlling.

For more information about the billing process and the requirements that must be fulfilled for billing, see:

- Billing
- Forwarding Billing Data to Financial Accounting.

Processing Actual Revenue of Price Items

Price items that are used as subitems of service items or execution order items inherit the account assignment object of the related higher-level item. So, the actual revenue that has occurred for the price item is posted to the higher-level item.

For more information, see Price Items in Service Orders and Service Quotations.

Finalizing the Service Order

Finalize the service order processing and close the service order for further processing.

Once the service has been finalized, you complete the service order. To do so, you first set the service order to status **Completed**. Having done this, you can perform the business completion for the service order to avoid any postings of costs and revenues to the related account assignment object and to make the service order archivable.

Setting Service Orders to Completed

To finalize the service processing, you set the service order to status **Completed**. In this status, follow-up processes in Financial Accounting, Controlling, and Sales Billing are still possible.

Business Completion of Service Orders

Performing the Business Completion

If no further costs and revenues are expected to be posted to the service order in status **Completed** and the related account assignment objects, you can set the service order to **Business Completed**. Having done this, you aren't able to process the service order anymore and to post costs and revenues to the account assignment objects.

You can perform the business completion for a service order if the following applies:

- All service confirmations that have been created as a follow-up to the service order have the status **Completed**.

  Once the related service order has been set to **Business Completed**, you can't cancel the related service confirmations afterwards.

- All intercompany orders that have been created as a follow-up to the service order have the status **Completed** and have already been set to **Business Completed**.

- All related billing documents have been created and posted to Financial Accounting.

  The billing documents have been created for the service order items or the follow-up service confirmations, or out of an ad hoc billing plan.

- All postings to the time sheets of the executing service employees that are related to the service order have been transferred to Financial Accounting and Controlling.

  You manage the CATS postings in the **Process Unposted Time Sheet Confirmations** app.

- There are no open supplier invoices that are related to the service order items.

- There are no open sales documents, such as credit memo requests, that are related to the service order items.

- There are no open billing documents, such as credit memos, that are related to the service order items.

- If Event-Based Revenue Recognition has been activated (see Event-Based Revenue Recognition for Service Orders): The clearing of the related balance sheet accounts for accruals and deferrals is done.

- If a mandatory settlement rule has been defined for the service order in the **Manage Settlement Rules - Service Documents** app: The settlement has been successfully performed for the service order in the **Run Settlement - Actual** app.

You can use the report **Set Service Orders to Business Completed** (`CRMS4_SET_BUSINESS_COMPLETED`) to schedule batch jobs that are used to perform the business completion for service orders and which haven't been reopened for business before.

Note

If the service order has been set to **Business Completed**, you can archive the service order. For more information, see Archiving Service Orders Using CRM_SERORD.

Reopening for Business

You might reopen the service order for business, for example, if:

- Further costs that are related to the service order have been occurred subsequently and have to be posted to Financial Accounting and Controlling.

- An unexpected debit/credit memo request is required.

- The invoice that has been created for the service order is to be canceled.

In this case, you revise the business completion of the service order and set it to **Business Reopened**. Having done this, you can make the required changes in the service order.

Once you've finalized the changes, you must set the service order to **Business Completed** again.

Event-Based Revenue Recognition for Service Orders

Use Event-Based Revenue Recognition to recognize costs and revenues in the service order process.

With event-based revenue recognition, costs and revenues that have been posted to the respective account assignment objects are recognized as they occur. Cost postings are continuously matched to revenues and immediately reported as expenses, while revenues are posted to an income statement account.

The following chapters in the product assistance for Finance provide further information about Event-Based Revenue Recognition:

- Event-Based Revenue Recognition: General information about Event-Based Revenue Recognition
- Event-Based Revenue Recognition for Service Orders: Overview about how costs and revenues for service orders are processed within the revenue recognition process
- Revenue Recognition Methods for Service Order Items: Overview of the revenue recognition methods that enable you to specify the conditions under which costs and revenues are recognized
- Customizing Requirements for Service Orders: Overview of the Customizing activities that are used to set up Event-Based Revenue Recognition

Posting Cost and Revenue Manually

Posting cost and revenue that aren't recorded in the service order items and aren't assigned to a service confirmation

If cost and revenue aren't recorded in the service order items and aren't assigned to a service confirmation, you can post cost and revenue incurred by service order items by means of the corresponding account assignment object. You can do this in the following apps that are provided in Financial Accounting:

- **Post General Journal Entries** (Post General Journal Entries)
- **Manage Direct Activity Allocation** (Manage Direct Activity Allocation)
- **Manage Supplier Invoices** (Manage Supplier Invoices)
- **Reassign Costs and Revenues** (Reassign Costs and Revenues)

Costs that aren't confirmed in a service confirmation aren't relevant for billing.

Item-Based Accounting for Service with Advanced Execution

Business process information about managing cost and revenue in Service with Advanced Execution

Process

Starting the Accounting Process

For services that are performed within Service with Advanced Execution (see Service with Advanced Execution), the processing of cost and revenue starts when you add execution order items to a service order. When you save your entries, the system creates an account assignment object for each execution order item.

Account Assignment for Service with Advanced Execution

For maintenance orders, cost postings are performed via the related account assignment object (see Working with the Order Header as Account Assignment Object), while execution order items use their own account assignment objects to post revenue. Both posting objects are related to each other using attributes that are provided by the account assignment object that is related to the execution order item (service document type, service document ID, and service document item) and market segment attributes, such as bill-to party or service contract type.

Profit Center Determination

Having added the execution order item to the service order, the system determines a profit center for the execution order item that is also copied to the related billable maintenance order. You can now maintain the same or different profit centers in the execution order item and the corresponding billable maintenance order based on the profit center Customizing option in the Customizing activity **Define Item Categories**. If you choose the settlement option **Settlement Without Hierarchy Allowed** or **No Settlement**, you can select one of the following:

- Dependent on Execution Order Item: The profit center in the execution order item and the corresponding billable maintenance order will be the same and cannot be changed.
- Independent of Execution Order Item: The profit center in the billable maintenance order can be different from the execution order item.

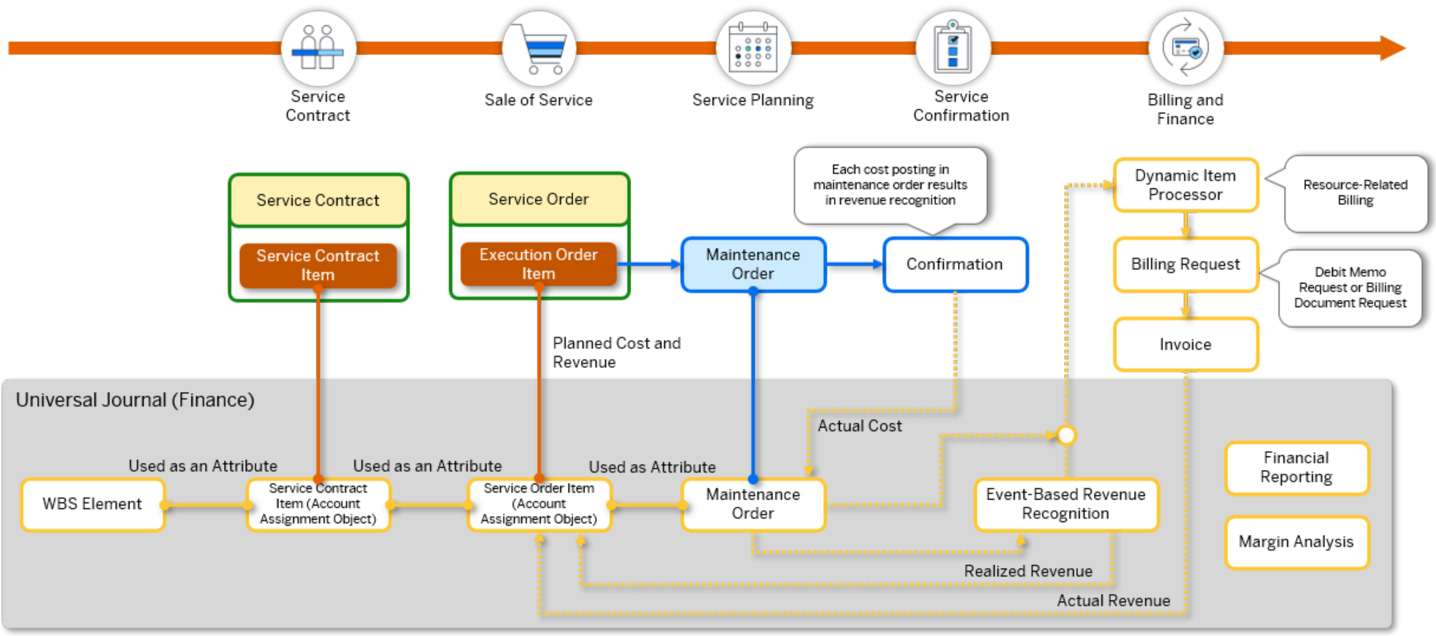
If you choose the settlement option **Settlement With Hierarchy Allowed**, the profit center in the execution order item and the corresponding billable maintenance order must be the same and can't be changed.

For more information about profit center determination, see Profit Center Determination.

i Note

If no profit center has been defined for the cost center, the system uses the standard profit center that has been defined for the controlling area.

Process Overview



Processing Planned and Actual Cost

Planned and actual cost occur in the maintenance order that is used to plan and perform the service. In the execution order item, the planned and actual cost that have been created for the maintenance order are displayed under **Maintenance Order**.

If a maintenance order is created as a follow-on to a billable maintenance order and added to an existing service order as an execution order item (see Distributed Execution), planned and actual cost of the follow-on order is also taken into account by the service order.

Planned Cost

Planned cost occurs in the planning phase of the service (see Planning of an Order).

In the execution order item and the related service order, you can check the planned cost under **Planned Cost and Revenue**:

- The baseline cost contains the planned cost that is caused when the maintenance order is released
- The ongoing cost contains the planned cost that is updated continuously when the planned cost in the maintenance order is updated.

Actual Cost

Actual cost occurs when the service is confirmed in the maintenance order and the confirmation is completed.

For more information about processing actual cost in Maintenance Management see:

- Maintenance Order Costing and Reporting
- Costs
- Completion of an Order
- Settlement of an Order

Processing Planned and Actual Revenue

Planned and actual revenue are created in the execution order item.

Planned Revenue

Planned revenue occurs in the service planning process. It derives from the price that includes, for example, any discounts or rebates that have been defined for the execution order item. You can check the planned revenue under **Planned Cost and Revenue**:

- The baseline revenue contains the planned revenue that results once the execution order item is released is displayed.

  The execution order item has the execution status **In Execution**.
- The ongoing revenue contains the planned revenue that is updated continuously once the planned revenue in the execution order item is updated.

Actual Revenue

Actual revenue is created when the billing document is created for the execution order item. It derives from the price of the performed service and that is charged to the customer in billing.

Each execution order item is related to an account assignment object that is used for cost postings.

For more information, see Billing: Managing Actual Revenue.

Finalizing the Advanced Execution Process

Completing Service Orders

To finalize the advanced execution process, you must set the status of the execution order item and the related service order to **Completed**. In this status, follow-up processes in Financial Accounting, Controlling, and Billing are still possible.

Completed for Business

If no further costs and revenues are expected to be posted to the service order in status **Completed** and to the related account assignment objects, you can set the service order to **Business Completed**. Having done this, you aren't able to process the service order anymore and to post costs and revenues to the account assignment objects.

If you set the service order that is related to the execution order to **Business Completed**, make sure that the following applies to the execution order item:

- All related billing documents have been created and posted to Financial Accounting and Controlling.

  The billing document has been created for the execution order item out of an ad hoc billing plan, or out of a billing document request or a debit memo request.

- There are no open billing documents, such as credit memos, that are related to the execution order item.

- If Event-Based Revenue Recognition has been activated: The clearing of the related balance sheet accounts for accruals and deferrals is done.

- In case of distributed execution, all billable maintenance orders need to be set to Completed.

For more information about setting service orders to **Business Completed**, see Business Completion of Service Orders.

Event-Based Revenue Recognition

Event-Based Revenue Recognition is used to recognize cost and revenue in real time. In this way, cost postings are continuously matched to revenue and immediately reported as expenses, while revenue is posted to an income statement account. Event-Based Revenue Recognition is controlled by revenue recognition methods.

For more information about Event-Based Revenue Recognition for Service with Advanced Execution and the provided methods, see:

- Event-Based Revenue Recognition for Service with Advanced Execution
- Revenue Recognition Methods for Execution Order Items
- Customizing Requirements for Execution Order Items
- Revenue Recognition Apps for Service Document Items

Item-Based Accounting for Service Contracts

Managing cost and revenue for service contracts

Procedure

Service contracts create revenue that is posted to Financial Accounting. Once you save at least one service contract item in status **Released** for the first time, the system creates an account assignment object, that is used to record the planned and actual revenue that has been incurred for the service contract:

- Planned revenue derives from the price, which includes, for example, any discounts and rebates that have been defined for the service contract items.
- Actual revenue derives from the price of the service that is charged to the customer in billing.

  For more information about how actual revenue occurs once billing has been finalized for a billing plan, see Billing of Service Transactions.

Profit Center Determination

Having added the service contract item to the service contract, the system first determines the cost center of the employee responsible that is assigned to the item and then determines the profit center that is defined for the cost center in the cost center master data.

You can use the **Manage Substitution/Validation Rules - Service Documents** app (see Substitution/Validation for Service Documents) to tailor the profit center determination to your business requirements.

For more information about profit center determination, see Profit Center Determination.

Event-Based Revenue Recognition

For service contracts, event-based revenue recognition enables time-based recognition of revenue based on the billing plan that has been defined for the service contract. Once the status of the service contract changes to **Released**, Event-Based Revenue Recognition calculates and posts the revenue based on the billing plan that has been created for the service contract items. Revenue is distributed evenly on a monthly basis according to the duration of the billing plan and realized with the period-end run.

For more information about the process and the methods that are provided for Event-Based Revenue Recognition, see Event-Based Revenue Recognition for Service Contracts.

Service Contract Determination in Service Orders

Service order items have their own account assignment objects. If a service contract has been determined for a service order item or if a service order is created as a follow-up to a service contract, the service contract is assigned as an attribute to the account assignment object of the service order item in the journal entry.

i Note

In the **Service Actuals** app, you can analyze the cost and revenue for a service order and the related service contract.

Finalizing the Service Contract

Finalize the service contract processing and close the service contract for further processing.

Once the service has been finalized, you complete the service contract. To do so, you first set the service contract to status **Completed**. Having done this, you can perform the business completion for the service contract to avoid any postings of costs and revenues to the related account assignment object and to make the service contract archivable.

Setting Service Contracts to Completed

To finalize the service processing, you set the service contract to status **Completed**. In this status, follow-up processes in Financial Accounting, Controlling, and Sales Billing are still possible.

Business Completion of Service Contracts

Performing the Business Completion

If no further costs and revenues are expected to be posted to the service contract in status **Completed** and the related account assignment objects, you can set the service contract to **Business Completed**. Having done this, you aren't able to process the service contract anymore and to post costs and revenues to the account assignment objects.

You can perform the business completion for a service contract if the following applies:

- All service orders that have been created as a follow-up to the service contract have the status **Completed**.

  Once the related service contract has been set to **Business Completed**, you can't cancel the underlying service orders in the root item in the financial hierarchy afterward.

- All related billing documents have been created and posted to Financial Accounting.

  The billing documents have been created for the service contract items or the follow-up service confirmations, or out of an ad hoc billing plan.

- All postings to the time sheets of the executing service employees that are related to the service contract have been transferred to Financial Accounting and Controlling.

  You manage the CATS postings in the **Process Unposted Time Sheet Confirmations** app.

- There are no open supplier invoices that are related to the service contract items.
- There are no open sales documents, such as credit memo requests, that are related to the service contract items.
- There are no open billing documents, such as credit memos, that are related to the service contract items.
- If Event-Based Revenue Recognition has been activated (see Event-Based Revenue Recognition for Service Contracts): The clearing of the related balance sheet accounts for accruals and deferrals is done.
- All assigned account objects to contract items are Technically Completed.

You can use the report **Set Service Contracts to Business Completed** (`CRMS4_SET_BUSINESS_COMPLETED`) to schedule batch jobs that are used to perform the business completion for service contracts and which haven't been reopened for business before.

Note

If the service contract has been set to **Business Completed**, you can archive the service contract. For more information, see Archiving Service Contracts Using CRM_SRCONT .

Reopening for Business

You might reopen the service contract for business, for example, if:

- Further costs that are related to the service contract have been occurred subsequently and have to be posted to Financial Accounting and Controlling.
- An unexpected debit/credit memo request is required.
- The invoice that has been created for the service contract is to be canceled.
- The child archived indicator is not set to true for any existing child objects.

In this case, you revise the business completion of the service contract and set it to **Business Reopened**. Having done this, you can make the required changes in the service contract.

Once you've finalized the changes, you must set the service contract to **Business Completed** again.

Related Information

[Hierarchy Settlement with Service Transactions]

Event-Based Revenue Recognition for Service Contracts

Learn more about how you can conduct Event-Based Revenue Recognition for service contracts.

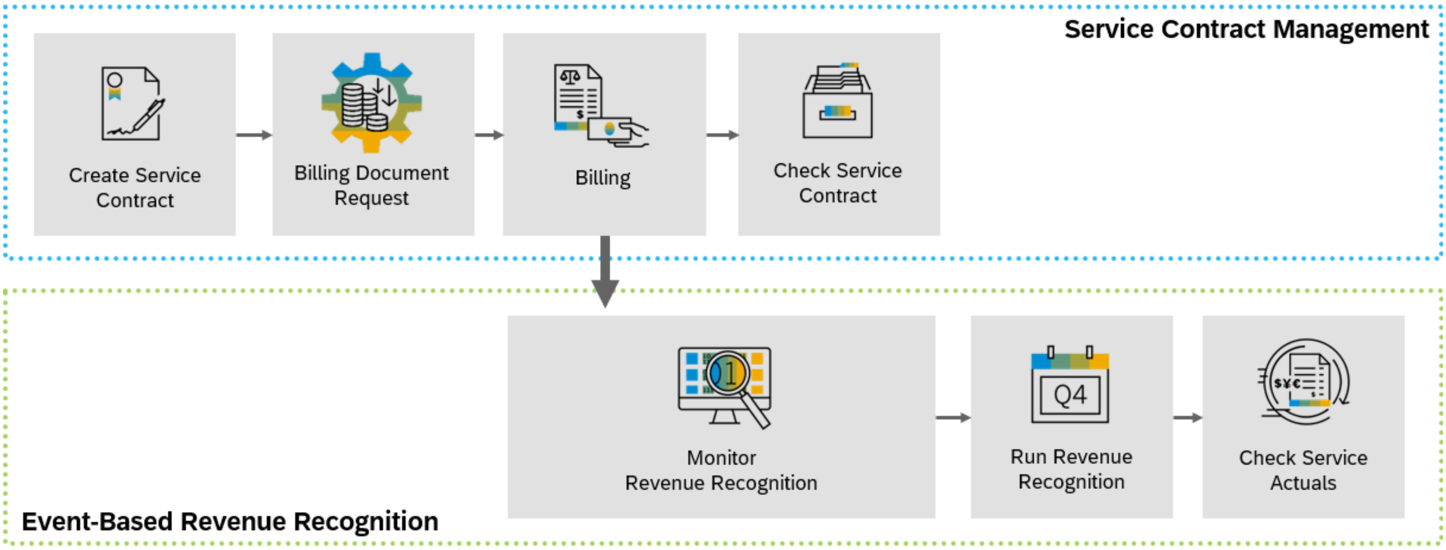
The standard **Service Contract** scenario enables handling of the end-to-end service contract management including billing, cost accounting, and revenue recognition. Service managers can manage the outline agreements with business partners that define the services offered/covered for a list of products and installations over a defined period. With the subsequent process that impacts revenue and costs, this scenario provides complete transparency of the service margin. It covers the processes including the creation of service contracts with billing plan, price adaptation, cancellation, and renewal. To learn more about service contracts in general, see Service Contracts.

Event-Based Revenue Recognition can be conducted for service contracts that are time-based. Furthermore, Event-Based Revenue Recognition can only be conducted for the service contract item categories `SCN`, `SCNA`, and `SCNB`.

Integration Overview

The following graphic illustrates the integration of the Service Contract Management with Event-Based Revenue Recognition.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

In addition to conducting revenue recognition for regular service contract items, you can also use Event-Based Revenue Recognition for service contract items that are part of a specific financial hierarchy. To learn more, see Event-Based Revenue Recognition for Service Contracts as Root Items in Financial Hierarchy.

Technical Completed Status

The status **Technically Completed (TECO)** indicates that the service associated with an item in a service order or service contract has been executed, billing has been finalized, and the primary cost and revenue have been posted. You need to set the account assignment object of an item to the status **TECO** if you want to start the period-end closing process, and meanwhile clear the balance of relevant cost and revenue with Event-Based Revenue Recognition. To learn more, see TECO Status of Account Assignment Objects in Service.

Related Information

Service Contract Management
Manage Service Contracts

Revenue Recognition Methods for Service Contracts

Learn more about the available revenue recognition methods for service contracts and how revenue recognition values are calculated.

During **Customizing**, your system administrator can either freely define revenue recognition keys that then have to be linked to the revenue recognition methods that are listed below, or you can use the predelivered revenue recognition keys. In the following, examples for the posting logic of the available revenue recognition methods are laid out.

> **Note**
> You might be able to select more methods than the ones listed below. However, the following methods are the only ones that are officially supported.

Revenue Recognition Method “Recognize Revenue Time-Based” (Method 10)

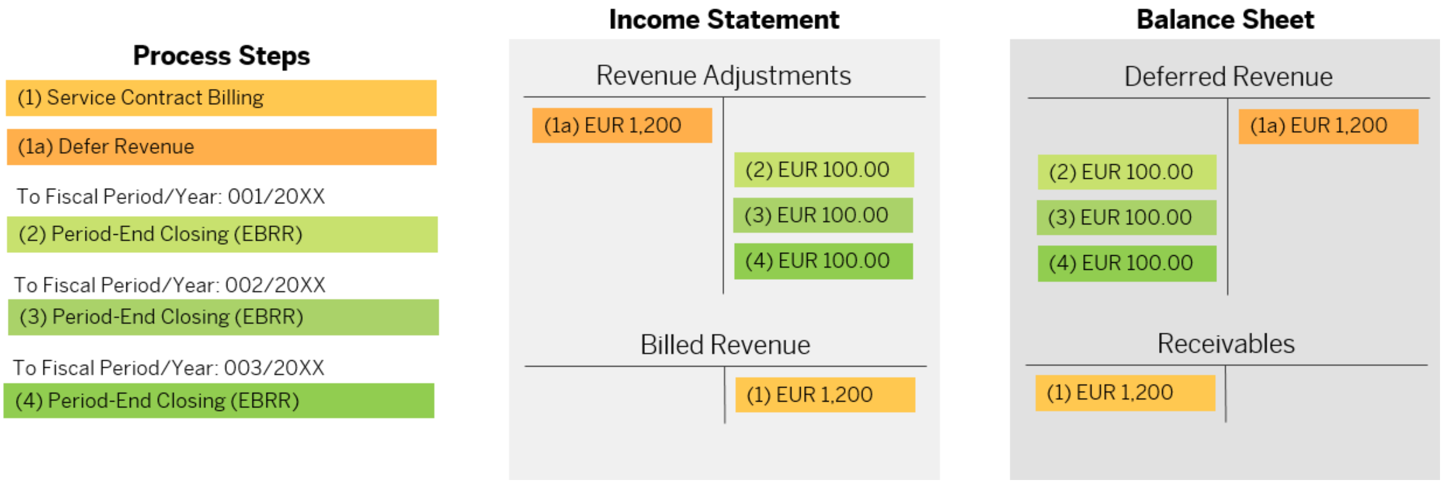
Here, we explain the posting logic behind the revenue recognition method 10 (corresponds to predelivered revenue recognition key CCS), which allows you to use time-based recognition of revenue based on the billing plan for the end-to-end process of the service contract.

Usually, after you assign the revenue recognition key to the service document, Event-Based Revenue Recognition triggers postings in Financial Accounting accordingly.

Let’s assume that the following parameters are defined for your service contract:

- Service contract for 1 year (January 1 – December 31)
- Net value: EUR 1,200.00
- Billing yearly
- Billing Amount: EUR 1,200.00 (Billing Date = Contract Start Date = January 1)

The following graphic shows you the process steps for this method and the corresponding postings:



In our example of this service contract, the contract's start date is January 1 with a contract duration of one year. Billing is carried out on a yearly basis with the invoice date January 1.

In event (1), the invoice is posted. The invoice refers to the whole duration of the service contract. As revenue can only be recognized with the period-end closing for each period, the whole revenue amount is deferred with the posting in event (1a).

With period-end closing, which can be done in the **Run Revenue Recognition - Service Documents** app, revenue is recognized periodically, meaning that the total billed amount is divided by the number of periods that fall into the billing period. In our case, this leads to equal postings each month that are represented in the postings (2), (3), and (4).

Revenue Recognition Method “Recognize All Costs and Revenues” (Method 2)

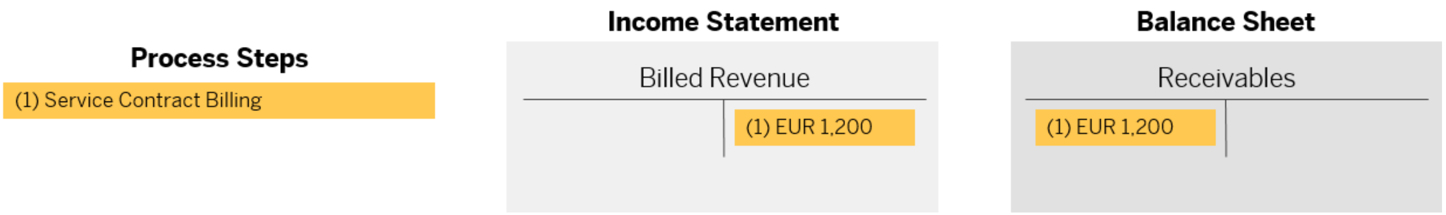
Here, we explain the posting logic behind the revenue recognition method 2 (corresponds to predelivered revenue recognition key CCSN), which allows you to recognize cost and revenue as occurred.

Usually, after you assign the revenue recognition key to the service document, Event-Based Revenue Recognition triggers postings in Financial Accounting accordingly.

Let’s assume that the following parameters are defined for your service contract:

- Service contract for 1 year (January 1 – December 31)
- Net value: EUR 1,200.00
- Billing yearly
- Billing Amount: EUR 1,200.00 (billing date = contract start date = January 1)

The following graphic shows you the process steps for this method and the corresponding postings:



In our example of this service contract, the contract's start date is January 1 with a contract duration of one year. Billing is carried out on a yearly basis with the invoice date January 1.

In event (1), the invoice is posted. The invoice refers to the whole duration of the service contract. As this revenue recognition method posts revenue as it occurs, no adjustment postings take place, and the process ends with the service contract billing.

Period-end closing is not relevant for this process as there are no cost or revenue adjustments nor any WIP postings on the balance sheet side.

Revenue Recognition Method “Recognize Revenue on Cost-Based POC” (Method 3)

You can use revenue recognition method 3, which allows you to recognize revenue using the cost-based percentage of completion (POC) method for service contract items in a specific hierarchy: The service contract item is considered as the root item in a hierarchy. It can have multiple service order items assigned. To learn more, see Event-Based Revenue Recognition for Service Contracts as Root Items in Financial Hierarchy.

The posting logic for this specific revenue recognition method and the respective key is similar to other integration scenarios. For an example of the cost-based POC method, see Revenue Recognition Methods for Service Order Items.

Hierarchy Settlement with Service Transactions

In Service, service contract item, service order item, service order item (execution order item with a follow-up maintenance order) are relevant for postings, and settlement can be executed in hierarchy. Costs and revenue are accumulated within the hierarchy and settled against the root object. Settlement rule maintenance and settlement run are allowed and required only at the root object.

The service settlement hierarchy can be structured in the following ways:

1. Service contract item as the root object, with service order item or service order item (execution order item with a follow-up maintenance order) as its child objects.
2. Service contract item as the root object, with service order item and service order item (execution order item) as a child, while maintenance order remains an independent object.
3. Service contract item as the root object with service order item as a child.
4. Service contract item as the root object with service order item (collective item) as child with further sub-hierarchy including execution order item with a follow-up maintenance order or service order item.
5. Service order item (execution order item) as the root object, with maintenance order as its child.

Example

You have a service contract item with a service order item (execution order item with a follow-up maintenance order). The cost is posted on confirmation to the maintenance order, and billing is done on the service order item or service contract item based on the billing customizations. All the costs and revenue are to be settled to an internal order or Profitability Segment (PSG). Instead of settling each object individually, you can run the settlement on the root object (the service contract item), which will settle all costs and revenue.

Prerequisites

1. Maintain the root item in the financial hierarchy.

   To set up the root item for hierarchy settlement in a service contract:

   a. Go to Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Item Categories**.

   b. Select the item category **Service Contract**.

c. Under Assignment of Business Transaction Categories, choose BUS2000112 - Service Contract.

d. Select Customizing and select the Root Item in Financial Hierarchy check box.

2. Select Settlement Options to allow a settlement run on the object.

There are 3 options: No Settlement Allowed, Settlement Without Hierarchy Allowed, and Hierarchy Settlement Allowed.

a. You can configure settlement options for service transaction items in Customizing for Service under Transactions > Basic Settings > Define Item Categories.

b. Select item category Service Item.

c. Under Assignment of Business Transaction Categories, choose BUS2000115 – Sales.

d. Under Customizing, go to Settlement Section > Settlement Options, then choose Hierarchy Settlement Allowed.

Note

If you use the process Service with Advanced Execution and want the maintenance order to be settled with the service order item, use Hierarchy Settlement Allowed.

For the category execution order item, select the option Hierarchy Settlement Allowed to include the maintenance order in the settlement hierarchy.

3. Make the following checks for hierarchy settlement run in the service transaction:

- Profit center and company code are the same for all objects in the hierarchy.
- Settlement rule is only created on the root object.
- In the settlement rule, for Settlement type, only FUL is allowed.

Features

Settlement Control Flag

The settlement control flag is assigned to a cost object and determines its settlement behavior. This flag is a technical field that is used by settlement runs to identify the hierarchy.

Based on the item category customizing for hierarchy and settlement options, the settlement control flag is set for the service transaction. The possible values for the settlement control flag are:

Settlement Control Flag	Description
N	No Settlement
S	Standalone Settlement
H	Hierarchy Settlement - Root
C	Hierarchy Settlement - Child
O	Hierarchy Settlement - Child and Parent
Blank value	Standalone Settlement (Existing)

The settlement control flag can have the following values:

- N (No Settlement): This object is excluded from settlement. Therefore, settlement rules can't be maintained, and settlement runs can't be executed for these objects.

- S (Standalone Settlement): Settlement runs can be executed for this object. As defined in the configuration of the settlement profile, all actual costs and revenue postings are settled to the receiver defined in the settlement rule.

- H (Hierarchy Settlement - Root): This is the root object in a hierarchy settlement. Settlement rules are maintained for this object. The settlement run accumulates all costs and revenue from all the objects in the hierarchy and settles them against the root object (sender) to the designated receiver. For hierarchical settlement, only the FUL settlement type is allowed.

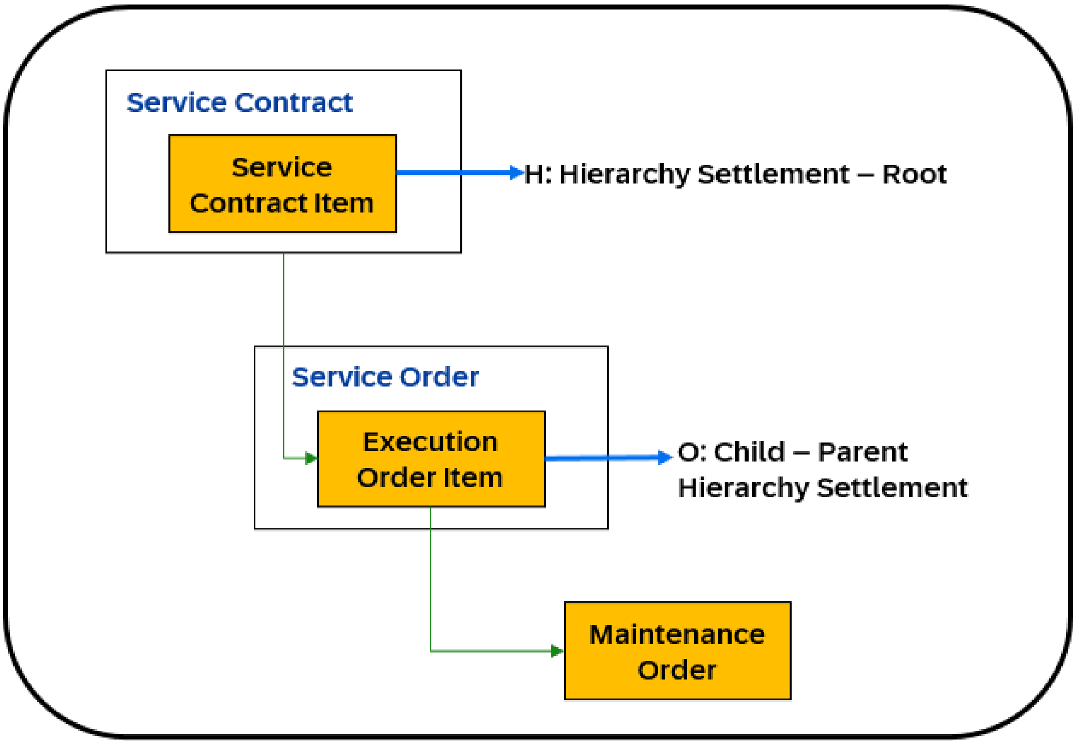
- C (Hierarchy Settlement - Child): This is a child object within a hierarchy, with no further child nodes. Settlement rules can't be maintained, and settlement runs can't be executed for this object. All costs and revenue are settled with the root node of the hierarchy.

- O (Hierarchy Settlement - Child and Parent): This is a child object within a hierarchy that also has its child objects. Settlement rules can't be maintained, and settlement runs can't be executed for this object. All costs and revenue are settled with the root node.

  Example

  In a service hierarchy, a service order with execution order item has the settlement control flag O, as it is a child of the service contract item and a parent to the maintenance order in an advanced execution. In this case, settlement can't be performed on the maintenance order.

  The following diagram illustrates the hierarchy settlement for a child object within a hierarchy that also has its child objects.



- Blank value (Standalone Settlement (Existing)): Service transactions created before SAP S/4HANA Cloud Private Edition 2025 FPS0 have a blank value. Their settlement behavior is the same as S (Standalone Settlement).

Settlement Rules

For a settlement hierarchy, if settlement rules can be maintained only on the root node, then rule maintenance is not allowed at any of the child nodes.

i Note

If there were rules maintained in the service order or maintenance order before assigning to the financial settlement hierarchy, the rules are not deleted but will not be used.

The settlement type in the rule allowed is **FUL – Full Settlement**.

Event-Based Revenue Recognition (EBRR) with Settlement

From service contract: EBRR and settlement are in the financial hierarchy based on the item category customization. Revenue recognition is performed at the root, which is the service contract item, collecting all the costs and revenue of the hierarchy. Similarly, the settlement collects all the costs and revenue in the hierarchy and settlement against the service contract item.

In EBRR, for an advanced execution scenario, the service order item and maintenance order are always in an EBRR hierarchy. Therefore, all cost and revenue postings for any maintenance orders or service orders are collected, and EBRR responds to the parent. But for the hierarchy settlement, there is an option to include the maintenance order as a child or make the service order item independent of the maintenance order for settlement.

Maintenance order as a child for settlement: Settlement runs in a hierarchy, and the costs at the maintenance order are settled against the root object. Direct settlement on a maintenance order is not allowed.

Maintenance order is not a child of service order item for settlement: Settlement run does not include the cost of maintenance order. Settlement can be executed on the maintenance order directly.

Activities

1. Go to the Fiori app **Run Settlement – Actual** (Fiori App ID - F4568) and select the cost object type as **Service Document**.
2. Under **Service Document ID**, enter the ID of the service transaction and other filter criteria.
3. Select the service document and run the settlement.

   The service document with settlement control flag H collects the cost from all the objects in the hierarchy and settles it collectively.
4. In the settlement results, you can see the settlement amount from the objects of the tree with the child object.

Related Information

Manage Settlement Rules - Service Documents
Run Settlement - Actual
Item-Based Accounting

Profit Center Determination

Obtain a detailed overview of the profit center determination for service transaction items.

A profit center is an organizational unit in **Accounting** that reflects a management-oriented structure of the organization for the purpose of internal control. The system determines a profit center for each service transaction item when you create a service order or a service contract. The profit center is then assigned to the service order item and the service contract item.

Access Sequence

The profit center is determined based on the following access sequence. If a method fails, the system automatically applies the next one in order to determine the profit center:

1. **Substitution rule**

   The system checks the app **Manage Substitution/Validation Rules - Service Documents** to determine a profit center based on a substitution rule, in case defined any.

   You can use this app to tailor the profit center determination function to your business requirements. For more information, see Substitution/Validation for Service Documents.

   If you select the option **Overwrite** when creating the rule, this has the following effects:

   - **Overwrite** flag **is set**:

     The profit center that was assigned manually or via an API (see 2.) will be overwritten by the value defined in the substitution rule.

   - **Overwrite** flag **is not set**:

     If there is no profit center maintained manually or derived via an API, the profit center from the substitution rule is derived.

     If there is a profit center maintained manually or derived via an API, the substitution rule will be ignored.

2. **Manual input**

   You enter the profit center manually or it is determined by a configuration in your system (via an API for example).

3. **Retrieval via cost center of the employee responsible**

   The system determines the cost center of the employee responsible for the item and then the profit center that is defined for this cost center.

4. **Default derivation**

   If no profit center is defined for the cost center, the system uses the default profit center defined for the controlling area.

i Note

You cannot change the profit center anymore if actual postings were already made. If actual cost or revenue postings have not been made and if the service transaction is in the **In Process** status the profit center field can be edited.

If you change the employee responsible for a service transaction item, the system redetermines the profit center.

i Note

Change of profit center is not allowed if there is a completed service confirmation for the order item. This also applies if a completed service confirmation is canceled. Even then changing the profit center is not allowed.

You can also manually change the profit center assigned to a service transaction item. Once you have made such a manual change, the system no longer redetermines the profit center when you change the employee responsible.

⚠ Caution

If you change the employee responsible, the system redetermines the profit center only if you have removed the manually assigned profit center.

Service Confirmations

The profit center determined for a service order item is automatically transferred to the corresponding service confirmation that has been created as a follow-up to the service order item.

- For service items and expense items in service confirmations, the system posts to the cost center and thereby to the profit center defined for the executing service employee. Should the profit center of the executing service employee differ from the profit center assigned to the respective service order item, the profit center of the executing service employee is credited and the profit center of the service order item is debited with the costs of the service confirmation.
- For service part items in service confirmations, the system posts costs to the plant-dependent profit center defined in the material master. If the profit center of the service part (material) differs from the profit center assigned to the respective service order item, the profit center of the material is credited and the profit center of the service order item is debited with the costs of the service confirmation.

Related Information

Substitution/Validation for Service Documents
Rules for Substitution/Validation

WBS Elements in Service

You can assign work breakdown structure elements (WBS elements) as attributes to service transactions.

In service transactions like service orders, you can view, choose, and assign WBS elements as attributes of profitability segments. A WBS element is a structural element in a work breakdown structure that represents the hierarchical organization of a specific enterprise project. In service transactions, a WBS element is used as an attribute of a profitability segment, which allows you to associate a project with the cost and revenue of service transactions.

Note

You can use the **Product and Service Margins** app to analyze your revenue, costs, and contribution margin across multiple dimensions, such as profitability segment, projects, and service orders. For more information, see Product and Service Margins.

Prerequisites

To use WBS elements as attributes of profitability segments in service transactions, you need to fulfill the following prerequisites:

- You need to set up item-based accounting in Customizing for **Service** under **Transactions** > **Settings for Service Transactions** > **Integration** > **Enable Item-Based Accounting for Service**.
- The project profile must be **Project with Revenue** when you create a project in Portfolio and Project Management. For more information, see SAP Portfolio and Project Management.
- You must set the billing element flag for the WBS elements that you plan to use.
- The company code and the controlling area in your project must be the same as in the sales organization of your service transaction.

Use

You can use WBS elements in the following service transactions:

- Service orders
- Service confirmations

- Service contracts

Service Orders

You can view the WBS elements of service orders at both header and item level.

You can assign WBS elements directly to service orders at both header and item level. If you assign WBS elements to service orders at header level, the system automatically copies the WBS elements to any items that you create in service orders. You can still overwrite the WBS elements inherited from the header when you edit the items.

Service Confirmations

The WBS elements of service confirmations are displayed at item level.

You can't assign WBS elements to the items in service confirmations. The WBS elements from the items in service orders are forwarded automatically to the items in service confirmations. However, if you add an unplanned item to service confirmations, you can edit the WBS element for the unplanned item.

Note

- **Service Orders**

  You can assign WBS elements to items in service orders as long as the items aren't assigned to service contract items.

- **Service Confirmations**

  You can assign WBS elements to the unplanned items added to service confirmations as long as these unplanned items aren't assigned to service contract items.

Service Contracts

The WBS elements of service contracts are displayed at both header and item level.

You can assign WBS elements directly to service contracts at both header and item level. If you assign WBS elements to service contracts at header level, the system automatically copies the WBS elements to any items that you create in service contracts. You can still overwrite the WBS elements inherited from the header when you edit the items.

TECO Status of Account Assignment Objects in Service

With item-based accounting, you can set the account assignment objects of service order items or service contract items to the status Technically Completed (TECO).

Use

The status Technically Completed (TECO) indicates that the service associated with an item in a service order or service contract has been executed, billing has been finalized, and the primary cost and revenue have been posted. You need to set the account assignment object of an item to the status TECO if you want to start the period-end closing process, and meanwhile clear the balance of relevant cost and revenue with event-based revenue recognition (EBRR).

Note

Do not confuse the status Technically Completed (TECO) with the system status Business Completed. Unlike the business completion of service transactions, the TECO status does not prevent further postings to Financial Accounting. For more information on the system status **Business Completed**, see Finalizing the Service Order and Finalizing the Service Contract.

Statuses of an Account Assignment Object

The status of an account assignment object specifies what follow-on process related to a service transaction item takes place in Financial Accounting and Controlling. The following table provides you with an overview of the following information:

- The statuses that an account assignment object can have
- The corresponding life cycle status of a service order item
- The corresponding life cycle status of a service contract item
- The follow-on processes in Finance

Status of Account Assignment Object	Status of Service Order Item	Status of Service Contract Item	Follow-On Process
Created	**Open** or **In Process**	**Open** or **In Process**	Service planning data can be created.
Released	**Released** or **Completed**	**Released** or **Completed**	Actual postings to Financial Accounting are possible.
Technically Completed (TECO)	**Completed** **Note** If the item is an execution order item and has been set to **Released**, its account assignment object can be set to the status **TECO**, provided that you have set the value of the field **TECO Determination** to **Set to Yes Manually**.	**Released** or **Completed**	Period-end closing starts, during which the balance of cost and revenue by means of EBRR can be cleared.
Closed	**Completed** **Note** The relevant service order has been set to **Business Completed** as well.	**Completed** **Note** The relevant service contract has been set to **Business Completed** as well.	No further postings to Financial Accounting are possible.

Report to Set TECO Status

To set the status **Technically Completed (TECO)** for account assignment objects and clear the balance of cost and revenue via EBRR during period-end closing, you need to use the report **Set Service Orders/Contracts to Technically Completed** (`CRMS4_SET_FIN_TECO`). This report allows you to schedule batch jobs to set the statuses of account assignment objects to the status **TECO**. You can find the report on the **SAP Easy Access** menu under **Service** > **Service Processes**.

TECO Determination Settings

You can use this function to instruct the system how the account assignment object of an **execution order item** or a **service contract item** is set to the status TECO when the report CRMS4_SET_FIN_TECO is run. The field TECO Determination can be used for this function and this field is displayed under Account Assignment in the item detail view of an execution order item or service contract item. In the field TECO Determination, you can select from the options described in the following table:

i Note

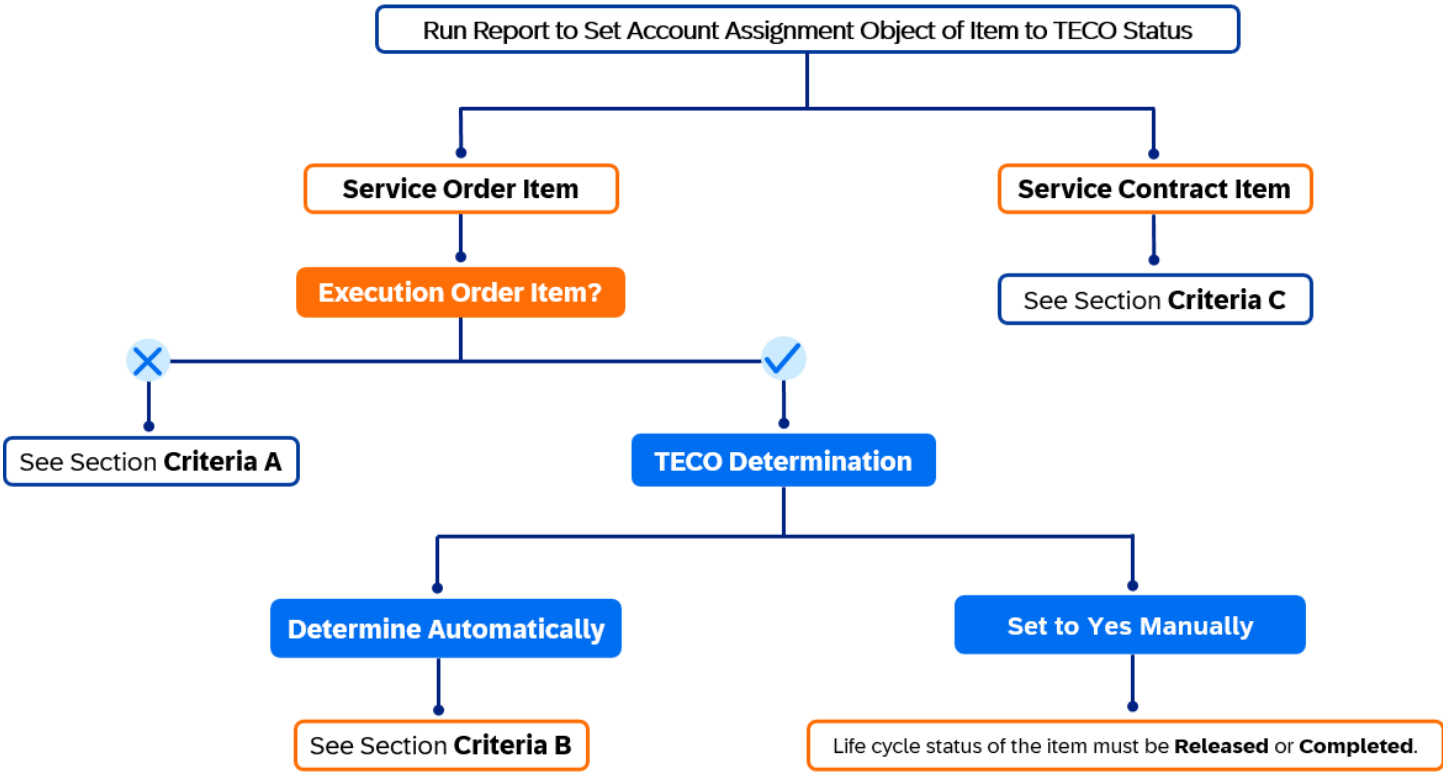
- The field TECO Determination is only supported for execution order items and service contract items, but not for the other service order item types.
- To use the field TECO Determination, the item must have the status Released or Completed.

Option	Action When Report Is Run	Use
Determine Automatically (Default Setting)	The system automatically sets the account assignment object of an item to the status TECO if the required criteria are met, as described below.	You use this option to start the period-end closing to clear the balance of cost and revenue using EBRR only for items that meet all the required criteria.
Set to Yes Manually	The system sets the account assignment object of an item to the status TECO regardless of whether the required criteria are met or not.	You select this option if you want to override the system and start the period-end closing to clear the balance of cost and revenue using EBRR for an item. For example, if you use itemized billing for an execution order item but the billing status is still Partially Billed, you can already use this option to realize the relevant revenue and indicate that costing is done so that the cost and revenue are reflected on the balance sheet.
Set to No Temporarily	The system does not set the account assignment object of an item to the status TECO even if the required criteria are met.	You use this option to prevent setting the status TECO for the account assignment object of an item. For example, you don't want to recognize the relevant revenue yet if the recording in Cross-Application Time Sheet (CATS) isn't completed and a part of the cost isn't recorded in the system yet.

Criteria to Set TECO Status

To set the Technically Completed (TECO) status with the report CRMS4_SET_FIN_TECO for the account assignment object of an item, there are certain criteria that must be met. The required criteria are different depending on which item type you use. For execution order items and service contract items, the required criteria to set the TECO status using the report also depend on the option that you select in the field TECO Determination.

The following diagram can help you find the corresponding criteria based on item types and options in TECO Determination field.



Criteria A

This section includes the criteria for a service order item that is **not** an execution order item. You can set the account assignment object of the item with the report to the status TECO if the following criteria are met:

- The life cycle status of the item has been set to Completed. If there is a follow-up service confirmation item, it must be set to Completed as well.
- There must be no open sales documents such as sales orders, debit memo requests, and credit memo requests.
- All the relevant billing documents have been posted to Financial Accounting.
- If the item is a service part, all the relevant supplier invoices must have been posted as well.
- If you use ad hoc billing for the item, all the billing request lines of the ad hoc billing plan must have been billed as well.

Criteria B

If you use an execution order item and the selection is Determine Automatically in TECO Determination, you can use the report to set the account assignment object of the item to TECO status if the following criteria are met:

- The life cycle status of the relevant service order has been set to Completed.
- All the relevant invoices and billing documents have been posted to Financial Accounting.
- The status of the relevant maintenance order has been set to TECO.
- If Contract Accounts Receivable and Payable (FI-CA) has been activated, all the financial postings must be in the general ledger.
- If you use **itemized billing** for the execution order item, all the billing documents in transaction DP92 must have been posted to Financial Accounting as well.
- If you use **fixed price billing** for the execution order item, the item must have been fully billed as well.

Criteria C

The account assignment object of a service contract item can be set by the report to the status **TECO** if the following criteria are met:

- All the relevant billing documents of the service contract item have been created and posted to Financial Accounting, or you have set the value of the field **TECO Determination** to **Set to Yes Manually**.
- If the item category is designated as a root item in financial hierarchy, the account assignment objects of all follow-up service order items must have the status **TECO** or **Business Completed**.
- For a contract that is not flagged as root item in financial hierarchy, the **TECO** status can be set manually if at least one invoice has been posted.

Related Information

Service Orders
Service with Advanced Execution
Service Contracts
Event-Based Revenue Recognition for Service Documents
Item-Based Accounting

Financial Reporting and Analytics

Use apps for financial reporting and analytics purposes in Service.

Analytics

You can use the following apps to report on cost and revenue that have occurred in service transactions:

- **Service - Actuals**

  You can use this app to report on actuals for service orders and service contracts. The reporting is based on different entities and multiple dimensions that you can define in the app.

  Note
  You can't report actuals of service order items that are processed in Service with Advanced Execution.

- **Service - Plan/Actuals**

  You can use this app to report on the following:

  - Planned and actual cost and revenue that have been recognized for service orders.
  - Actual cost and revenue that have been recognized for service contracts.

For more information about the apps, see Analytical Apps for Management Accounting.

Profitability

You can use the **Product and Service Margins** app to analyze cost and revenue that have been posted for service transactions (service document) and the margins. This allows a margin reporting and shows the balance sheet value that has been posted by Event-Based Revenue Recognition.

For more information, see Product and Service Margins.

Setting Up Item-Based Accounting

Information for key users to adapt Service to their business requirements

For item-based accounting, the configuration expert has to make the following settings that are used to transfer cost and revenue from Service to Financial Accounting and Controlling.

Enabling Item-Based Accounting

Activate item-based accounting in the **Enable Item-Based Accounting for Service** Customizing activity in Customizing for **Service** under **Transactions > Settings for Service Transactions > Integration**.

In a test system, you can use the user parameter `CRMS4_SRV_CO` to enable item-based accounting by setting the value to X. To disable item-based accounting, set the value to `N`. To use this parameter, go to the menu bar and select **System > User Profile > User Data**.

Once you've activated item-based accounting in a productive system, don't revert this setting. This will lead to inconsistencies in the financial data in your system.

Settings for Cost and Revenue Postings

Posting Cost and Revenue for Service Items

The following settings are required to post cost and revenue that are caused by service items (service product item):

1. Set up the following Customizing activities in Customizing for **Service** under **Transactions > Settings for Service Transactions > Integration > Set Up Time Sheet Integration**:
    - **Define Derivation of Activity Type**

      The activity type is used to post the confirmed labor to the time sheet of the executing service employee and to the universal journal. For more information, see [Activity Type].
    - **Assign Data Entry Profile**: Enter the data entry profile that is used to transfer data from service confirmation items to the cross-application time sheet (CATS).
2. Maintain the cost rates for the activity types that are used to post actual costs that have been incurred by the executing service employee under **Change Activity Type/Price Planning - Change** (transaction code `KP26`).

   For more information, see [Cost Rates].
3. Activate the derivation of profitability characteristics for postings with account assignment to service orders and service contracts in the **Activate Derivation for Items without Profitability Segment** Customizing activity in Customizing for **Controlling** under **Profitability Analysis > Master Data**.

   Note

   This setting is mandatory if you've implemented Service with Advanced Execution.

Posting Cost and Revenue for Expense Items

Map the products that are used as expense items to G/L accounts. You use this mapping to allocate cost and revenue that are caused by expense items in the **Assign G/L Accounts to Expense Product** Customizing activity (in Customizing for **Controlling** under **Product Cost Controlling > Cost Object Controlling > Product Cost by Service Document**).

Event-Based Revenue Recognition

Set up Event-Based Revenue Recognition in the following Customizing activities in Customizing for **Controlling** under **Product Cost Controlling > Cost Object Controlling > Product Cost by Sales Order > Period-End Closing > Event-Based Revenue Recognition**. For more information, see:

- Customizing Requirements for Service Orders
- Customizing Requirements for Execution Order Items
- Event-Based Revenue Recognition for Service Documents

Substitution and Validation

You use the **Manage Substitution/Validation Rules- Service Documents** app to display, change, and create substitution and validation rules for your selected business contexts and events for service transactions (service documents). These rules are used in the respective business processes to validate, derive, or replace values at the time of entry for the profit center, functional area, costing sheet, settlement profile, or costing variant. For more information about setting up your system, see Substitution/Validation for Service Documents.

Contract Accounts Receivable and Payable (FI-CA)

You can assign contract accounts to items in service transactions if you've activated item-based accounting. For more information, see Contract Accounts in Service Transactions.

Substitution/Validation for Service Documents

You can use the app **Manage Substitution/Validation Rules - Service Documents** to define substitution rules for service documents. They signify which profit center, costing sheet, functional area, settlement profile, business area, and costing variant is applied.

App ID: `F4406`

When you create a service transaction (service document), the system uses the conditions that you have defined in the substitution rule to derive the available target field for the item. When you make any changes to the item, the system automatically redetermines the profit center, costing sheet, functional area, settlement profile, business area, or costing variant using the conditions which were defined for the item.

For more information about creating the substitution rules, see Rules for Substitution/Validation and Manage Substitution/Validation Rules.

Note

In the **Execution Sequence** customized rules always take priority over the default rules predelivered by SAP.

For more information, see the in-app (F1) help in **Manage Substitution/Validation Rules - Service Documents**.

Features

Profit Center Substitution

When you create a substitution rule to determine the profit center based on the material (service product) and the plant, the system uses the plant-dependent product master to determine the profit center. If you change the product, the system redetermines the profit center.

If no profit center is defined in the product master, the system determines the profit center based on the employee responsible for the service transaction item.

For more information, see Profit Center Determination.

If you have changed the profit center manually, the system does not redetermine the profit center when you change the product. If you want the system to redetermine the profit center when you change the product, you must remove the manually assigned profit center.

Access to Manage Substitution/Validation Rules

For more information, see Authorization Concept.

Customizing

Check and adopt the following default derivation rules according to your business requirements:

> **i Note**
>
> The default settings for functional area and costing sheet as described below are only available if SAP Best Practices content is installed in your system. You can also create your own master data as per your business requirements.

1. **Costing Sheet Rule**: The default rule is `/0SAP/SV_COSTING_SHEET` (**SAP Default Costing Sheet Derivation**) which has costing sheet `SV001` assigned. This rule is active by default. If you create your own substitution rule for costing sheet derivation, you need to activate it after creation.

   > **i Note**
   >
   > You need to update the costing sheet `SV001` with your required master data in Customizing under **Controlling** > **Cost Center Accounting** > **Actual Postings** > **Period-End Closing** > **Overhead** > **Define Costing Sheets**.

2. **Functional Area Rule**: The default rule is `/0SAP/SV_FUNCTIONAL_AREA` (**SAP Default Functional Area Derivation**) which has functional area YB25 assigned. This rule is active by default. If you create your own substitution rule for functional area derivation, you need to activate it after creation.

   > **i Note**
   >
   > If you create your own rule for the functional area, you leave the section **Precondition** blank. In the **Substitution** section, select **Functional Area** as the **Target Field** and specify the functional area in the **Source** field and activate the rule. Your custom rule will then take priority over the predelivered rule.

3. **Settlement Profile Rule**: The default rule is `/0SAP/SV_SETTLEMENT_PROFILE` (**SAP default Settlement Profile derivation**) which has the settlement profile YNS assigned. If you create your own substitution rule for settlement profile derivation, you need to activate it after creation.

   > **i Note**
   >
   > It´s mandatory to derive a settlement profile. In case you do not work with the standard content and you do not want to use settlement, you need to create your own **No Settlement** substitution rule. You have two options:
   >
   > - Under **Precondition** you select the **Field/Function RevenueRecognitionIsActive** and assign the **Value** X.
   > - Under **Precondition** you select **ResultAnalysisInternalID** and specify an ID for which the revenue recognition key is active.

You need to cover all possible cases for service documents that have no revenue recognition key. To do this, you can create a rule without specifying a **Precondition** and select the target field **Settlement Profile** YNS (**No Settlement**). You can also create a custom settlement profile for the **No Settlement** case in transaction OKO7. We recommend you use the same parameters as those that are delivered with rule YNS.

4. **Profit Center Rule**: There is no default rule provided for profit center derivation. You can define it according to your business requirements.

   For more information about the profit center derivation logic, see [Profit Center Determination](#).

   > **Note**
   >
   > Deriving a profit center with the accounting indicator as a prerequisite is not recommended.

5. **Costing Variant Rule**: Defining a costing variant rule is not mandatory. However, if no costing variant is defined and you also don't use the standard content, no planned costs will be calculated. You can use the predelivered rule YS02 or create your own rule according to your business requirements.
6. **Business Area Rule**: Defining a business area rule is not mandatory.

> **Note**
>
> Once a confirmation is created and a journal entry is posted for a service transaction, no change or update of the substitution rule can be applied anymore.

Related Information

Rules for Substitution/Validation

Rules for Substitution/Validation

You can create substitution rules to determine the profit center, costing sheet, functional area, settlement profile, business area, and costing variant for service documents.

Make the following selections in the app **Manage Substitution/Validation Rules - Service Documents** if you want to create substitution rules in Service:

- Context: **Substitution in Service Document**
- Event: **Create Service Document**

Available Fields

You can create substitution rules in the app **Manage Substitution/Validation Rules - Service Documents** for the following target fields using the **Substitution in Service Document** context:

Target Field	Description
`ProfitCenter`	**Profit Center**
`CostingSheet`	**Costing Sheet**
`FunctionalArea`	**Functional Area**

Target Field	Description
ControllingSettlementProfile	Settlement Profile
BusinessArea	Business Area
CostingVariant	Costing Variant

You can use the following fields for profit center derivation for service order items and service contracts:

- Sales Organization (SalesOrganization)
- Distribution Channel (DistributionChannel)
- Division (Division)
- Plant (Plant)
- Product (Product)
- Profit Center (ProfitCenter)
- Sales Office (SalesOffice)
- Sales Group (SalesGroup)
- Responsibility Management (RespyMgmtTeamID)
- Equipment (Equipment)
- Functional Location (FunctionalLocation)
- Customer Group (CustomerGroup)
- Billable Control (BillableControl)
- Bill-to Party (BillToParty)
- Ship-to Party (ShipToParty)
- Sold-to Party (SoldToParty)
- Accounting Indicator (BillableControl)

For more information about the substitution rules, see Manage Substitution/Validation Rules.

For information about the profit center determination, see Profit Center Determination.

Authorization Concept

Access to Manage Substitution/Validation Rules

As a configuration expert for the Service area, to access the app **Manage Substitution/Validation Rules - Service Documents** from the SAP Fiori launchpad, you must have a business role that contains the business catalog **Accounting Configuration** (SAP_CA_BC_ACC_CONFIG). SAP delivers the business role template **General Ledger Accountant** (SAP_BR_GL_ACCOUNTANT) for your reference.

App Extensibility: Manage Substitution/Validation Rules - Service Documents

As a key user, you can extend the **Manage Substitution/Validation Rules - Service Documents** app according to your business needs.

Custom Fields

You can create and edit custom fields to enhance applications using the **Custom Fields** app. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	Navigation Path	UI Element Type
Service Header (CRMS4_SERV_H)	SAP Fiori	You need to add the field first in the app **Manage Service Orders** under **Adapt UI** > **Settings** > **Custom Codelists** > **Service Order Details** > **Edit Group**. Then, create your substitution rule including the new field in the app **Manage Substiution/Validation Rules - Service Documents** under **Create Rule** > **Select: Precondition Field / Function** > **Items**.	Header
Service Item (CRMS4_SERV_I)	SAP Fiori	You need to add the field first in the app **Manage Service Orders** under **Adapt UI** > **Settings** > **Custom Codelists** > **Items** > **Add Column**. Then, create your substitution rule including the new field in the app **Manage Substiution/Validation Rules - Service Documents** under **Create Rule** > **Select: Precondition Field / Function** > **Items**.	Column

For more information, see:

- Custom Fields
- Adding Extension Fields to Apps with Different UI-Technologies

Related Information

Extensibility

Manage Substitution/Validation Rules

With this app, you can display, change, and create substitution and validation rules for your selected business contexts and events. These rules can be used in the respective business processes to validate, derive, or replace values at the time of entry for the relevant fields.

Key Features

The app provides the following key features:

- Create Validation Rules
- Create Substitution Rules
- Change Rules
- Display Rules

Applicable Business Contexts

You manage substitution and validation rules by business context. A context represents the situation in which the validations and substitutions are defined and applied. It determines the fields or functions that are available in the rule definition. The following business contexts are supported for rule definition:

Solution Area	Business Context	Substitution/Validation	App Variant	Context-Specific Documentation
Management Accounting and Margin Analysis	**Market Segment**	Not supported	Manage Substitution/Validation Rules - Journal Entries	
Management Accounting and Margin Analysis	**Financial Planning**	Both rule types	Manage Substitution/Validation Rules - Journal Entries	Substitution/Validation Rules for Financial Planning
General Ledger Accounting	**Journal Entry Item**	Both rule types	Manage Substitution/Validation Rules - Journal Entries	Rules for Substitution/Validation
Group Reporting	**GR Journal Entry Item**	Both rule types	Manage Substitution/Validation Rules - Group Journal Entries	Rules for Substitution/Validation
Service	**Service Document**	Substitution rules only	Manage Substitution/Validation Rules - Service Documents	Substitution/Validation for Service Documents

i Note

The applicable business contexts are defined by SAP and cannot be changed. However, you can create your own fields for these contexts and these custom fields are supported by the **Manage Substitution/Validation Rules** app. For more information about custom field creation, see Custom Fields App and Custom Logic App.

Apply Rules at Runtime

At runtime, business users make postings or record data using various apps, such as the **Post Group Journal Entries** app. Based on the respective context and event, your defined rules apply automatically as follows:

- **Substitution**: Derives, replaces, or clears values for the relevant fields or functions defined in the substitution rules. The substitutions take place at the time of data entry with no system messages.
- **Validation**: Validates the values entered for the relevant fields or functions defined in the validation rules. Depending on the **Control Level** of each validation rule, when an entered value doesn't comply with the rule, a warning or an error message is raised. You can follow the link provided in the message to check the rule details and correct the entered values as necessary. Note:
    - A **warning** message is indicated by the ⚠ icon. If there are only warnings, postings can still be completed.
    - An **error** message is indicated by the ⊗ icon. Errors must be resolved before postings can be completed.

i Note

To navigate to the rule display, you must have the necessary authorization.

Supported Device Types

- Desktop
- Tablet

i Note

This app contains in-app help for key fields and concepts. To display the help while working in the app, press F1 or click the question mark displayed in the app header.

More Information

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

Create Validation Rules

Validation rules are used to check values as they are being entered in runtime applications. Follow the procedure below to create a validation rule:

1. Choose **Create Rule**.
2. In the dialog box that appears, choose a context and an event, then select **Validation Rule** as the rule type.

    i Note

    The available contexts and events are defined by SAP. Context determines which fields can be included in the rule definition. Event specifies the exact time and location for the validation to occur, for example, when users execute a particular step in a business process.

3. The detail input page is divided into three sections: **General Information**, **Precondition**, and **Validation**. In the **General Information** section, do the following:

   a. Specify a rule name and a meaningful description.

   b. Choose a control level, **Error** or **Warning**. It controls how strict the validation rule is. When data fails the validation at runtime, the system responds differently as follows:

      - **Error**: An error occurs and prevents users completing the posting until they correct the erroneous entries.

      - **Warning**: A warning occurs but doesn't prevent users completing the posting.

4. You can define fields or functions as conditions or statements in the **Precondition** and **Validation** sections. The following functions are predelivered by SAP:

   a. **CONCATENATE**

      Use this function to concatenate a sequence of field values or constants.

   b. **SUBSTRING**

      Use this function to calculate a substring of a given input string. It consists of three parameters:

      i. `STRING`

         The parameter string is a mandatory input value.

      ii. `OFFSET`

         This parameter is optional. The character offset in the input string must be a positive integer and only literal values are allowed. 0 is the default offset. If the given offset is too long for a specific input, an empty string is returned.

      iii. `LENGTH`

         The length of the substring must be a positive integer. Only literal values are allowed.

   You can also define your own functions using the **Custom Function** interface (`IF_FIN_RE_CUSTOM_FUNCTION`). For more information, see Developer Extensibility.

5. In the **Precondition** section, set one or more conditions for the validation to take place.

   - If the conditions are met, the values entered for the relevant fields or functions are to be checked based on the check statements defined in the **Validation** section.

   - If the conditions are not met, the values entered are not subject to the validation.

   If you leave this section empty, the values entered are always to be checked.

6. In the **Validation** section, define one or more check statements using **Field**, **Operator**, and **Value**.

   Note that the relationships between conditions (or statements) are as follows:

   - Filters for different fields or functions have an AND logic between each other.

   - Filters for a same field or function have an `OR` logic between each other. There is an exception: Filters using a negative operator, such as `Not equal to`, have an AND logic with the filters that use positive operators.

   ### Example

   The following rule requires when a business user posts data using the G/L account between `4164000` and `41649999` for the company code `1000` that the cost center is between `200` and `300` but not equal to `210`, and the distribution channel is `01`:

Section	Field	Operator	Value
Precondition	`GLAccount`	`Between`	`41640000` And `41649999`
	`CompanyCode`	`Equal to`	`1000`
Validation	`CostCenter`	`Between`	`200` And `300`
	`CostCenter`	`Not equal to`	`210`
	`DistributionChannel`	`Equal to`	`01`

7. Save the rule. The rule now has the status **New**. You must activate the rule before applying it at runtime. For more information about various rule statuses, see Change Rules.

Matches Operator

When defining validation rules and the **Precondition** part of substitution rules, you can use the **Matches** comparison operator to match a string-type field value against a regular expression that is based on the POSIX standard 1003.2. A regular expression is made up of literal characters and special characters following the syntax of regular expressions. It provides a concise and flexible means to represent a set of character strings.

For example, you can define the following validation rule using the **Matches** operator:

Section	Field	Operator	Value (Regular Expression)
Precondition	`ProjectExternalId`	**Matches**	`12345.*`
Validation	`CostCenter`	**Matches**	`[A-Z][A-Z][[:digit:]]*999999`

When a project ID starts with 12345, the rule requires that the cost center value starts with two capital letters and be followed by a number that ends with 999999.

For more information about how to write regular expressions, see Syntax of Regular Expressions.

Create Substitution Rules

Substitution rules are used to replace or derive values when values are being entered in runtime applications. Follow the procedure below to create a substitution rule:

1. Choose **Create Rule**.

2. In the dialog box that appears, choose a context and a event, then select **Substitution Rule** as the rule type.

   ### i Note

   The available contexts and events are defined by SAP. Context determines which fields can be included in the rule definition. Event specifies the exact time and location for the substitution to occur, for example, when users execute a particular step in a business process.

3. The detail input page is divided into three sections: **General Information**, **Precondition**, and **Substitution**. In the **General Information** section, specify a rule name and a meaningful description.

4. You can define fields or functions as conditions or statements in the **Precondition** and **Substitution** sections. The following functions are predelivered by SAP:

   a. **CONCATENATE**

      Use this function to concatenate a sequence of field values or constants.

   b. **SUBSTRING**

      Use this function to calculate a substring of a given input string. It consists of three parameters:

      i. `STRING`

         The parameter string is a mandatory input value.

      ii. `OFFSET`

         This parameter is optional. The character offset in the input string must be a positive integer and only literal values are allowed. 0 is the default offset. If the given offset is too long for a specific input, an empty string is returned.

      iii. `LENGTH`

         The length of the substring must be a positive integer. Only literal values are allowed.

   You can also define your own functions using the **Custom Function** interface (IF_FIN_RE_CUSTOM_FUNCTION). For more information, see Developer Extensibility.

5. In the **Precondition** section, set one or more conditions for the substitution to take place.

   - If the conditions are met, values for the relevant fields or functions are substituted based on substitution lines defined in the **Substitution** section.
   - If the conditions are not met, values are not substituted.

   If you leave this section empty, values for the relevant fields or functions are always to be substituted.

   Note that the relationships between conditions are as follows:

   - Filters for different fields or functions have an `AND` logic between each other.
   - Filters for a same field or function have an `OR` logic between each other. There is an exception: Filters using a negative operator, such as `Not equal to`, have an `AND` logic with the filters that use positive operators.

   ### Note

   When defining a filter condition, you can use the **Matches** operator to compare a field value with a text string represented with a regular expression. For more information, see Matches Operator.

6. In the **Substitution** section, define one or more substitution lines with the following settings:

   - **Target Field**: Select a field whose values are to be substituted.
   - **Substitution Type**: Select how the values for the target field are to be substituted. You can choose from the following types:
     - **Clear value**: Clears any existing value for the target field.
     - **Substitute with Constant**: Fills in the target field or replaces its existing value with a constant value.

- **Substitute with Field / Function**: Fills in the target field or replaces its existing value with the value from the source field or source function.
- **Table Lookup**: Looks in a custom business object for a source field or source function and uses its value to fill in the target field. For more information, see Table Lookup.

- **Source**: Depending on your selected substitution type, either enter a constant value or select a source field or source function whose value is to substitute the target field.
- **Overwrite**: Select the checkbox if you want to overwrite any existing value of the target field with the value you specified in **Source**. If you want to keep any existing value of the target field, leave the checkbox unselected.

Note

Depending on the business context setting, some fields cannot be overwritten. In this case, the checkbox is disabled.

Example

When a business user uses the billing document type `F1` to post data for the company code `1000`, the following rule fills in the **Product Sold Group** field with the value `A001` and the **Functional Area** field with the value from the **Functional Area** field of the relevant WBS element:

Section	Field	Operator	Value
Precondition	`BillingDocumentType`	`Equal to`	`F1`
	`CompanyCode`	`Equal to`	`1000`

Section	Target Field	Substitution Type	Source
Substitution	`SoldProductGroup`	**Substitute with Constant**	`A001`
	`FunctionalArea`	**Substitute with Field / Function**	`...\_WBSElementBasicData-FunctionalArea`

7. Save the rule. The rule now has the status **New**. You must activate the rule before applying it at runtime. For more information about various rule statuses, see Change Rules.

→ Tip

You can also create a rule by copying from an existing one and making any necessary changes.

Change Rules

For an existing validation or substitution rule, you can edit, activate, disable, or delete it.

When editing a rule, a draft of the rule is generated. Any changes are first saved to the draft and can only be saved to the active version of the rule after you activate the draft. The possible actions to a rule or its draft are as follows:

- **Edit new rule**: After creation, a rule has the **New** status. You can continue editing the rule, the **New** status is kept.
- **Activate new rule**: You can activate a new rule so that it has the **Active** status.

> **Note**
>
> Only active rules can be used in validation or substitution at runtime.

- **Edit draft or active rule**: After activation, when you choose **Edit** for the rule for the first time, a draft is generated. Any changes you made thereafter are first saved to the draft. The rule displays as the **Active** status with a **Draft changed by: <user name>** link underneath. For this kind of rule, you can take either of the following actions:
    - Edit the draft again by choosing the draft link and making changes. Draft rules have the **Modified** status.
    - Activate the draft by choosing **Activate**. The draft rule changes to the **Active** status and **overwrites** the original active version.
    - Edit the active rule by choosing the row or the arrow icon (>), it will discard the existing draft and generate a new draft. Any changes are saved to the new draft.
- **Disable rule**: You can disable an active rule so that it cannot be used in validation or substitution until you activate it again. The rule status changes to **Disabled**. This is useful if you want to invalidate a rule temporarily.
- **Delete rule**: You can delete a draft or an active rule. Note that when deleting an active rule, its draft version is also deleted.

> **Note**
>
> To ensure auditability, the system keeps track of all changes to the substitution and validation rules on a database table level. The user names and the timestamps of the changes are also persisted.

Display Rules

View available rules on the list page or navigate to the detail screen of a rule.

You can view all your available rules on the list page or navigate to the detail screen of a rule. When doing so, you have the following options:

- Search for Rules
- Personalize List Layout
- Show Rule Execution Sequence
- Check Rule Errors
- Show Rule Script

Search for Rules

You can search for specific rules as follows:

- In the **Search** field, enter the text contained in the rule name and description.
- Use various filters available in **Adapt Filters**.
- In the **Search by Field** dialog, select the precondition fields or functions, validation fields or functions, or substitution fields or functions contained in the rules.

Personalize List Layout

The rule list displays all substitution and validation rules of the supported contexts and events. You can choose **Settings** (⚙ icon) to personalize the list layout. For example, you can add or remove columns, group by **Context** or **Event**, and sort your selected columns in ascending or descending order. You can also save your personalized settings to a new view, set the new view as default, and select **Apply Automatically** for the view.

Show Rule Execution Sequence

During runtime, the available rules for a given event are run in a certain sequence. This execution sequence is determined automatically by the system. To check the sequence, choose **Analyze Rules** on the list page. In the dialog box that appears, specify an event and choose **Go**. The rules are listed in the **Enabled Rules** tab in prioritized order, that is, the rules to be applied first are listed first.

Generally, rules that can change field values are run before the rules that only read the field values. However, the following conflicts between rules can occur when determining the sequence:

- Two or more rules could change values for the same field. These rules are run in alphabetical order of the rule name.
- One rule reads the value of field A and fills field B with a value, while a different rule reads the value of field B and fills field A with a value. In this case, the execution sequence cannot be determined, and an error is raised.

i Note

The execution sequence of rules that don't have a logical dependency is undefined and cannot be influenced.

Check Rule Errors

When activating a rule, the system checks if your input is correct or complete for that single rule. However, conflicts between rules aren't raised at this point.

To check rule conflicts, choose **Analyze Rules** on the list page. In the dialog box that appears, specify an event and choose **Go**. The errors and warnings are listed in the **Messages** tab.

Show Rule Script

For a created rule, you can choose **Show Script** to convert its conditions and validations or substitutions to a script with logic in a format that is both processable by computer and comprehensible to humans. This is especially helpful for complex rules.

Transport Rules

Substitution and validation rules are configured in a customizing client. These customizing changes are recorded and transported to the production client via customizing requests. In the production client, users only have read access to the configured rules.

Note that any changes made to rules can only be transported in this way after **activation** of the rules. In addition, **deletion** of rules can also be transported.

i Note

To enable the transport as described above, **Automatic recording of changes** must be selected as the option for **Changes and Transports for Client-Specific Objects** in transaction `SM30` (table/view `T000`) for your customizing client.

Developer Extensibility

You can extend the **Manage Substitution/Validation Rules** app according to your business needs. For this purpose, the following development objects were released for developer extensibility purposes.

Interfaces

The following interface is available for development in the SAP S/4HANA ABAP Environment:

- **Custom Function** (`IF_FIN_RE_CUSTOM_FUNCTION`)

  In addition to the SAP-defined functions `CONCATENATE` and `SUBSTRING` that are available in the **Manage Substitution/Validation Rules** app, you can use this interface to extend the app with customer-defined functions.

  Implementing the **Custom Function** interface allows you to integrate custom ABAP logic into the **Manage Substitution/Validation Rules** app. Once the implementing ABAP class is activated, the implementation is made available as a function inside the **Manage Substitution/Validation Rules** app, which allows you to create custom substitution/validation rules.

  i Note

  When implementing the **Custom Function** interface, make sure to assign the functions to your own namespace.

  Additionally, a demo implementation of this interface is available with the class **Demo Customer Function** (`CL_SUBVAL_WEEKDAY`). It can be used as a starting point to implement custom functions in the SAP S/4HANA Cloud ABAP Environment. Therefore, the actual logic that is provided with the class is very simple: a weekday calculation.

  For more detailed information about the interface and the class, see the **ABAP Doc** documentation that is directly available within the coding.

For more information about developer extensibility in SAP S/4HANA, see Set Up Developer Extensibility.

Table Lookup

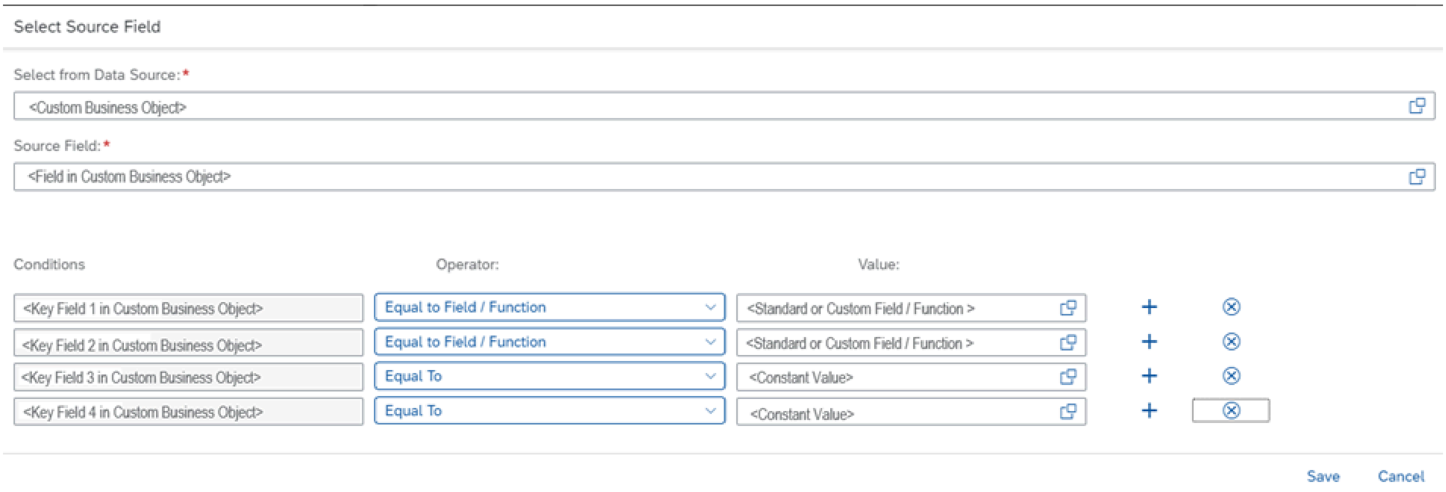
The **Table Lookup** allows you to choose a field from a custom business object and it uses this value to either substitute a target field or to use this value in a comparison condition.

If you selected **Table Lookup** as the substitution type or as a righthand side of a comparison, the **Select Source Field** dialog opens and you are prompted to specify the following:

- A data source that is defined using the Custom Business Objects app.
- A field from the data source you want to use in your substitution or validation.
- Under the **Conditions** section, all the key fields of your data source are listed. You must specify the filter conditions by comparing values of the key fields with those of standard or custom fields. You can also choose to compare values of the key fields with functions. Furthermore it is possible to use constant literal values as part of this condition. By choosing the corresponding compare operator, the input field on the right provides a value help to select possible values from the current custom business object in the system.
- If the condition is met by one of the rows in the custom business object, the value for the indicated column is used accordingly by the substitution or validation. If no row is met by the condition, an empty value is returned by the table lookup and no substitution is executed. When the table lookup is used in a compare operation, an empty result returned by the table lookup always means that the comparison is evaluated as FALSE. If the specified condition is met by more than

one row in the custom business object, the result is sorted in ascending order according to the primary key and the first row is selected.

The following figure illustrates the main aspects of the table lookup:



How to Substitute with Time-Dependent Data

In some cases, values in the source field change frequently across periods, which are known as “time-dependent data”. You may want your target field filled with those varied values accordingly at runtime, for example, during document creation. Let's see how it works in the following example:

Example

In your custom data source, you have the Company Code and Cost Center fields as key fields. The value of Business Unit constantly changes, for example, the business unit for a same combination of company code and cost center is `RD10` in year YYYY and it changes to `OP20` from the beginning of YYYY+1. In this case, you can also define a validity field as a key field in your data source, for example, the Valid To field of Date type in your data source. So your data source looks like the data in the following table:

Company Code (Key)	Cost Center (Key)	Valid To (Key)	Valid From	Business Unit
C100	CC01	20191231	20190101	RD10
C100	CC01	20201231	20200101	OP20

Now, you want the business unit value from your data source to substitute the Functional Area field. You can define your substitution rule as follows:

Section	Target Field	Substitution Type	Source
Substitution	`FunctionalArea`	Table Lookup	• Data Source: `YY1_<Your custom BO ID>` • Source Field: `BusinessUnit` • Conditions: ◦ `CompanyCode` Equal To

Section	Target Field	Substitution Type	Source
			CompanyCode ◦ CostCenter Equal To CostCenter ◦ ValidTo Greater Than or Equal To <Your specified date field> ◦ ValidFrom Less Than or Equal To <Your specified date field>

In this example, depending on which period range the value of your specified date field falls within, the corresponding `BusinessUnit` field value is used to fill the Functional Area field when you make an entry at runtime.

Substitution/Validation Logs

With this app, you can display in-depth logs for substitution or validation rules at runtime for a specific event.

Key Features

You can use this app to:

- View the substitution or validation rule log list and details.
- Filter the logs by event, user, or runtime.
- Select and combine multiple logs.
- Export a list of logs to a spreadsheet.

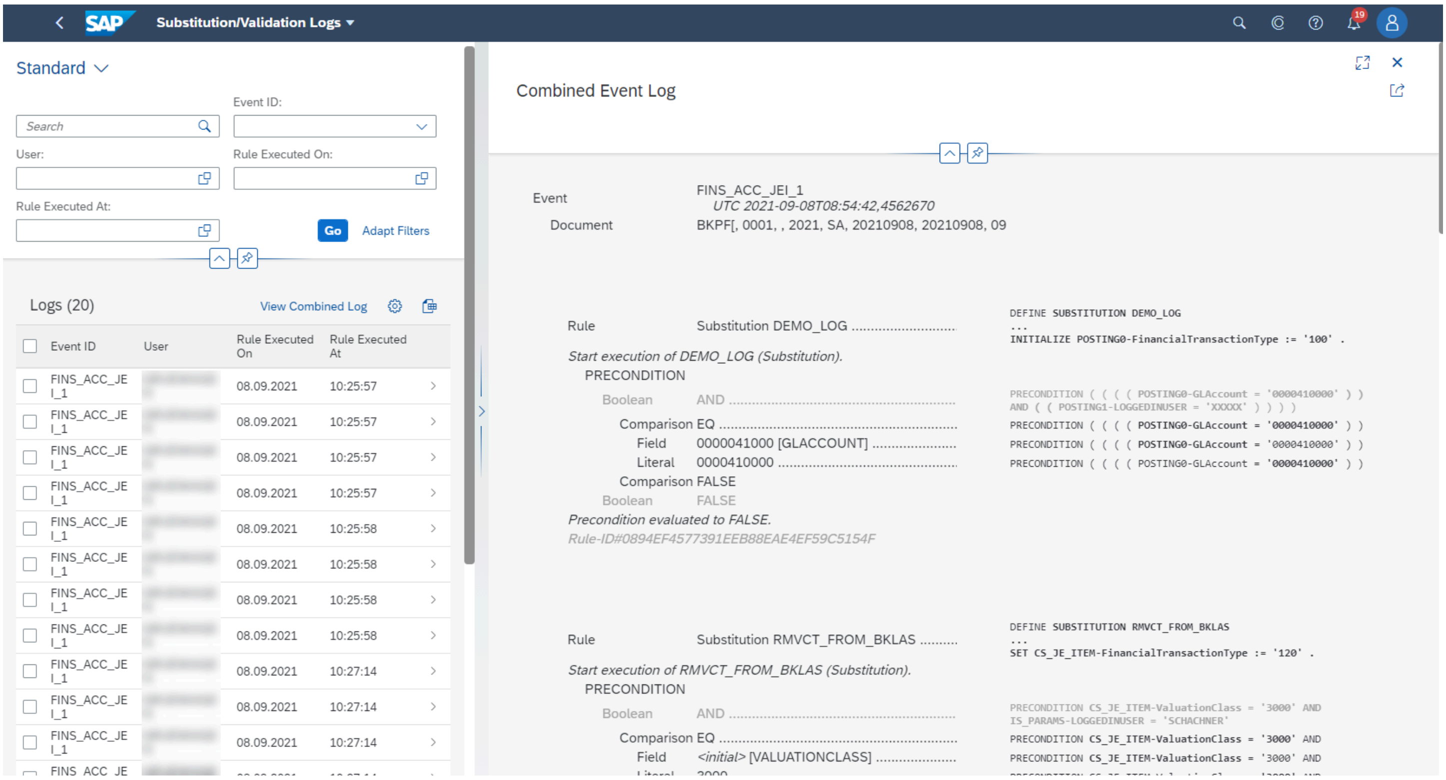
By selecting a row in the log from the list view, you can open the log detail page. It displays, in a tree-like structure, the processing flow when a set of rules for a specific event are triggered. Each log includes the following information from top to bottom:

- A line depicting the event being processed, such as `FINS_ACC_JEI_1` for Journal Entry Item.
- A line indicating the input document that the substitution or validation applies to, `ACDOCU` for Group Reporting Journal Entries.
- Substitution or validation rule content structured in a way that shows the internal processing flow of how the rule is applied. The third column of this part additionally shows an excerpt of the substitution or validation script, which represents the rule in the back-end process.
- The result of the rule application shows, for example, whether the precondition was met, which fields were subject to validation or substitution, or whether validation was successful or the substitution was performed.

i Note

Log lines formatted in gray color are less important and are included simply for completeness reasons.

The following screenshot (English only) shows the main aspects described above:



Supported Device Types

- Desktop
- Tablet

i Note

This app contains in-app help for key fields and concepts. To display the help while working in the app, press F1 or click the question mark displayed in the app header.

More Information

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

Set Substitution/Validation Logs

With this app, you can enable or disable logging substitution or validation rule execution for a certain user or a time period and set logging level.

Key Features

You can use this app to:

- View or filter all enabled log settings by event, rule execution user, or logging level

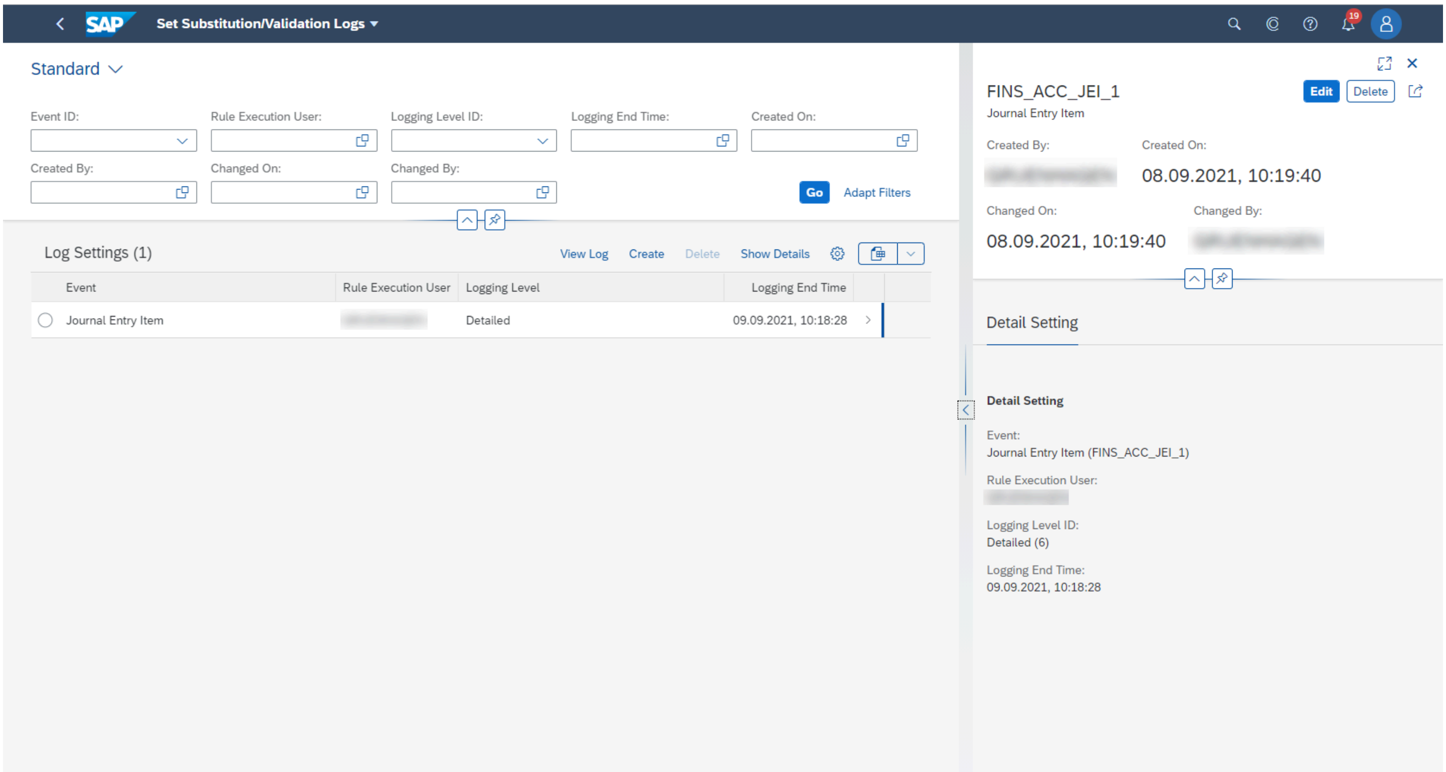
- Create new log settings for a specific event or user. The logging is enabled instantly after creation.
- Delete any log settings to disable the logging.
- Navigate to substitution and validation rule execution logs.

Logging Settings

When setting logs, you can specify the follows:

- Event: The logging only applies to the rules specific to the event.
- Logging Level: Represents the level of log detail, such as recording only the rule execution failures or all rule execution instances.
- Logging End Time: Specify until when the rule execution is logged. The logging starts from the current time to your specified end time. The default logging end time is 24 hours from your current time. You can change it to any future time point.
- Rule Execution User: Restrict the logging to a certain user. If you leave it empty, all users will be logged.

The following screenshot (English only) shows the main aspects described above:



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Integrating with Account Assignment Manager

Business process information about the accounting scenario that integrates Service with the account assignment manager

Use

The integration of the account assignment manager with Service enables you to transfer controlling-relevant data from Service to Controlling using account assignment objects.

The postings of actual costs to the account assignment objects (controlling objects) in Controlling use the following data from the service confirmations:

- **Activities** are automatically transferred to a time sheet. The sender cost center is determined from the personnel number of the service employee who creates the confirmation. The activity type is determined from Customizing. Time sheet data must be transferred to Controlling from the time sheet. Activity allocation from the service employee's cost center to the account assignment object then takes place in Controlling.
- **Service parts** automatically generate goods issues in Inventory Management and Physical Inventory, with a posting in Accounting and to the account assignment object in Controlling.

Sales items aren't confirmed. Instead, a sales order is created for the service item in Sales. Goods issue for delivery in Inventory Management and Physical Inventory triggers the posting of costs in Accounting and to the account assignment object.

Controlling Types

For each transaction type and service organization, you can choose one of the following controlling types in Customizing for **Service**:

- Mass-Object Controlling

  Unlike single-object controlling, you can immediately analyze the values (without period close) in Profitability Analysis (`CO-PA`). However, you can't calculate work in process (WIP).

- Single-Object Controlling

  Enables you to analyze individual transactions in detail using internal orders, and to calculate WIP.

- Controlling for account assignment objects

  Enables you to analyze individual transactions in detail using work breakdown structure (WBS) elements or internal orders and to calculate WIP.

  For more information about account assignment objects (controlling objects), see Account Assignment of Controlling Objects.

Revenue Accounting

Revenue accounting enables you to manage revenue recognition for service orders and service contracts. For more information about the integration of Service with Revenue Accounting, see Revenue Accounting Integration.

Integration with Business Transactions

Business Transaction Categories

Integration is available for the following business transaction categories:

- Service Process (`BUS2000116`)
- Service Confirmation (`BUS2000117`)

Item Categories

Integration is also available for item categories (provided in the standard system) to which you've assigned the following item object types in Customizing for **Service** under **Transactions > Basic Settings > Define Item Categories**:

- Service Product Item (`BUS2000140`)
- Service Product Confirmation Item (`BUS2000143`)
- Service Part Item (`BUS2000146`)
- Service Part Confirmation Item (`BUS2000142`)

Mass-Object Controlling

Usage

You can post the costs and revenues from a business transaction to a profitability segment. You can analyze the values immediately in Profitability Analysis (CO-PA).

Integration

With service transactions, the costs are posted when the goods issues for delivery are entered in the SAP Materials Management component (MM-IM). With service activities, the costs are posted to the profitability segment when the confirmation is entered.

Revenues are calculated in Billing, transferred to Controlling, and posted directly to the profitability segment.

The profitability segments are part of Profitability Analysis.

Features

A profitability segment is created for each business transaction item automatically.

- You can view and evaluate the data by product category or service organization, for example.

The following additional characteristics are available in Profitability Analysis:

- Business transaction
- Business transaction type
- Product category ID
- Service organization

- Responsible organizational unit (service)
- Group of the service employee
- Customer hierarchy

Single-Object Controlling

Note

The following documentation is based on the assumption that you have selected **transaction** as the controlling level in Customizing (**Integration with Other SAP Components** > **Customer Relationship Management** > **Settings for Service Processing** > **Controlling Integration**) under **Establish Controlling Type, Controlling Level, and Controlling Scenarios**. However, the documentation also applies when the controlling level is **item**. This was done to simplify the documentation and because SAP recommends that you use the controlling level **transaction**.

Use

You can use this function to report and analyze the planned and actual costs and revenues of business transactions in Service. In contrast to mass-object controlling, with single-object controlling you can analyze the data for each individual transaction and its follow-up documents.

Prerequisites

You are using Cost Center Accounting (CO-OM-CCA).

You have selected the controlling type **Single-Object Controlling** in Customizing under **Establish Controlling Type, Controlling Level, and Controlling Scenarios** and have carried out the activities under **Settings for Single-Object Controlling**.

Features

When you release a transaction in Service, an internal order with the order type SAPS is generated automatically. The profit center for this internal order is determined using the following sequence:

1. Alternative default profit center

   The dummy profit center `TKA01-DPRCT` is first assigned to the internal order with the order type `SAPS`.

2. SAP enhancement `PCACRM01`

   This SAP enhancement provides a user exit to determine the profit center in the case of a CRM integration.

   For more information, see SAP Note 386391.

3. Profit center substitution

   During CRM integration, you can create a PCA (profit center accounting) substitution under **EC-PCA: Maintain substitutions** (transaction `0KEM`). You then assign the substitution to the controlling area under **PCA: Substitutions CRM Integration** (transaction `KECRM_0KEL`).

   You can define and assign the substitution rules in Customizing for **Profit Center Accounting** under **Integration with Other SAP Components** > **Customer Relationship Management** > **General Settings** > **Settings for Profit Center Accounting** > **Define Substitution Rules for CRM Processes**.

The following parameters are taken from the **controlling scenario** and transferred to the order master data:

- Costing sheet
- Overhead key
- Results analysis key
- Settlement profile
- Object class
- Functional area

The settlement rule is generated automatically when you release the transaction. For more information, see Settlement in Single-Object Controlling. If you have specified a costing variant in Customizing, a standard cost estimate is generated automatically.

Actual costs are posted to the internal order in Controlling when the confirmations assigned to a transaction are entered, or in the case of sales items when the goods issue is entered for the delivery. Revenues are posted to the internal order by billing in Service.

All period-end closing functions are available for the internal order:

- You can assign overhead costs to the internal orders by defining overhead rates or process costs.
- You can evaluate the activities at actual prices.
- You can calculate work in process (WIP) and pass it on to financial accounting.
- You can settle the internal order to the following receivers:
    - Profitability segment
    - WBS element
    - Sales order item
    - Another internal order
    - Other settlement receivers (through a customer enhancement)

      You can take accounting indicators into account when you settle.

The Cockpit for Controlling Integration and the service process report are available for analysis purposes.

Activities

To access the cockpit in the SAP Easy Access menu, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Cost Object Controlling** > **CRM Service Processes** > **Cockpit for Controlling Integration (CRM Service)**.

To run the period-end close, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Cost Object Controlling** > **CRM Service Processes** > **Period-End Closing**.

To access a report with characteristics in Service, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Cost Object Controlling** > **CRM Service Processes** > **Service Process Report**.

Standard Cost Estimates for Single-Object Controlling

Use

In single-object controlling, a cost estimate that calculates the planned costs and revenues can be generated for each service order automatically.

The planned costs are calculated in Controlling using the transferred quantity structure. In addition, the revenues are transferred to the items of the service order.

Overhead rates can be calculated based on the plan data.

⚠ Caution

Planned costs and revenues can only be calculated for service orders.

Integration

The only conditions in the service order that are used to calculate planned revenues are those that transfer actual revenues to SAP FI with Billing in Service.

For service products, the cost center is specified in Service during planning and transferred to Controlling.

Prerequisites

You are using Product Cost Planning (CO-PC-PCP).

The item category of the items in Service must be relevant to the expense account.

Planned revenues are only transferred for conditions that fulfill the following prerequisites:

- The condition belongs to an item whose item category is relevant for billing or confirmation.
- The condition has a condition key that is not statistical and consists of a currency amount.

A costing variant must be defined in Customizing in the controlling scenario.

Features

The service order is costed if the status **Released** is saved at the header level. The cost estimate then includes all items that also have the status **Released**.

The planned costs and planned revenues for the internal orders are determined automatically and assigned to the internal orders.

If a change is made in the service order that is relevant to costing, the planned data in Controlling is updated automatically.

You can analyze the planned data in detail using reports.

If an error occurs during costing, the error message is not sent to Service but you can analyze the error in Controlling in the Cockpit for Controlling Integration , correct the error, and then restart costing. To access the Cockpit, in the SAP Easy Access menu, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Cost Object Controlling** > **CRM Service Processes** > **Cockpit for Controlling Integration (CRM Service)**.

Settlement in Single-Object Controlling

In the period-end close, costs and revenues from the internal orders are posted to other account assignment objects (such as profitability segments) and the internal orders are credited. If you are using results analysis, you can settle the work in process and reserves to Financial Accounting.

You can control which account assignment objects are used in settlement. In Controlling, this information is stored in distribution rules in the settlement rule. The settlement rule is generated automatically. The distribution rules are based on strategies defined by SAP. The strategy sequences are defined in Customizing and assigned to the controlling scenarios. For more information, see Customizing (Integration with Other SAP Components) under **Settings for Single-Object Controlling** > **Strategy Sequences for Automatic Generation of Settlement Rules**. The strategies have the following effects:

- Strategy 02: The costs and revenues are settled to a profitability segment. The profitability segment is selected based on the standard characteristics and additional characteristics in Service. You must have added the additional characteristics to your operating concern and activated them.

  If you want to define the settlement receiver in the transaction by entering an account assignment there, one of the following strategies must exist in the assigned strategy sequence:

- Strategy 30: The costs and revenues are settled to the transferred account assignment object. If the account assignment object specified in the transaction does not exist in Controlling, you receive an error message. In this case, the internal order is created without a settlement rule.

- Strategy 31: As with strategy 30, the costs and revenues are settled to the transferred account assignment object. However, if the account assignment object does not exist, a rule for distribution to a profitability segment is generated. There is no error message.

  If you are using accounting indicators and want to have a different settlement receiver for each accounting indicator:

- Strategy 35: The costs and revenues are settled to the receivers specified in the controlling scenario. For more information, see Accounting Indicators in Single-Object Controlling.

  We recommend the following strategy sequences, whereby each strategy must have priority 1:

  - 02 and 35: The costs and revenues are settled to a profitability segment and also to the receivers specified in the controlling scenario.

  - 30 and 35: The costs and revenues are settled to the transferred account assignment object and also to the receivers specified in the controlling scenario. If the account assignment object specified in the transaction does not exist in Controlling, you receive an error message.

  - 31 and 35: The costs and revenues are settled in the same way as with strategy sequence 30 and 35. However, if the account assignment object does not exist, a rule for distribution to a profitability segment is generated. There is no error message.

If an error occurs when you try to create the settlement rule, the error message is not sent to Service but you can analyze it in Controlling in the Cockpit for Controlling Integration and create the missing settlement rule. To access the Cockpit, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Cost Object Controlling** > **CRM Service Processes** > **Cockpit for Controlling Integration (CRM Service)**.

Changing the **account assignment** in the transaction results in the settlement rule of the internal order being created again or changed.

i Note

If the account assignment or accounting indicator fields are not available when you enter the transaction, you cannot settle to a transferred account assignment object or settle separately based on the accounting indicator.

Accounting Indicators in Single-Object Controlling

Use

You can enter accounting indicators for business transactions in Service in Service.

Note

You cannot enter accounting indicators for sales items.

If the Accounting Indicator field is not visible when you enter the transaction, this function is not available.

The accounting indicator gives you the following options:

- You can increase the level of detail of the costs on internal orders from Cost Element to Cost Element/Accounting Indicator. This enables you to collect and analyze costs separately by accounting indicator.
- In results analysis, you can assign the costs with accounting indicators to a separate line ID. You can then decide whether to include the costs of that line ID when capitalizing the work in process. For more information, see Customizing for Integration with Other SAP Components under Customer Relationship Management > Settings for Service Processing > Controlling Integration > Settings for Single-Object Controlling > Settings for Use of Accounting Indicators > Create and Edit Source Structures.
- You can assign costs with accounting indicators to a separate value field in CO-PA.
- You can settle costs with accounting indicators to a settlement receiver defined through the source assignment.

Prerequisites

You have carried out the customizing activities under Settings for Use of Accounting Indicators.

If you want to settle costs with accounting indicators to different receivers, the following prerequisites must be met:

- In the source structure of the settlement profile, all cost element/accounting indicator combinations must be defined.
- Settlement receivers must be assigned in the controlling scenario for these source assignments.
- The strategy sequence for the controlling scenario must include strategy 35.

Features

You can enter the accounting indicator at the header or item level. If you enter the accounting indicator at the header level, it is inherited by all items. The accounting indicator is inherited by the follow-up documents as a default value, and can be changed in the documents. The accounting indicator that is relevant for posting the costs is the accounting indicator transferred from the service confirmation to Controlling.

When the settlement rule is generated for the internal order automatically, an additional distribution rule is created for each assignment in the controlling scenario, which defines the receiver and the apportionment of the costs. These distribution rules are generated regardless of whether they are actually needed later.

Example

Suppose you want to collect all goodwill costs on an internal order. You create the **Internal order for goodwill costs** and the accounting indicator **Goodwill** . In the source structure, you assign the accounting indicator **Goodwill** to an assignment. In the controlling scenario, you assign the settlement receiver **Internal order for goodwill costs** to this source assignment.

When you confirm goodwill costs, you specify the accounting indicator **Goodwill**:

Confirmation for repair of a photocopier:

Item 1: 3 hours (accounting indicator: Goodwill)

Item 2: 3 hours (no accounting indicator)

When the internal order on which the costs of the confirmation were posted is settled, the confirmed goodwill costs are settled to the **Internal order for goodwill costs** .

Cockpit for Controlling Integration

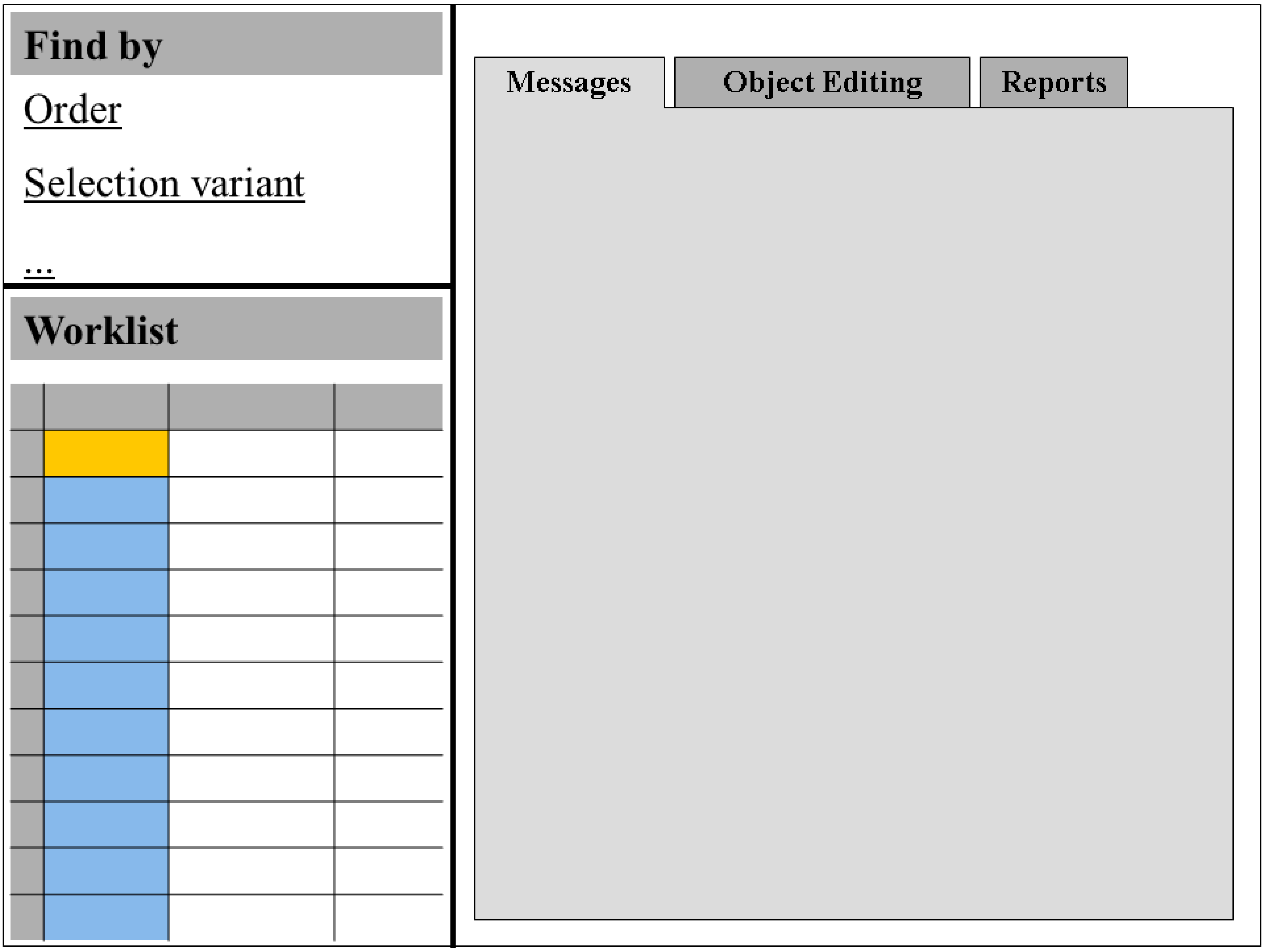
Use

The cockpit provides a central point of entry for all cost controlling tasks:

- Error correction
- Editing the master data and the settlement rule of the internal orders
- Analysis of costs and revenues in reports

Features

The Cockpit is divided into the following areas:



Find by Area

To edit internal orders, correct errors in them, and display reports, you must first select the internal orders. A number of different search functions are available:

- You can search for individual internal orders based on selection criteria. The Cockpit provides a number of search helps for this purpose, which give you all characteristics of the underlying object as selection options.
- If you often search using the same selection criteria, you can save the criteria in a selection variant.

The search results are displayed in the worklist:

Worklist Area

The worklist displays the number of the internal order and of the underlying object, plus the characteristics of the object and other information. You can define you own custom layouts to control which information is displayed.

When you exit the cockpit, the last worklist displayed is saved automatically and will appear the next time you start the cockpit.

Messages Tab

The Cockpit collects the messages issued in the Controlling processes in a separate log. This tab displays all messages on the internal orders in the worklist. You have the following options:

- Display long text of messages

- Change internal orders and cost estimates, for example
- Restart subprocesses containing errors (applications)

  When the errors are corrected, the error messages disappear from the message list. If new errors occur, the old messages are replaced by new ones.

Once you have finished processing a message, you can set a check indicator showing that you are finished with the message. Messages for which the check indicator is set can be deleted from the log.

Object Editing Tab

Here you can change the master data and settlement rules of the internal orders in a separate session. It is possible to edit multiple internal orders by selecting more than one order in the worklist.

Reports Tab

The following reports are available:

- Order: Actual/Plan/Variance
- Order: Breakdown by Partner
- Order: Result Analysis/Category

You can also add additional reports to the list and run them.

Activities

Searching for Internal Orders with a Selection Variant

1. Click on one of the selection variant search helps in the Find area, such as Selection Variant Messages .

   Choose an existing selection variant or create a new one.

   If you want to create a new selection variant, carry out the following steps:

   Enter a name for the selection variant and choose with quick info text Create Selection Variant .

   The screen ABAP: Variants - Initial Screen appears.

   Choose Create . Enter the selection criteria.

   Choose Attributes .

   For more information on creating selection variants, refer to Getting Started > Reports > Report Variants > Creating Report Variants .

   Save your variant and return to the dialog box.

2. Choose with quick info text Continue .

   The selected internal orders are displayed in the worklist. The error messages are displayed on the Messages tab. You can use functions on this tab such as the following:

Access the Master Data of an Internal Order

Click on the order number. A new session opens.

Restart Faulty Subprocesses

On the Messages tab, click on the symbol in the Check Automatically column.

Set Check Indicator

Click on the symbol in the Message Checked column. The check indicator is set. To delete the indicator, click on it again.

ReportsTab

Unless you enter a Version Plnnd/Accrued , the reports are displayed with plan/actual version 0.

If you want to specify selection criteria (such as the fiscal year) to be filled automatically when the reports are called, choose Extras User Settings - Reports .

If you want to access reports other than the standard reports, carry out the following steps:

1. Choose Extras Assign Reports .

   The Assign Reports dialog box appears.

2. Choose with quick info text Insert Row .

3. Choose Other Report Type .

4. Select Report Writer .

5. Select a report type for internal orders using the input help.

6. Save your entries.

Example

The error list contains a message on the cost estimate that G/L account 800000 does not exist as a cost element.

1. Correct the error by creating the cost element.

2. Select the error message.

3. Click on the symbol in the Check Automatically column.

The system re-costs the internal orders with the faulty cost estimates. If there were no other error messages on these internal orders, or if additional errors occurred, these internal orders disappear from the message list.

Revenue Accounting Integration

Use

You can integrate Service with Revenue Accounting and Reporting (Revenue Accounting). This integration supports you in recognizing revenue that results from service contracts and service orders according to the rules framework of the International Accounting Standards Board (IASB) and Financial Accounting Standards Board (FASB).

Revenue Accounting and Reporting specifically targets the International Financial Reporting Standards (IFRS) 15 standard for revenue recognition (Revenue from Contracts with Customers).

Note

Use the integration with Revenue Accounting and Reporting in Service as described for the standard implementation.

Setting Up the Revenue Accounting Integration

When setting up the integration with Revenue Accounting and Reporting, the following applies:

- The activation and configuration of the Sales integration to Revenue Accounting and Reporting is a prerequisite to enable revenue accounting for service contracts.
- Perform the steps to integrate Revenue Accounting and Reporting with Service as described in Customizing for **Service** under **Transactions** > **Settings for Service Transactions** > **Integration** > **Revenue Accounting Integration**.

Prerequisites

Service Contracts

You must set up service contracts as follows so that you can use the integration with Revenue Accounting and Reporting:

- Enter a contract start date and end date for the service contract.
- Provide the information about the billing plan in the service contract item.

Service Orders

You must set up service orders as follows so that you can use the integration with Revenue Accounting and Reporting:

- Enter the planned billing date.
- Enter the billing relevance for the service order items.

Process

Once the service contract or service order has been released, a notification (revenue accounting item) is sent to Revenue Accounting and Reporting.

The revenue accounting item contains the following information:

- Revenue accounting relevance
- Reference type and ID for revenue accounting contract bundling
- Duration of the contract (specifically the information about billing) or the planned billing date of service order items
- Conditions for service contract items or service order items
- Controlling data
- Revenue account

Once the revenue accounting item has been processed, the system creates a revenue contract that contains performance obligations.

Invoice items for service contracts are generated in Sales (Sales and Distribution). The Sales integration is used to send them to Revenue Accounting and Reporting.

Based on the information provided from Service, Revenue Accounting and Reporting corrects the invoice information, calculates and posts revenue adjustments and balance sheet effects (for example, contract asset and contract liability).

Note

- When the revenue accounting item is processed in Revenue Accounting and Reporting, the information from Service is enhanced using the rules that were set up in BRFplus. For example, you derive a standalone selling price.
- The generation of revenue accounting items as fulfillment items from service confirmations is not supported.

Migration

If you need to migrate your existing service transactions to Revenue Accounting and Reporting, you can use the operational load program **Service: Revenue Accounting Operational Load** (`CRM_SRV_REVACC_OL`) that creates RAIs for optimized inbound processing. For more information, see Operational Load in Service.

Related Information

Revenue Accounting and Reporting

Integration with Service

Operational Load in Service

Service Transactions with SEPA Mandate

In Service, you can process SEPA mandates in service transactions such as service contracts, service orders, and service confirmations.

Use

You can choose and assign SEPA mandates in service transactions. To be able to assign a SEPA mandate in Service, the transaction being processed has to be SEPA-relevant. The system uses the payment method, which you have entered manually at header level or item level, to determine whether the transaction is SEPA-relevant. The system can also determine the payment method automatically from the terms of payment defined in the customer master data. If this is the case, the terms of payment must be connected to a SEPA-relevant payment method.

Note

You can only use SEPA mandates that are created for the Financial Accounting (FI) component **Accounts Receivable** (FI-AR). For more information, see SEPA Mandates.

You can only assign valid mandates to a service transaction. A mandate is valid if the following applies:

- The mandate has the status **Created**, **Active**, or **To Be Confirmed**.
- The mandate has a current validity date.

Prerequisites

You have assigned incompleteness group SEPA to relevant service transaction types and item categories in Customizing for Service under **Service > Transactions > Basic Settings > Incompleteness Check > Define Incompleteness Procedures**.

Note

Depending on to which transaction type and item category you have assigned the incompleteness group SEPA, SEPA mandate becomes a mandatory field for the corresponding service transactions on header and item level.

For more information about the prerequisites for SEPA mandates, see Sales Documents with SEPA Mandate.

Related Information

SEPA Direct Debits

Contract Accounts in Service Transactions

You can assign contract accounts to items in service contracts, service orders, and service confirmations.

If you have integrated Contract Accounts Receivable and Payable (`FI-CA`) and Convergent Invoicing, you can assign contract accounts in your service contracts, service orders, and service confirmations to control processes in invoicing, contract accounts receivable and payable, taxation, and correspondence.

Prerequisites

1. You use Convergent Invoicing.
2. You use Contract Accounts Receivable and Payable (`FI-CA`). Additionally, you must set the active accounting indicator (**Active Account.**) to **FI-CA Active** in your Customizing settings under **Service > Master Data > Contract Account > Define Basic Settings**.
3. You have created contract accounts in the master data. To create, edit, or view a contract account, use the **Manage Contract Accounts** app with the business role **Accounts Payable and Receivable Accountant (FI-CA)** (`SAP_BR_APR_ACCOUNTANT_FICA`). Alternatively, you can use the transaction code `CAA1` to create contract accounts, `CAA2` to change them, and `CAA3` to display them.

Features

Contract Account Assignment

You can assign contract accounts to items in the respective service contracts, service orders, and service confirmations directly.

Note

If you assign contract accounts to sales items in service orders, the contract accounts of the sales items are forwarded to the follow-up sales orders in Sales and Distribution (SD).

If you add service contract items, service order items, or service confirmation items to a solution quotation, you can also assign contract accounts to those items in the solution quotation. When the solution quotation is accepted, contract accounts are forwarded to the follow-up service contract items, service order items, and service confirmation items.

Payment Data

When you assign contract accounts, certain data becomes obsolete and is removed from the service transaction. For example, the terms of payment determined by the payer.

If you have activated FI-CA, you cannot edit the payment method.

i Note

The integration of credit management with service transactions is supported in `FI-CA`.

Related Information

[Convergent Invoicing](#)
[Contract Accounting](#)

Transition to Item-Based Accounting for Production System

This topic describes the transition for Service from **Integration with Account Assignment Manager** to **Item-Based Accounting** in a production system.

Context

In Service, there are two concepts to integrate with Financial Accounting (FI) and Controlling (CO):

- **Integration with Account Assignment Manager**: This concept provides single-object controlling or a mass controlling. Depending on your system configuration, an internal order, a WBS element, or a profitability segment is used as an account assignment object. The account assignment object is used to post costs and revenue that have occurred in Service.
- **Item-Based Accounting**: This concept is based on dedicated account assignment objects that are created for service order items and service contract items. The items are recognized as the account assignment objects to which actual costs and revenue are posted. The planned and actual costs and revenue are analyzed based on the account assignment objects. You use event-based revenue recognition to recognize revenue when processing service orders and service contracts.

Transition to Item-Based Accounting

If you use a production system in which there are already service transactions that use **integration with account assignment manager**, and you decide to activate **item-based accounting** for this system, there are ways for you to move from **integration with account assignment manager** to **item-based accounting**. The following scenarios are supported for this transition:

- Assign an existing service contract that uses **integration with account assignment manager** to a newly-created service order item that uses **item-based accounting**.
- Assign an existing service order item that uses **integration with account assignment manager** to a newly-created service contract that uses **item-based accounting**.
- Ensure that the existing service orders that use **integration with account assignment manager** continue with this concept and their follow-up transactions use the same concept as well.
- Ensure that the existing service orders that use **item-based accounting** continue to use **item-based accounting** and their follow-up transactions use the same concept as well.
- Automatic assignment of a settlement rule to settle service order items using **item-based accounting** to a service contract using **integration with account assignment manager**. With this, all costs and revenue from the service order are assigned to the service contract and the costs and revenue are available for further processing.

i Note

- If you use a service order with **item-based accounting**, Revenue Accounting and Reporting (Revenue Accounting) is not supported because revenue accounting integration is only supported in **integration with account assignment manager**.
- You cannot create an execution order item in a service contract that uses **integration with account assignment manager**. The reason is that the settlement does not consider the costs from the follow-up maintenance order, which can cause inconsistency in Profitability Analysis (CO-PA) on the service contract.
- If a service order uses **integration with account assignment manager**, you may not use the item categories that are only supported when **item-based accounting** is active in the service order. The item categories that are only supported when **item-based accounting** is active are as follows: execution order items, price items, items for advance shipment and items for service hierarchy with collective accounting.

Automatic Generation of Settlement Rules for Transition Scenario

If you assign an existing service contract that uses **integration with account assignment manager** to a newly-created service order item that uses **item-based accounting**, you can enable the automatic generation of settlement rules for this scenario. In this scenario, the service contract has an account assignment object created from **integration with account assignment manager** (for example, internal order, WBS element, or profitability segment), but the follow-up service order has another account assignment object from item-based accounting. To settle the costs and revenue from the service order to the account assignment object of the service contract, you can use the Customizing activities described in the following section. The settlement rule you set up in Customizing sets the account assignment object of the service contract as the receiver of the costs and revenue from the service order automatically.

Customizing

To generate a settlement rule automatically, you carry out the following procedures:

1. Define the parameters for a settlement profile using the Customizing activity Maintain Settlement Profiles under Service > Transactions > Settings for Service Transactions > Integration > Settlement.
2. Create a strategy sequence using the Customizing activity Strategy Sequences for Automatic Generation of Settlement Rules under Service > Transactions > Settings for Service Transactions > Integration > Settlement > Automatic Generation of Settlement Rules:
   a. Use the user-defined strategy `CB1` for the transition scenario with the Customizing activity User-Defined Strategies for Automatic Generation of Settlement Rules under Service > Transactions > Settings for Service Transactions > Integration > Settlement > Automatic Generation of Settlement Rules.

      Note

      For more information on how to use the user-defined strategy, see the exemplary implementation described in Customizing for User-Defined Strategies for Automatic Generation of Settlement Rules.

   b. Prioritize the user-defined strategy in the sequence.
3. Assign the strategy sequence to a service order.
4. Define a substitution rule to assign the predefined settlement profile to the service order.

You can then use the app Run Settlement – Actual to run the settlement manually. Alternatively, the settlement can be run automatically during period-end closing by using the Financial Closing Cockpit.

Note

- To support Profitability Analysis (CO-PA) for a service contract, the costs and revenue from the service order must be settled to the profitability segment of the service contract.

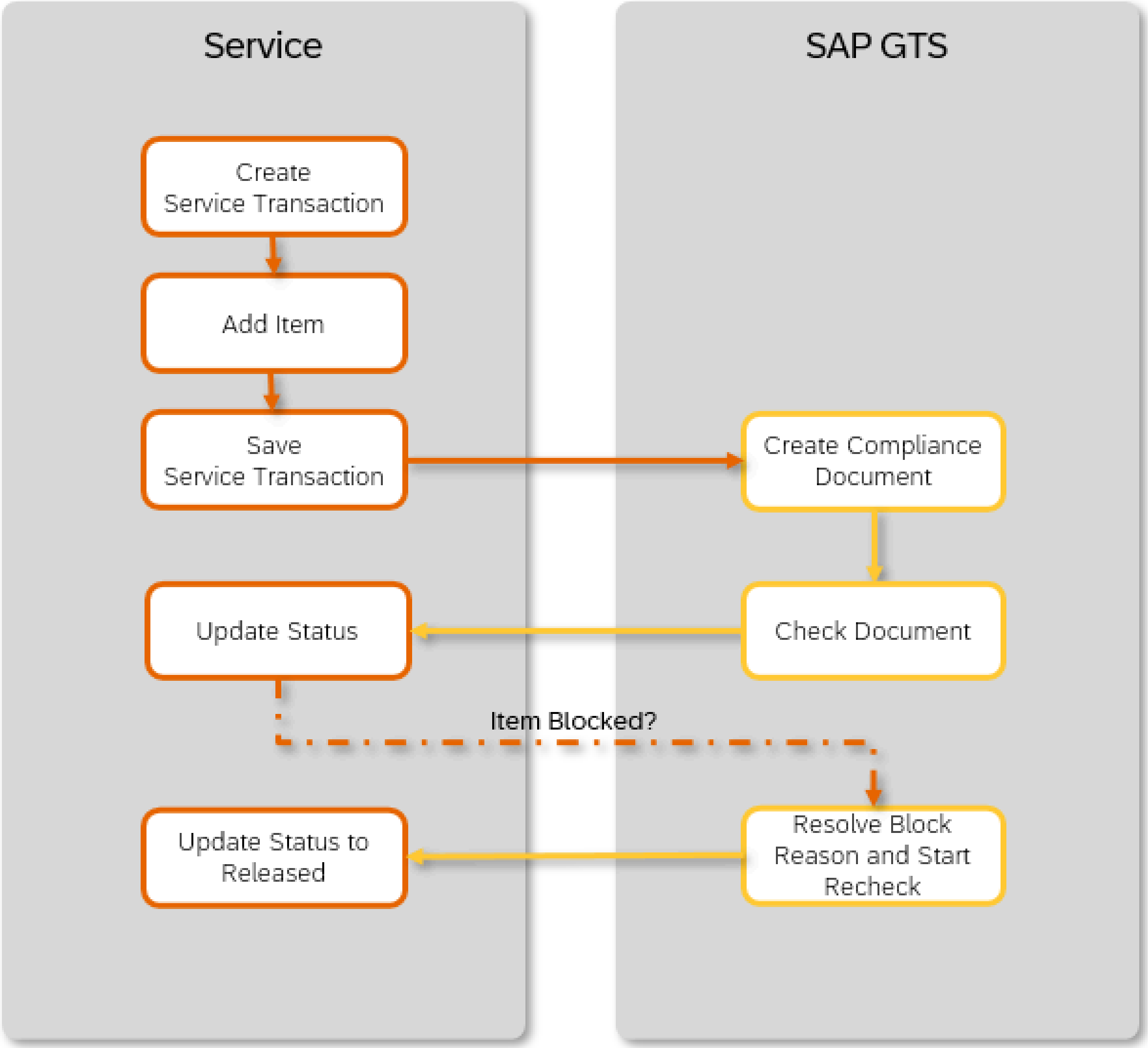
- If you assign an existing service order item that uses **integration with account assignment manager** to a newly-created service contract that uses **item-based accounting**, you need to settle the service order to the service contract manually.

Integration with SAP Global Trade Services

Nowadays, businesses have many challenges in following global trade rules. To help you comply with complex regulations, handle customs procedures, and reduce non-compliance risks, SAP provides SAP Global Trade Services (SAP GTS). The solution tightly integrates with Service to make your global service operations more efficient and ensure that they don't violate any global trade regulations.

Compliance Check Process

The integration solution performs embargo checks, sanctioned party list screening, and legal controls for service transactions, such as service orders and service quotations. The following diagram illustrates a compliance check process for a service transaction:



Please note that image maps are not interactive in PDF outputs.

When you change the service transaction and save it, it's transferred to SAP GTS again and a new check is performed.

Compliance Check Statuses

For each item in a service transaction, the SAP GTS system executes trade compliance checks and sets corresponding statuses – embargo status, screening status, and legal control status. You can use these statuses to search and filter the list of service transactions to find all transactions that have trade compliance issues. For example, you might want to find all service quotations in which at least one item is blocked by an embargo check. In addition, you can use the **GTS Status** field to conveniently search either for all transactions in which at least one of the three checks has found an issue, or for transactions that aren't relevant for the check at all.

For more information on trade compliance statuses for service transactions, see Compliance Check Statuses.

Configuration

To activate the transfer of service transactions to SAP GTS, you use the Customizing activity **Activate Transfer of Service Transactions (SRVA)**. This activity activates the BAdI `SLL_PI_SRVA_TRANSFER` responsible for the mapping of the data transferred. You can access the activity in Customizing under **Integration with Other SAP Components** > **Integration with Governance, Risk and Compliance** > **SAP Global Trade Services** > **Control Data for Transfer to SAP Global Trade Services** > **Technical Activation of Document Transfer**.

To configure the document types and company codes relevant for the transfer to SAP GTS, you use the Customizing activity **Configure Control Settings for Document Transfer**. Using this Customizing activity, you can also configure the system to prevent subsequent processing of service transactions, for example, to prevent creation of a purchase order or a maintenance order, based on the trade compliance check result. For more information, see the activity documentation in Customizing under **Integration with Other SAP Components** > **Integration with Governance, Risk and Compliance** > **SAP Global Trade Services** > **Control Data for Transfer to SAP Global Trade Services** > **Configure Control Settings for Document Transfer**.

Situation Handling

If you want the system to inform you automatically about service transaction items that are blocked or for which the trade compliance check is pending, you can use the situation template Trade Compliance Check Blocked/Pending.

Related Information

Integration with SAP Global Trade Services

Compliance Management

Compliance Check Statuses

For each item in a service transaction, SAP GTS executes a trade compliance check. The prerequisite is that the service transaction and the item are relevant for the check. Which items are relevant depends on the settings configured in Customizing for the SAP GTS system. The status at the transaction header level is determined by the statuses of the individual items.

For each of the three compliance checks – embargo check, sanctioned party list screening, and legal control – the system sets one of the following statuses both on header and item level depending on the check result:

Compliance Check Status	Description
Not Relevant, No Check	The system hasn't performed a compliance check because the service transaction item is not relevant for the check.
Released	The system sets a service transaction or item to **Released** if no trade compliance issues are found.
Blocked	The system sets a service transaction item to **Blocked** if there's a trade compliance issue. The trade compliance specialist can resolve the block reason, for example, by maintaining

Compliance Check Status	Description
	the missing data such as an export license, and start the recheck of the service transaction.
Partially Blocked	This status is only relevant for the **header level** of a service transaction. The system sets it if at least one of the items of the service transaction is blocked. Refer to the examples in the table below for details of when this status is set.
Pending – Not Yet Checked	The compliance check of an item is still pending.

Trade Compliance Status of Service Transactions

On the header level of a service transaction, the system sets a cumulative status depending on the statuses of the service transaction items. The following examples explain how the status of a service transaction with three items is set:

Status of Item 1	Status of Item 2	Status of Item 3	Resulting Transaction Status	Comment
Not Relevant, No Check	Not Relevant, No Check	Not Relevant, No Check	Not Relevant, No Check	Only if all three items are not relevant for a check, the status at transaction level is Not Relevant, No Check.
Released	Released	Released	Released	The transaction status is set to Released if there are no issues in any of the items.
Blocked	Blocked	Blocked	Blocked	The transaction status is set to Blocked if all of its items are blocked.
Released	Released	Blocked	Partially Blocked	The transaction is set to Partially Blocked if at least one of its items is blocked and at least one relevant item is not blocked.
Released	Blocked	Not Relevant, No Check	Partially Blocked	The transaction is set to Partially Blocked if at least one of its items is blocked and at least one relevant item is not blocked.
Released	Pending – Not Yet Checked	Blocked	Partially Blocked	The transaction is set to Partially Blocked if one of its items is blocked and at least one other relevant item is not blocked.
Released	Pending – Not Yet Checked	Released	Pending - Not Yet Checked	The transaction is set to Pending – Not Yet Checked if for one of its items the check is pending and none of the items are blocked.

Related Information

[Configuration Guide for SAP Global Trade Services, edition for SAP HANA](#)
[Compliance Management](#)

Trade Compliance Check Blocked/Pending

Situation Template ID: `SRVC_GTS_BLOCKED_OR_PENDING`

This is a situation template delivered by SAP for Situation Handling.

Consuming App: Manage Service Orders, Manage Service Quotations

The consuming app is a business app for which this template is designed and which displays a situation message.

Business Value

You can use this template to inform your customer service managers automatically if a service transaction is blocked by a trade compliance check or the check for the service transaction is pending.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the Manage Situation Types app with which you can display and use the template.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Object	Service Transaction Item
CDS View for Trigger Object	`I_SrvcDocItmGTSStsBlkdPndgSitn`
Anchor Object	Service Transaction Item
CDS View for Anchor Object	`I_SrvcDocItmGTSStatusAnchor`
Trigger Type	Batch

The situation is triggered when the system finds service order items or service quotation items that have been blocked by the trade compliance check or for which a compliance check is pending.

> **i Note**
> These settings aren't visible in the template in the **Manage Situation Types** app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this template defines that a situation is created with the status **Open** and a notification is sent to the defined recipients if the system detects a service transaction item that meets the criteria specified above.

For more information, see Conditions.

This template provides some filters, such as **Service Transaction Type**, **Sales Organization**, **Distribution Channel**, that you can use to define your own conditions when you create your situation types.

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the My Situations app.

The predefined settings in this template determine that the notification addresses only members of the **Service** team category.

In addition, the recipients of the notifications can also be determined by the predefined rule **Notify Employee Responsible for Service Order**. This rule determines that if a service order or a service quotation contains an item that is blocked by the trade compliance check or for which the check is pending, the user who is specified as the employee responsible for this service transaction is informed about this critical business situation.

For more information, see Recipients.

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of business situation events. You can view the data in the Monitor Situations app.

Related Information

Integration with SAP Global Trade Services
Situation Handling

Data Exchange in Service

For implementing data exchange processes between service transactions and any other preceding transactions or follow-up transactions, for example, between sales orders and service orders, you can use custom handler classes in the following configuration activities:

- **Define Custom Handler Classes for Forward Data Exchange**

  In this activity, you can specify the handler classes you want to use for the forward data exchange process (for example, from a service order to a sales order) of business transactions.

- **Define Custom Handler Classes for Backward Data Exchange**

  In this activity, you can specify the handler classes you want to use for the backward data exchange process (for example, from a sales order to a service order) of business transactions.

In both activities, you have to specify the identifier and the name of the handler class. The handler class inherits the methods from the standard handler class so that you only need to program the specifics you require. The settings you make override the standard handler classes and affect the standard behavior of business objects.

You access the activities in Customizing under **Service** > **Transactions** > **Data Exchange in Service**.

Situation Templates in Service

Get an overview of situation templates available in Service, together with the apps that display the situations. You can filter them according to various criteria.

Note

For more information about the Situation Handling framework in SAP S/4HANA, see the Product Assistance for Situation Handling.

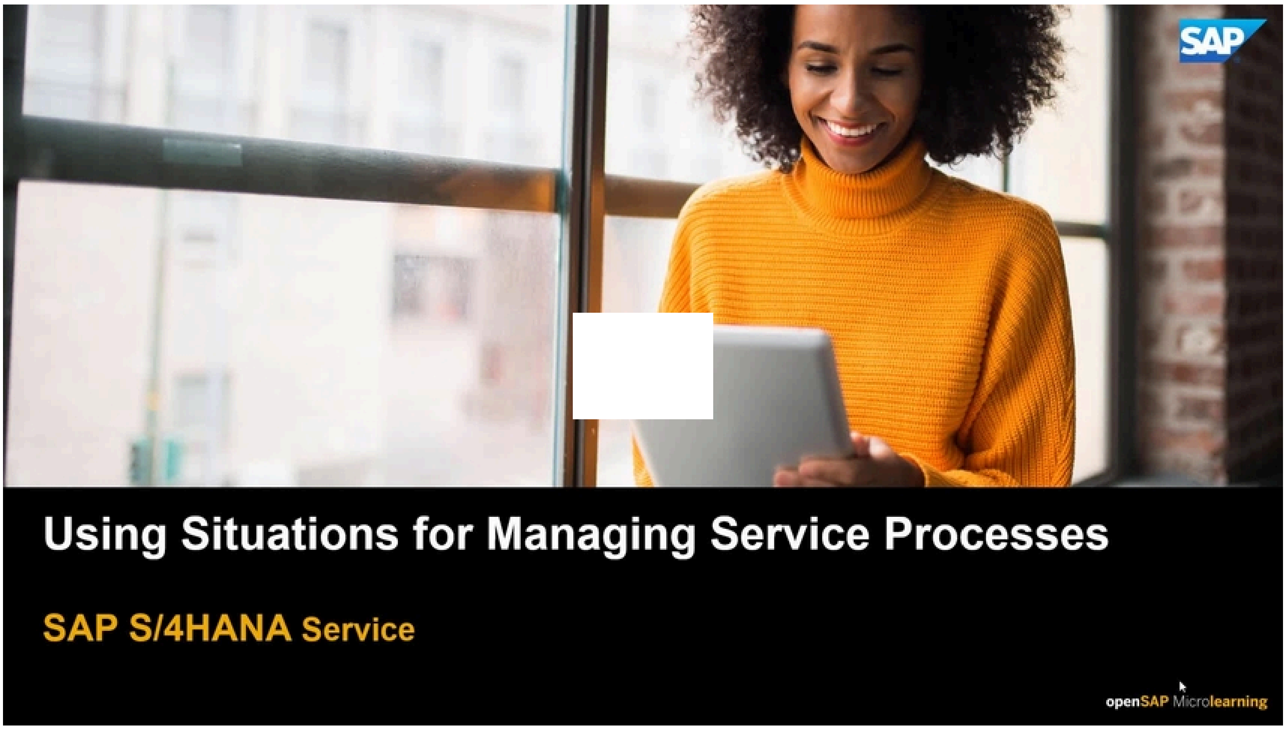
Situation Templates in Service

Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	In-House Service	Situation Template: In-House Service Not Confirmed	SRVC_DELAY_IN_IHR_CONFIRMATION	Standard object-based	No	Manage In-House Services	F402
Service	In-House Service	Situation Template: Processing of In-House Service Overdue	SRVC_DELAY_IN_IHR_PROCESSING	Standard object-based	No	Manage In-House Services	F402
Service	In-House Service	Situation Template: Follow-Up Step of Precheck Pending	SRVC_IHRITM_PRECHECK_DELAYED	Standard object-based	No	Manage In-House Service Objects	F769
Service	Service and Maintenance Plan	Service Maint. Plan Mass Schedule Issue	SRVC_MPLAN_MASS_SCHEDULING_ISSUE	Standard object-based	No	Mass Schedule Maintenance Plans - Service	F492
Service	Service Contract Management	Service Contract Due to Expire Soon	SRVC_EXPIRING_SERVICE_CONTRACT	Standard object-based	No	Service Contracts	TBT1 (Web UI ID)
Service	Service Contract Management	Service Contract Item Due to Expire Soon	SRVC_EXPIRING_SRVC_CONTR_ITEM	Standard object-based	No	Service Contracts	TBT1 (Web UI ID)
Service	Service Contract Management	Service Contract Item Not Released	SRVC_CONTRACT_ITEM_RELEASE_DUE	Standard object-based	Yes	Service Contracts	F376
Service	Service Contract Management	Service Contract with Zero Billing Value	SRVC_CONTR_WITH_EXP_BILL_VALUE	Standard object-based	No	Service Contracts	TBT1 (Web UI ID)
Service	Service Contract Management	SrvcContr Item Blocked for Billing	SRVC_CONTRACT_ITEM_BILLG_BLOCKED	Standard object-based	Yes	Service Contracts	F376
Service	Service Contract Management	Price Check for Automatic Renewal	SRVC_CONTR_AUTO_RENEW_PRICE	Standard object-based	No	Service Contracts	F376
Service	Service Contract Management	Service Contract GTS Blocked/Pending	SRVC_CTR_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Service Contracts	F376
Service	Subscription Order Management	Subscription Contract Item Due to Expire	SOM_EXPIRING_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F619

Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	Subscription Order Management	Subscription Contract Item Grace Period	SOM_GRACE_PERD_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Subscription Order Management	Subscription Contract Item Distribution	SOM_DISTR_ISSUE_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Service Order Management	Trade Compliance Check Blocked/Pending	SRVC_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Manage Service Orders	TBT1 TBT1
Service	In-House Service	Situation Template: Stock Inconsistency for Service Object	SRVC_IHRITM_PROD_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695
Service	In-House Service	Situation Template: Inconsistency in Logistics Information for Service Object	SRVC_IHRITEM_EQUIP_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695

Using Situations for Managing Service Processes (English Only)

Watch this video at learning.sap.com.



Related Information

Use Cases for Situation Handling

Data Management in Service

Data management includes data archiving, blocking, unblocking and deleting personal data, as well as data destruction.

Data Archiving in Service
Blocking, Unblocking, and Deletion of Personal Data in Service

Related Information

Data Management in SAP S/4HANA

Data Archiving in Service

Data archiving is used to remove mass data that is no longer required but must be kept in a format that can be analyzed from the database. This section includes archiving objects for Service.

For more information about the archiving objects that are used for leads, opportunities, and activities, see Data Archiving in Sales.

Prerequisites for Archiving Service Transactions

Depending on the Finance integration scenario, you should consider the following when archiving service orders and service contracts:

- If you use item-based accounting, the service orders and service contracts to be archived must be set to **Business Completed**.
- If you use the integration with account assignment manager, the service orders and service contracts to be archived must be set to **Completed**.

Regardless of the Finance integration scenario used, all other transactions, such as service requests, service quotations, and service confirmations, must be set to **Completed** to be archived.

Related Information

Data Archiving
Data Management in Global Trade Management
Item-Based Accounting
Finalizing the Service Order
Finalizing the Service Contract

Archiving Service Contracts Using CRM_SRCONT

You can use archiving object `CRM_SRCONT` to archive service contracts.

Tables

CRM_SRCONT archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SRCONT can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SRCONT is delivered with the following programs:

Program	Name
Write	CRM_ARC_SRCONT_SAVE
Delete	CRM_ARC_SRCONT_DELETE_2
Preprocessing	CRM_ARC_SRCONT_CHECK_2

When archiving service contracts, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SRCONT

Prerequisites

The following prerequisites must be fulfilled before you can archive a service contract:

- The service contract has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service orders and service confirmations are archivable.
- If item-based accounting has been enabled: The service contract has been set to **Business Completed**.

  For more information about item-based accounting, see Item-Based Accounting.

The following checks have been carried out:

- You've executed an archivability check for service contracts by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service contracts that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SRCONT ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SRCONT, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SC for service contract.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service contract was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_SRCONT assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation of the following BAdIs for CRM_SRCONT:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146 .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SRCONT, there is at least one active archive information structure that contains the following selectable field:
    - Transaction ID (object_id)

  This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SC that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SRCONT archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service contracts.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Contracts Archived With CRM_SRCONT

You have the following options for displaying archived service contracts:

- From within **Archive Administration** (transaction SARA), you can search for archived service contracts using the read program CRM_ARC_SRCONT_READ
- You can also use the **Archive Information System** (transaction SARI) to display archived service contracts by entering the archiving object CRM_SRCONT and the archive infostructure SAP_CRMS4_SC

In both cases, the system provides a range of criteria that enable you to select the service contracts that you want to display.

Variants for CRM_SRCONT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SRCONT.

Defining Write Variants

The write variant contains the parameters for the service contracts that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service contracts that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you’re processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_SRCONT_CHECK_2` determines archivable service contracts.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It’s possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Requests Using CRM_INCDNT

You can use archiving object `CRM_INCDNT` to archive service requests.

Tables

`CRM_INCDNT` archives data from several tables. To check which tables these are, call up transaction `SARA`, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_INCDNT` can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

`CRM_INCDNT` is delivered with the following programs:

Program	Name
Write	`CRM_ARC_SRQM_INCIDENT_SAVE`
Delete	`CRM_ARC_SRQM_INCIDENT_DELETE_2`
Preprocessing	`CRM_ARC_SRQM_INCIDENT_CHECK_2`

When archiving service requests, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_INCDNT.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service request:

- The service request has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.

- Any follow-up service orders and service confirmations are archivable.

The following checks have been carried out:

- You've executed an archivability check for service requests by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service requests that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_INCDNT` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_INCDNT`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance, SRVR for a service request.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization the rule applies to.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT`)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service request was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object `CRM_INCDNT` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for `CRM_INCDNT`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)

- BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity Maintain Purposes (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity Personalize Data Collection via Profiles (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_INCDNT, there is at least one active archive information structure that contains the following selectable field:
    - Transaction ID (object_id)

  This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRM_INCDNT that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_INCDNT archiving object, you must first check the settings in Customizing for Service under Transactions > Data Archiving.

In the activity Configure Archiving Preselection, you define parameters for the check and preprocessing programs. In the activity Define Residence Times, you define the residence time of service requests.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Requests Archived With CRM_INCDNT

- From within Archive Administration (transaction SARA), you can search for archived service requests using the read program CRM_ARC_SRQM_INCIDENT_READ
- You can also use the Archive Information System (transaction SARI) to display archived service requests by entering the archiving object CRM_INCDNT and the archive infostructure SAP_CRMS4_SRVR

In both cases, the system provides a range of criteria that enable you to select the service requests that you want to display.

Variants for CRM_INCDNT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction `SARA`, you can create variants of the write and preprocessing programs available for `CRM_INCDNT`.

Defining Write Variants

The write variant contains the parameters for the service requests that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SRQM_INCIDENT_CHECK_2 determines archivable service requests.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Orders Using CRM_SERORD

You can use archiving object CRM_SERORD to archive service orders.

Tables

CRM_SERORD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SERORD can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SERORD is delivered with the following programs:

Program	Name
Write	CRM_ARC_SERORD_SAVE
Delete	CRM_ARC_SERORD_DELETE_2
Preprocessing	CRM_ARC_SERORD_CHECK_2

When archiving service orders, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SERORD.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service order:

- The service order has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service confirmations are archivable.
- If item-based accounting has been enabled: The service order has been set to **Business Completed**.

  For more information about item-based accounting, see Item-Based Accounting.

The following checks have been carried out:

- You've executed an archivability check for service orders by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service orders that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SERORD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SERORD, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SRVO for a service order.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.

Condition	Description
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service order was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_SERORD assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRM_SERORD:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SERORD, there is at least one active archive information structure that contains the following selectable field:
  - **Transaction ID** (object_id)

  These fields are used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRVO that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SERORD archiving object, you must first check the settings in Customizing in **Service** under **Transactions** > **Data Archiving**.

In the activity Configure Archiving Preselection, you define parameters for the check and preprocessing programs. In the activity Define Residence Times, you define the residence time of service orders.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Orders Archived With CRM_SERORD

You have the following options for displaying archived service orders:

- From within Archive Administration (transaction SARA), you can search for archived service orders using the read program CRM_ARC_SERORD_READ.
- You can also use the Archive Information System (transaction SARI) to display archived service orders by entering the archiving object CRM_SERORD and the archive infostructure .

In both cases, the system provides a range of criteria that enable you to select the service orders that you want to display.

Archiving Service Orders in Hierarchical Settlements

It is possible to archive service orders even when one or more items are part of a financial hierarchy with a service contract and when hierarchical settlement is used. It is, however, required that all service contracts involved are already set to Business Completed.

If you revoke a completed service contract, you can also revoke any completed follow-up service orders associated with it, provided they are within the same financial hierarchy. The following applies when doing so:

- Lower levels in the hierarchy cannot have a status that precedes the status of the higher levels.
- If the service contract is archived, the associated service orders and PM orders must also be archived.
- If the service contract is Business Completed, the associated service orders and PM orders must be at least Business Completed as well. They cannot be reopened for business.

Variants for CRM_SERORD

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SERORD.

Defining Write Variants

The write variant contains the parameters for the service orders that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It’s possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  i Note

  Only generate detail logs if you’re processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_SERORD_CHECK_2` determines archivable service orders.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It’s possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  i Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving In-House Services Using CRMS4_REPA

You can use archiving object CRMS4_REPA to archive in-house services.

Tables

CRMS4_REPA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRMS4_REPA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRMS4_REPA is delivered with the following programs:

Program	Name
Write	CRMS4_ARC_REPA_SAVE
Delete	CRMS4_ARC_REPA_DELETE
Preprocessing	CRMS4_ARC_REPA_CHECK

When archiving in-house services, you must use these programs or custom variants. For information about creating variants, see Variants for CRMS4_REPA.

Prerequisites

The following prerequisites must be fulfilled before you can archive an in-house service:

- The in-house service has the status **Complete**.
- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for in-house services by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). You can use the program to run several check jobs in parallel, which is especially useful when archiving mass data. During the archivability check, in-house services that meet the archivability criteria, are assigned the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case, parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the ILM object `CRMS4_REPA` as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

You can select several of the following condition fields and put them into a specific order for the ILM object `CRMS4_REPA`. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type to which the rule applies, for instance, REPA for in-house service.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization to which the rule applies.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, to which the rule applies.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT`)	Use this field to define a country/region.

Once you've selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the in-house service was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management].

Enablement for Information Retrieval

The ILM object `CRMS4_REPA` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data for a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRMS4_REPA:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in the Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRMS4_REPA, there's at least one active archive information structure that contains the following selectable fields:
  - **Contact Person** (CONTACT_PERSON)
  - **Employee Responsible** (PERSON_RESP)
  - **Ship-To Party** (SHIP_TO_PARTY)
  - **Sold-To Party** (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_CRMS4_REP1 that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRMS4_REPA archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time for in-house services.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying In-House Services Archived With CRMS4_REPA

You have the following options for displaying archived in-house services:

- From within **Archive Administration** (transaction SARA), you can search for archived in-house services using the read program CRM_ARC_REPA_READ.
- You can also use the **Archive Information System** (transaction SARI) to display archived in-house services by entering the archiving object CRMS4_REPA and the archive infostructure SAP_CRMS4_REPA.

In both cases, the system provides a range of criteria that enable you to select the in-house-services that you want to display.

Archiving of Attachments

For more information about archiving attachments using the Document Management Service (DMS), see Archiving Document Info Records Using CV_DVS.

Variants for CRMS4_REPA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRMS4_REPA.

Defining Write Variants

The write variant contains the parameters for the in-house services that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the in-house services that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

> **i Note**
>
> Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRMS4_ARC_REPA_CHECK` determines archivable in-house services.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Confirmations Using CRM_SRVCON

You can use archiving object CRM_SRVCON to archive service confirmations.

Tables

CRM_SRVCON archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SRVCON can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SRVCON is delivered with the following programs:

Program	Name
Write	CRM_ARC_SRVCON_SAVE
Delete	CRM_ARC_SRVCON_DELETE_2
Preprocessing	CRM_ARC_SRVCON_CHECK_2

When archiving service confirmations, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SRVCON.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service confirmation:

- The service confirmation has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up transactions are archivable.

The following checks have been carried out:

- You've executed an archivability check for service confirmations by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service confirmations that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SRVCON ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_SRVCON`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance, `SVC1` for a service confirmation.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization the rule applies to.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT`)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service confirmation was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object `CRM_SRVCON` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for `CRM_SRVCON`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)
- **BAdI: Archiving Information of Tables** (`BADI_DTINF_ARCHIVE_DETAILS`)

For more information, see SAP Note [2939146](#).

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction `DTINF_MAINTAIN_PURP`), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction `DTINF_MAINT_PROFILE`) defining that archived personal data must be included in the search.
- Ensure that for archiving object `CRM_SRVCON`, there is at least one active archive information structure that contains the following selectable field:
  - **Transaction ID** (`object_id`)

This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRVC that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SRVCON archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service confirmations.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Confirmations Archived With CRM_SRVCON

You have the following options for displaying archived service confirmations:

- From within **Archive Administration** (transaction SARA), you can search for archived service confirmations using the read program CRM_ARC_SERORD_READ
- You can also use the **Archive Information System** (transaction SARI) to display archived service confirmations by entering the archiving object CRM_SRVCON and the archive infostructure SAP_CRMS4_SRVC

In both cases, the system provides a range of criteria that enable you to select the service confirmations that you want to display.

Variants for CRM_SRVCON

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SRVCON.

Defining Write Variants

The write variant contains the parameters for the service confirmations that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service confirmations that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_SRVCON_CHECK_2` determines archivable service confirmations.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Knowledge Articles Using CRM_KA

You can use archiving object CRM_KA to archive knowledge articles.

Tables

CRM_KA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_KA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_KA is delivered with the following programs:

Program	Name
Write	CRM_ARC_KA_SAVE
Delete	CRM_ARC_KA_DELETE_2
Preprocessing	CRM_ARC_KA_CHECK_2

When archiving knowledge articles, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_KA.

Prerequisites

The following prerequisites must be fulfilled before you can archive a knowledge article:

- The knowledge article has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for knowledge articles by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, knowledge articles that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_KA` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_KA`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (BUKRS)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance KNAR for a knowledge article.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT`)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the knowledge article was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object `CRM_KA` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for `CRM_KA`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)
- **BAdI: Archiving Information of Tables** (`BADI_DTINF_ARCHIVE_DETAILS`)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes**DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_KA, there is at least one active archive information structure that contains the following selectable field:
    - **Employee Responsible** (PERSON_RESP)

  This field is used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_CRMS4_KNAR that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_KA archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of knowledge articles.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Knowledge Articles Archived With CRM_KA

You have the following options for displaying archived knowledge articles:

- From within **Archive Administration** (transaction SARA), you can search for archived knowledge articles using the read program CRM_ARC_KA_READ
- You can also use the **Archive Information System** (transaction SARI) to display archived knowledge articles by entering the archiving object CRM_KA. You must create an archive infostructure for this purpose. For more information, see Providing Archive Infostructures for Data Access

Variants for CRM_KA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for `CRM_KA`.

Defining Write Variants

The write variant contains the parameters for the activities that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_KA_CHECK_2` determines archivable knowledge articles.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  i Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Cases Using SCMG

You can use archiving object `SCMG` to archive cases.

Tables

`SCMG` archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`SCMG` can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

`SCMG` is delivered with the following programs:

Program	Name
Write	`CASE_WRITE_PROGRAM`

Program	Name
Delete	CASE_DELETE_PROGRAM
Preprocessing	CASE_PRE_PROGRAM

When archiving cases, you must use these programs or custom variants. For information on creating variants, see Variants for SCMG.

Prerequisites

The following prerequisites must be fulfilled before you can archive a case:

- The case has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- All follow-on transactions related to the case must be archivable.

The following checks have been carried out:

- You've executed an archivability check for cases by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, cases that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SCMG ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object SCMG, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Case Type (CASE_TYPE)	Use this field to define the case type.
Area ID (AREA_ID)	Use this field to define the area the rule applies to, for instance case management.
RMS ID (RMS_ID)	Use this field to define the records management system ID. The RMS ID is a logical grouping of all objects that have been created for a specific business unit.
System Status (STATUS)	Use this field to define the case status the rule applies to, for instance, **Completed**.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The available time references are the end date of the case as well as the month, quarter or year of the end date. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object SCMG assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for SCMG:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object SCMG, there is at least one active archive information structure that contains the following selectable fields:
  - **Case Key** (CASE_GUID)
  - **Customer** (FIN_KUNNR)

  These fields are used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_FSCM that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the SCMG archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of cases.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Cases Archived With SCMG

You have the following options for displaying archived cases:

- From within **Archive Administration** (transaction SARA), you can search for archived cases using the read program CASE_READ_PROGRAM
- You can also use the **Archive Information System** (transaction SARI) to display archived cases by entering the archiving object SCMG and archive infostructure SAP_CRM_CMG.

In both cases, the system provides a range of criteria that enable you to select the cases that you want to display.

Related Information

BAdis for SCMG

Variants for SCMG

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for SCMG.

Defining Write Variants

The write variant contains the parameters for the cases that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CASE_PRE_PROGRAM` determines archivable cases.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

BAdis for SCMG

You can enhance the archiving functions provided for cases using Business Add-Ins (BAdi).

Business Add-In	Use
ARC_SCMG_CHECK	You can use this BAdi to define custom checks that must be met before a case can be archived.
ARC_SCMG_WRITE	You can use this BAdi to write additional data to the archive that isn't directly associated with the case.
ARC_SCMG_DELETE	You can use this BAdi to delete additional data that isn't directly associated with the case.

For more information on the BAdis, see the system help in Customizing under **Service** > **Case Management** > **Extended Customzing** > **Case Archiving** > **BAdis for Case Archiving**.

You can access the BAdIs using transaction SE18.

Archiving of Subscription Product-Specific Data Using SOM_PROD

You can use archiving object SOM_PROD to archive subscription product-specific data.

Tables

SOM_PROD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_PROD may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before subscription product-specific data can be archived:

- The related product master is marked for deletion.
- No drafts exist for the subscription-product specific data.
- The product is not used in any subscription order/contract.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SOM_PROD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- PRODUCT: Product number

The following time references are available:

- CREATION DATE: Date when the product was created
- LAST_CHANGE_DATE: Date when the product was last changed

For more information, search for SAP Information Lifecycle Management in the relevant version of SAP S/4HANA on the SAP Help Portal.

Performing Application-Specific Configuration

There is no specific Customizing for the SOM_PROD archiving object.

Defining Preprocessing Variants

There is no preprocessing program for SOM_PROD.

You can define variants for this program using the following parameters:

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

This program analyzes the subscription product-specific data for archiving and then writes the data to an archive file. The program selects all subscription product-specific data that fits in with the selected products. You can execute the report in either test or production mode. Data, however, is written to the archive file in production mode only.

In the **Comment for Archive Run** field, you can enter your own comment for the archive run.

You can archive subscription product-specific data only if the following prerequisites are met:

- The related product master is marked for deletion.
- No draft exists for the subscription product-specific data.
- The product is not used in any subscription order or subscription contract.

A write variant contains the parameters for the subscription product-specific data that you want to archive.

SAP delivers the following parameters:

- Product

  This is the product number for the related product.

The write program is CRMS4_SOM_PROD_ARCHIVE_WRITE.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRMS4_SOM_PROD_ARCHIVE_DELETE for SOM_PROD deletes the subscription product-specific data based on an existing archive. The program first reads the subscription product-specific data belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program CRMS4_SOM_PROD_ARCHIVE_READ displays the archived subscription product-specific data. This data must be archived and deleted. The selected product IDs are displayed using ALV. You can navigate to the details of each product by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Subscription product-specific data UUID
- Product ID
- Key of archive file

Defining Postprocessing Variants

There is no postprocessing program for SOM_PROD.

Defining Reloading Variants

There is no reloading program from SOM_PROD.

Dependencies of SOM_PROD

Subscription orders and subscription contracts must be archived before you can archive subscription-product specific data.

The related material master (archiving object MM_MATNR) can only be archived after the subscription-product specific data is archived.

Authorizations for SOM_PROD

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_PROD
ACTVT: 01 - Add or create ARCH_OBJ: SOM_PROD APPLIC: IS	S_ARCHIVE

Activity	Required Authorization Object

Enhancements for SOM_PROD

The SOM_PROD archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
ARC_SOM_PROD_CHECK	Archiving subscription product-specific data: Check archivability	Check if an object can be archived
ARC_SOM_PROD_WRITE	Archiving subscription product-specific data: Write additional table	Write additional tables for archiving

Displaying Subscription Product-Specific Data Archived with SOM_PROD

You can use program CRMS4_SOM_PROD_ARCHIVE_READ to display archived and deleted subscription product-specific data.

Archiving of Allowance Definition Groups Using SOM_ADG

You can use archiving object SOM_ADG to archive allowance definition groups.

Tables

SOM_ADG archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_ADG may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before an allowance definition group can be archived:

- Status of the allowance definition group must NOT be **In Preparation**.

  As long as the allowance definition group has this status, it can be deleted directly.

- No draft exists for the allowance definition group.
- The allowance definition group, to be deleted, is not used in any subscription product-specific data.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `SOM_ADG` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Allowance Definition Group ID

The following time references are available:

- CREATION_DATE: Date when the allowance group was created.
- LAST_CHANGE_DATE: Date when the allowance group was last changed.

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Performing Application-Specific Configuration

There is no specific Customizing for the `SOM_ADG` object.

Defining Preprocessing Variants

There is no preprocessing program for `SOM_ADG`.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the allowance definition group that you want to archive. This program covers the following features:

- The analysis of allowance definition groups with respect to archivability.
- Writing allowance definition groups to an archive file based on the analysis in the previous step.

The following database tables are included:

- CRMS4_SOM_ADG
- CRMS4_SOM_ADGT
- CRMS4_SOM_AD

Additionally, change documents related to the archived allowance definition group are also archived.

SAP delivers the following parameters:

- Allowance Definition Group ID

  The write program is `CRMS4_SOM_ADG_ARCHIVE_WRITE`.

Defining Delete Variants

The program CRMS4_SOM_ADG_ARCHIVE_DELETE is used to delete the allowance definition groups based on an existing archive.

The program first reads the allowance definition groups belonging to an archiving run from the archive, and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program `CRMS4_SOM_ADG_ARCHIVE_READ` displays the selected allowance definition groups using ALV. You can navigate to the details of each allowance definition group by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Allowance Definition Group UUID
- Allowance Definition Group ID
- Key for Archive File

Defining Postprocessing Variants

There is no postprocessing program for `SOM_ADG`.

Defining Reloading Variants

There is no reloading program for `SOM_ADG`.

Dependencies of `SOM_ADG`

Authorizations for `SOM_ADG`

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	CRM_ISX_AD
ACTVT: 01 ARCH_OBJ: SOM_ADG APPLIC: IS	S_ARCHIVE

Enhancements for `SOM_ADG`

The `SOM_ADG` archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
`ARC_SOM_ADG_CHECK`	Archiving Allowance Definition Group: Check Archivability	Check if an object can be archived.

BAdI / User Exit	Description	Used to
ARC_SOM_ADG_WRITE	Archiving Allowance Definition Group: Write Additional Table	Write additional tables for archiving

Displaying Allowance Definition Groups Archived with SOM_ADG

You can use the program CRMS4_SOM_ADG_ARCHIVE_READ to display archived and deleted allowance definition groups.

Archiving Subscription Orders Using CRM_PRVO

You can use archiving object CRM_PRVO to archive subscription orders (BUS2000265) in transaction SARA.

Tables

CRM_PRVO archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRVO may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a subscription order can be archived:

- The subscription order status must be set to Completed at the header level.
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- If you have set up ILM integration for the CRM_PRVO ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_PRVO ILM object as part of SAP Information Lifecycle Management. In the transaction IRMPOL, you can create policies for residence time and retention rules, depending on the available policy category. Here, you can also see the available time reference and the existing condition fields. You can decide which of these will be used to define your rule structure. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- OBJECT_TYPE - Business Transaction Category

- PROCESS_TYPE - Business Transaction Type
- BS_CRM_COUNTRY_OF_ORGUNIT - Country of Org Unit

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_PRVO assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVO:

- BAdI Implementation BADI_CRM_SUBSCR_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRVO_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVO, there is at least one active archive information structure that contains the following selectable fields:
  - Field name (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.
- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_PRVO archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide** > **Service** > **Transactions** > **Data Archiving**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_PRVO_CHECK_2` for `CRM_PRVO` checks the prerequisites discussed above so that subscription orders can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction `SARA`.

The write program CRM_ARC_PRVO_SAVE for `CRM_PRVO` writes the data to the archive file. The program contains the parameters for the subscription orders that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

- Changed On

  Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction `SARA`.

The delete program CRM_ARC_PRVO_DELETE for `CRM_PRVO` deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

  Changed On

  Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Read Program Variants

The read program CRM_ARC_PRVO_READ for `CRM_PRVO` reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

> **i Note**
>
> The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVO and field catalog SAP_CRMS4_PRVO, you can maintain your own infostructure and field catalogs.

Archiving Subscription Contracts Using `CRM_PRVC`

You can use archiving object `CRM_PRVC` to archive subscription contracts (BUS2000266) in transaction SARA.

Tables

`CRM_PRVC` archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_PRVC` may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction `AOBJ`, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a `subscription contract` can be archived:

- The subscription contract statuses must be set to Completed and Finished at the header level
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- All linked subscription orders must be archived and deleted from the system.
- The FI-CA provider contracts that correspond to the subscription contracts must be archived and deleted from the system. For more information about archiving a FI-CA provider contract, see Archiving of Provider Contracts (FI_MKKCAVT) on the SAP Help Portal.
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- Sharing Contracts can only be archived when no more linked Shared Contracts exist, i. e. Shared Contracts have to be archived and deleted before the Sharing Contracts they refer to.
- If you have set up ILM integration for the `CRM_PRVC` ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM-PRVC ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- OBJECT_TYPE - Business Transaction Category
- PROCESS_TYPE - Business Transaction Type
- BS_CRM_COUNTRY_OF_ORGUNIT - Country of Org Unit

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Enablement for Information Retrieval

The ILM object CRM_PRVC assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVC:

- BAdI Implementation BADI_CRM_SUBSCR_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRVC_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVC, there is at least one active archive information structure that contains the following selectable fields:
  - Field name (SOLD_TO_PARTY)

These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction `DTINF_START_COLL` - Start Data Collection.
- You can view the collected data using the transaction `DTINF_PROC_COLL` - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the `CRM-PRVC` archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide** > **Service** > **Transactions** > **Data Archiving**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVC_CHECK_2 for `CRM_PRVC` checks the prerequisites discussed above so that subscription contracts can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction `SARA`.

The write program CRM_ARC_PRVC_SAVE FOR `CRM_PRVC` writes the data to the archive file. The program contains the parameters for the subscription contracts that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

  Changed On

  Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction `SARA`.

The delete program CRM_ARC_PVC_DELETE for `CRM_PRVC` deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVC_READ for `CRM_PRVC` reads data from archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

> **Note**
> The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVC and field catalog SAP_CRMS4_PRVC, you can maintain your own infostructure and field catalogs.

Archiving Master Agreements Using `CRM_PRVMA`

You can use archiving object `CRM_PRVMA` to archive Master Agreements (BUS2000267) transaction SARA.

Tables

`CRM_PRVMA` archives data from several tables. To check which tables these are, call up transaction `SARA`, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_PRVMA` may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction `AOBJ`, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a `master agreement` can be archived:

- The header status must be set to Completed.
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- All subscription orders, subscription contracts, and solution quotations that refer to the master agreement must be archived.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_PRVMA` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- PROCESS_TYPE - Business Transaction Type

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On
- CONTRACT_END_DATE - Contract End Date

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Enablement for Information Retrieval

The ILM object CRM_PRVMA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVC:

- BAdI Implementation BADI_CRM_MA_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRMA_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVMA, there is at least one active archive information structure that contains the following selectable fields:
  - Field name (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.
- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_PRVMA archiving object, you must first check the settings in Customizing under SAP Customizing Implementation Guide > Service > Transactions > Data Archiving.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVMA_CHECK for CRM_PRVMA will check the prerequisites discussed above so that BUS2000267 can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Transaction type/Process type of a document to be archived.

- Transaction ID

  Object ID of a document to be archived.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the <business objects> that you want to archive.

SAP delivers the following parameters:

The write program CRM_ARC_PRVMA for CRM_PRVMA writes the data to the archive file. The program contains the parameters for the master agreements that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Transaction type/Process type of a document to be archived.

- Transaction ID

  Object ID of a document to be archived.

- Changed On

  Last changes made to the document.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRM_ARC_PRVMA_DELETE_2 for CRM_PRVMA deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVMA_READ for `CRM_PRVMA` reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVMA and field catalog SAP_CRMS4_PRVMA, you can maintain your own infostructure and field catalogs.

Archiving Sharing Group Using SOM_SHGR

You can use archiving object `SOM_SHGR` to archive sharing groups.

Tables

`SOM_SHGR` archives data from several tables. To check which tables these are, call up transaction `SARA`, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`SOM_SHGR` may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction `AOBJ`, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction `SARA` enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a sharing groups can be archived:

- No draft exists for the sharing group.
- The sharing group is not used in a subscription order, subscription contract, or a master agreement.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `SOM_SHGR` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Sharing group ID

The following time references are available:

- CREATION_DATE: Date on which the sharing group was created
- LAST_CHANGE_DATE: Date on which the sharing group was last changed.

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Performing Application-Specific Configuration

There is no specific Customizing for the `SOM_SHGR` archiving object.

Defining Preprocessing Variants

There is no preprocessing program for `SOM_SHGR`.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction `SARA`.

A write variant contains the parameters for the sharing object that you want to archive. The program CRMS4_SOM_SHGR_ARCHIVE_WRITE is used to write the archive.

This report carries out the first step in archiving sharing groups and covers the following features:

- Analyzing sharing groups with respect to archivability
- Writing sharing groups to an archive file based on the results of the analysis

The following database tables are included:

- `CRMS4_SOM_SRG`
- `CRMS4_SOM_SRGT`

Additionally, change documents related to the archived sharing group are also archived.

SAP delivers the following parameters:

- Sharing Group ID

  The write program is CRMS4_SOM_SHGR_ARCHIVE_WRITE

Defining Delete Variants

The program CRMS4_SOM_SHGR_ARCHIVE_DELETE is used to delete sharing groups based on an existing archive, and is the second step of the archiving process.

The report first reads the sharing groups belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program CRMS4_SOM_SHGR_ARCHIVE_READ is used to display the archived and deleted sharing groups.

You must ensure the following prerequisites are fulfilled before executing the read program:

- Data is archived and deleted

You can select the archived data by using the following selection parameters:

- Sharing Group UUID
- Sharing Group ID
- Key for Archive File

The selected sharing groups are displayed using ALV. You can navigate to the details of each sharing group by double clicking on the relevant record.

Defining Postprocessing Variants

There is no postprocessing program for `SOM_SHGR`.

Defining Reloading Variants

There is no reloading program for `SOM_SHGR`.

Dependencies of `SOM_SHGR`

Subscription orders, subscription contracts, and master agreements using a specific sharing group ID must be archived before you can archive the sharing group.

Authorizations for `SOM_SHGR`

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_SHGR
ACTVT: 01 ARCH_OBJ: SOM_SHGR APPLIC: IS	S_ARCHIVE

Enhancements for `SOM_SHGR`

The `SOM_SHGR` archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
`ARC_SOM_SHGR_CHECK`	Archiving Sharing Group: Check Archivability	Check if an object can be archived

BAdI / User Exit	Description	Used to
ARC_SOM_SHGR_WRITE	Archiving Sharing Group: Write Additional Table	Write additional tables for archiving

Displaying Sharing Groups Archived with SOM_SHGR

You can use the program CRMS4_SOM_SHGR_ARCHIVE_READ to display the archived and deleted sharing groups.

Blocking, Unblocking, and Deletion of Personal Data in Service

You can use SAP Information Lifecycle Management (SAP ILM) to control the blocking, unblocking, and deletion of business partner, customer, and supplier master data. According to the business partner approach, blocking and deletion of customer and supplier master data is part of the blocking and deletion of the business partner master data. This means that the business partner considers all residence periods of linked customer or supplier master data. Personal data collected in master data can be blocked as soon as the business activities for which this data was collected have been completed and the residence time for this data has elapsed.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, customer and supplier master data must be archived and deleted from the database before the associated business partner master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction IRMPOL).

The following scenarios for blocking business partners are supported:

1. You want to block business partner master data for which the associated customer and supplier data is blocked. The business partner master data satisfies the EoP check criteria and therefore can be blocked.
2. You want to block business partner data for which the associated customer and supplier master data is not blocked. The associated customer and supplier master data satisfies their EoP checks as does the business partner master data. All of the associated customer and supplier master data as well as the business partner master data can be blocked.
3. You want to block business partner master data for which the associated customer and supplier master data is not blocked and where at least one customer or supplier does not meet the EoP check criteria. The business partner master data cannot be blocked until all of the associated customer and supplier master data meet the EoP check criteria and can also be blocked.
4. Independent blocking of the customer and supplier master data is possible. The corresponding roles of the business partner are also blocked.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, associated customer and supplier master data must be archived and deleted from the database before the associated business partner master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction IRMPOL).

Specifics Business Partner Master Data

The functions required for blocking and unblocking or viewing blocked business partner master data are provided through the data protection specialist roles:

- SAP_CA_BP_DP_DISPLAY
- SAP_CA_BP_DP_BLOCK
- SAP_CA_BP_DP_UNBLOCK
- SAP_CA_BP_DP_BLOCK_REQUEST

Specifics for Customer and Supplier Master Data

The functions required for blocking and unblocking or viewing blocked customer and supplier master data are provided through the data protection specialist roles:

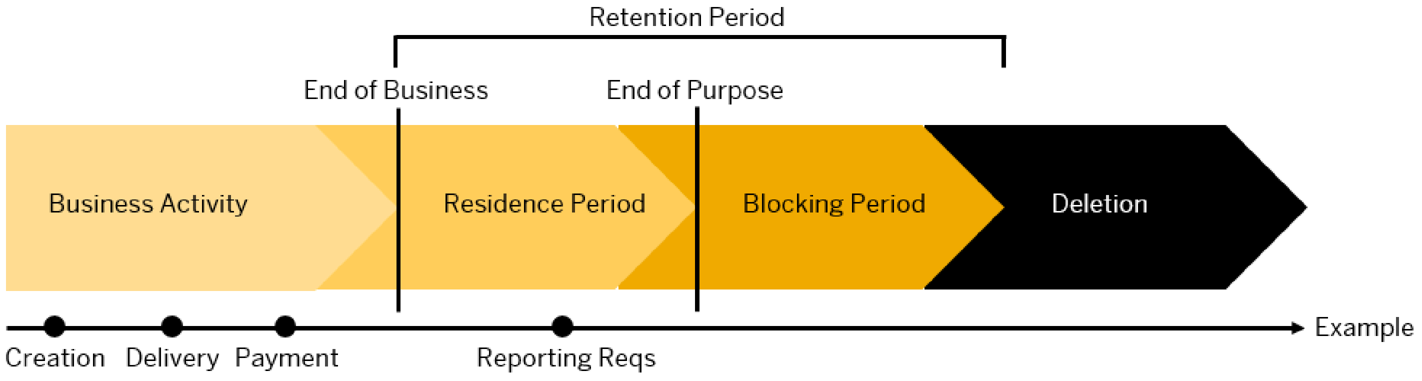
- SAP_LO_MD_CVP_DISPLAY
- SAP_LO_MD_CVP_BLOCK
- SAP_LO_MD_CVP_UNBLOCK

You can copy these template roles to custom roles, adjust them to meet your requirements, and assign them to your users.

Business Partner End of Purpose (EoP) Check in Service

An end of purpose (EoP) check determines whether data is still relevant for business activities based on the retention period defined for the data. The retention period is part of the overall lifecycle of personal data, which consists of the following phases:

- **Business activity:** The relevant data is used in ongoing business, for example contract creation, delivery or payment.
- **Residence period:** The relevant data remains in the database and can be used in subsequent processes related to the original purpose, for example reporting obligations.
- **Blocking period:** The relevant data must be retained for legal reasons. During the blocking period, business users of SAP applications are prevented from displaying and using the data. However, it can be processed, if so required due to mandatory legal provisions.
- **Deletion:** The data is deleted and no longer exists in the database.



Personal Data Lifecycle

Blocking of data impacts system behavior in the following ways:

- **Display:** Only users with specific auditing authorizations can display blocked data.
- **Change:** It is not possible to change a business object that contains blocked data.

- **Create:** It is not possible to create a business object that contains blocked data.
- **Copy or follow-up:** It is not possible to copy a business object or perform follow-up activities for a business object that contains blocked data.
- **Search:** Only user with specific auditing authorizations can search for blocked data or search for a business object using blocked data.

Customizing for the End of Purpose Check

Activity	Customizing Path
Define the settings for authorization management in Customizing for Cross-Application Components under:	Data Protection > Authorization Management
Configure the settings related to the blocking and deletion of business partner master data in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner
Configure the settings related to the blocking and deletion of customer and supplier master data in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Customer Master/Supplier Master Deletion
Define whether you want to use no blocking or dual control for blocking in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner > Set Activation Status of No Blocking/Dual Control for Role Group
Define whether you want to use dual control for unblocking in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Reasons for Unblocking Business Partner
	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Registered Function Modules for Unblock BP
	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Dual Control Setting for Unblocking
	Data Protection > Blocking and Unblocking of Data > Business Partner > Set Status for Automatic Unblocking
Configure the settings for transactional data specific to service solution in the Customizing for Service under:	Cross-Application Components > Data Protection

For more information about the possible Customizing settings, refer to the system help of the respective activity.

Process Flow

Activity	Additional Information
Enable the end of purpose check function for the central business partner by running transaction BUPA_PRE_EOP	System help for the Blocking of Business Partner Data report
Enable the end of purpose check function for the customer and supplier master data by running transaction CVP_PRE_EOP	System help for the Block Customer and Supplier Master Data report
Request the unblocking of blocked data	Request Unblocking of Master Data
Unblock master data	Unblocking Master Data
Delete master data	Deletion of Master Data

Activity	Additional Information
Display application logs for the business partner, customer, and supplier	Logging

ILM-enabled Objects in Service

Service uses the central business partner CA_BUPA.

Authorizations

Users who have a role based on the SAP_CA_BP_DP_DISPLAY or SAP_LO_MD_CVP_DISPLAY template roles, can display blocked business partner master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Users who have a role based on the SAP_LO_MD_CVP_DISPLAY template role with the B_BUP_PCPT authorization object including the assignment of activity 03, DISPLAY can display blocked customer and supplier master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Business Partner Master Data in Multisystem Landscapes

In a multisystem landscape, the EoP for business partner master data is checked for an application in the local system as well as the remote systems. The EoP check is used to ensure data integrity in case of potential blocking. The EoP determines whether there are any data dependencies, that is, whether the data is still required for business activities. If there is such a dependency, the system does not block the data. If you still want to block the data, the dependent data must be blocked or deleted using archiving and deletion tools. If the EoP criteria are met, the business partner master data is blocked in the local and remote systems.

Blocking Master Data

You can use the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP) to block data related to a business partner including associated customer and supplier master data or the **Block Customer and Supplier Master Data** report (transaction CVP_PRE_EOP) to block data related to customer and supplier master data. These reports determine whether data can be blocked using the End of Purpose (EoP) check. If the data is still in use and business activities have not been completed, the system does not block the data. If business is complete and the residence period has expired, the reports block the business partner, customer or supplier master data as defined in the report parameters. To block business partner master data, any associated customer or supplier master data must first be blocked in your system.

Business Partner Master Data

Business partner master data is blocked completely in one step.

Customer and Supplier Master Data

Data is blocked by the system in the following order:

1. Block contact persons and/or the company codes
2. Block general data

You can perform both steps at the same time. However, performing the first step separately allows the contact person and/or company code data to be blocked and the residence time of the general data to be reset. The general data can then be blocked when the revised residence time is met.

More Information

- Report documentation for the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP)
- System help for the **Block Customer and Supplier Master Data** report (transaction CVP_PRE_EOP)

Request Unblocking of Master Data

You can request the unblocking of business partner, customer or supplier master data using the **Request Unblocking** report (transaction BUP_REQ_UNBLK). You may do this because new business starts with a historic business partner, customer or supplier and their master data has not yet been deleted.

More Information

System help for the **Request Unblocking** report (transaction BUP_REQ_UNBLK).

Unblocking Master Data

You can use the **Request Unblocking** report (transaction BUP_REQ_UNBLK) to unblock business partner data. If the dual control setting for unblocking data has not been set, you can unblock data from your own unblocking request. If your system has dual control for unblocking set-up, another user must process the request to unblock data.

You can unblock customer and supplier master data with the **Unblock Customer and Supplier Master Data** report (transaction CVP_UNBLOCK_MD).

Blocked data can only be unblocked, if the master data has not been archived and it is still within the retention period.

Unblocking is performed in the opposite order of the blocking process. That means that business partner master data is unblocked in one step.

Blocked customer or supplier master data that extends across several company codes and has several contact persons, is unblocked as follows:

1. Unblock the general data
2. Unblock the contact persons and the company codes

More Information

- Report documentation for the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP)
- System help for the **Request Unblocking** report (transaction BUP_REQ_UNBLK)
- System help for the **Unblock Customer and Supplier Master Data** report (CVP_UNBLOCK_MD)
- For more information on unblocking, see [Request Unblocking of Master Data]
- For more information on dual control and unblocking, see the product assistance on SAP Help Portal and search for *Unblocking*.

Deletion of Master Data

After the residence period has been completed and data is blocked, it either remains in the database or is stored in an external archive. If the blocked data remains in the database, the data manager can delete the blocked business partner, customer, and supplier master data from the database using the corresponding archiving objects. The business partner master data can be deleted once all associated customer and supplier master data has been deleted from the database.

To destroy data from the database or archives, you can use the Data Destruction report (transaction `ILM_DESTRUCTION`). Subsequently, the data can no longer be retrieved.

Logging

You can use the Analyze Application Log report (transaction `SLG1`) to view the centralized business partner log. To do so, use the `BDT_DATAARCHIVING` object with the `BUPA_EOP` subobject.

You can use the Display Stored Application Logs for Customer and Supplier report (transaction `CVP_DISPLAY_LOG`) to display the application log for one or various subobjects including:

Subobject	Subobject Description
`CUSTOMER_BLOCKING`	Customer blocking
`CUSTOMER_UNBLOCKING`	Customer unblocking
`SUPPLIER_BLOCKING`	Supplier blocking
`SUPPLIER_UNBLOCKING`	Supplier unblocking

This report provides more contextual details at the level of the log items compared to the Analyze Application Log report.

More Information

- System help for the Display Stored Application Logs for Customer and Supplier report (transaction `CVP_DISPLAY_LOG`)
- System help for the Analyze Application Log report (transaction `SLG1`)

Blocking and Deleting Personal Data in Subscription Order Management

The processes and functions of Subscription Order Management use personal data of the SAP business partner. If, from the point of view of these processes and functions, there is no longer any reason to store the data (for example, for a possible tax audit), you can block this data from the perspective of Subscription Order Management and ultimately destroy it.

For destroying personal data, use the functions of Information Lifecycle Management (ILM). Using the ILM functions, you first block the personal data of business partners for whom there is no further business. Then you delete this data as soon as it is no longer needed.

All archiving objects in Subscription Order Management support data protection notifications (see **Deletion of Personal Data in SAP BW/4HANA and SAP BW**).

Related Information

ILM Objects in Subscription Order Management

Use

In Subscription Order Management, the following archiving objects are enabled for use with SAP Information Lifecycle Management (SAP ILM):

ILM-Enabled Archiving Objects

Technical Name	Description	Available Fields of Retention Period	Available Start Times
SOM_PROD	Subscription product-specific data	PRODUCT - Material ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On
SOM_ADG	Allowance definition group	GROUP_ID - Allowance definition group ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On
CRM_PRVO	Subscription order	BS_CRM_COUNTRY_OF_ORGUNIT - Standard field *Country* from Org Unit OBJECT_TYPE - Business Trans. Cat. PROCESS_TYPE - Business Transaction Type	LAST_CHANGE_DATE - Last Changed On
CRM_PRVC	Subscription contract	BS_CRM_COUNTRY_OF_ORGUNIT - Standard field *Country* from Org Unit OBJECT_TYPE - Business Trans. Cat. PROCESS_TYPE - Business Transaction Type	LAST_CHANGE_DATE - Changed On
CRM_PRVMA	Master agreement	OBJECT_TYPE - Business Trans. Cat CRMT_SUBOBJECT_CATEGORY_DB PROCESS_TYPE - Business Transaction Type CRMT_PROCESS_TYPE_DB	CONTRACT_END_DATE - Contract End Date LAST_CHANGE_DATE - Last Changed On
SOM_SHGR	Sharing group ID	GROUP_ID – Sharing Group ID	CREATION_DATE - Created On

Technical Name	Description	Available Fields of Retention Period	Available Start Times
			LAST_CHANGE_DATE - Last Changed On
CRMS4_SOM_MR_DESTRUCTION	Mass Runs	SERVICEMASSRUNTYPE – Mass Run Type SERVICEMASSRUNSTATUS – Mass Run Status	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On

Prerequisites

To use the blocking and delete functions, you must:

- Set up SAP Information Lifecycle Management. For more information, see SAP Information Lifecycle Management.
- Activate the business function ILM. For more information, see Activating SAP ILM.

ILM Notifications for Data Protection

The archiving objects in Subscription Order Management support data protection notifications. The table below highlights the specific ILM objects and the relevant fields that are included in the notifications.

ILM Object	Source Fields
Subscription Order (CRM_PRVO)	OBJECT_ID OBJTYPE_H
Subscription Contract (CRM_PRVC)	OBJECT_ID OBJTYPE_H NUMBER_INT CI_CONTRACT_ID
Master Agreement (CRM_PRVMA)	OBJECT_ID OBJTYPE_H
Allowance Definition Group (SOM_ADG)	ALLOWANCEDEFINITIONGROUPID ALLOWANCEDEFINITIONGROUPUUID
Subscription Product-Specific Data (SOM_PROD)	PRODUCT SUBSCRPNPRODSPECUUID
Sharing Group (SOM_SHGR)	SUBSCRPNCONTRACTSHARINGGROUPID SUBSCRPNCONTRACTSHARINGGROUPUUID

Notification Modes

ILM notification logs the following processes categorized as notification modes:

Notification Mode	Process Step
01	Archiving Write
02	Archiving Delete
03	Data Destruction Run Using Archiving Write Report
04	Data Destruction of Archive Files
05	Destruction Run Using Data Destruction Object
*	All

Authorization Objects

The authorization object `S_ILM_NOTI` is created with the Display, Delete, and Distribute activities. This object is used for the following authorizations required for performing actions on the notification logs:

- `S_ARCHIVE` is checked for creation of log entries during archive process.
- `S_ARCD_RUN` is checked for ILM destruction mode during deletion of archive files.
- `S_ARCD_OBJ` is checked when the Create API (SO Create SOAP API?) Is called from the destruction object program.
- `S_ILM_NOTI` with activity = Distribute is required to extract the notification log data.

Veto Check for HR Master Data in Service

Perform a veto check for HR master data in Service.

Use

Service provides a veto check for the deletion of the HR master data. The veto check enables the application to prevent HR master data being deleted for as long as transactions requiring HR master data information exist.

The veto check is relevant to the deletion of personnel numbers. This reference indirectly extends the residence and retention policy of the HR master data. As long as such references exist, HR master data is not being deleted.

Application Name	Function Module Type	Function Module for Check
`S4CRM`	For Deleting Personnel Numbers	`CRMS4_HR_PER_VETO_CHECK`

You have to activate all applications that should participate in the veto check for blocking data destruction

If not already available, create an entry for `S4CRM` in Customizing under **Personnel Management** > **Personnel Administration** > **Tools** > **Data Privacy** > **Data Destruction** > **Block Data Destruction (Veto)** > **Activate Applications for Veto Check**. To activate the veto checks, set the entry with `S4CRM` to **Needs to be checked** (**Application Active** checkbox).

If necessary, enter the logical system for `S4CRM` as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- `CRM_SERORD` (service orders)
- `CRM_SRVCON` (service confirmations)
- `CRM_SRCONT` (service contracts)
- `CRMS4_REPA` (in-house services)

Veto Check Implementation

Service provides the following functionality for the veto check of the HR master data:

- The application searches for the following data in relation to the HR master data:
    - Service orders, service contracts, and in-house services that do not have the status **Completed**.

      Service confirmations that have the status **Completed** but have not been released for billing.
- In the case of deletion of the personnel numbers, the application returns the personnel number to be deleted.

Based on this information, the function modules can return information at personnel number level about whether the data in the applications is still required. The **Veto** checkbox is selected at personnel number level.

If the **Veto** checkbox is selected, the system prevents both the destruction of infotype and subtype data and the deletion of personnel numbers.

Analysis

In the **ILM Data Destruction** app, once the job that performs the veto check has finished, the results of the check are written to the application log.

If the veto check has failed, the log provides information about any service transactions that triggered the veto and are therefore preventing the deletion of HR master data.

The log also provides the personnel numbers for the affected employees.

Related Information

[Destruction of Data in HR]
[Destroying Personnel Numbers Using HRPA_PERNR]

Veto Check for HR Master Data in Subscription Order Management

Use

Subscription Order Management provides a veto check for the deletion of the HR Master Data. The veto check enables the application to prevent the deletion of the HR Master Data as long as transactions requiring HR Master Data information exist.

The veto check is relevant for the following situations:

- For deletion of employee data. A veto check is required to ensure that the employee data remains accessible for users (for audit and revision reasons, for example) with special authorization to the application and its processes.

These references indirectly extend the residence/retention policy of the HR Master Data. As long as such references exist, HR Master Data will not be deleted.

Application Name	Application Description	Function Module Type	Function Module for Check
Register the application and implementation class in view cluster VC_WFD_EOP_APP. Application Name: `S4CRM` Application Component: `CRM-BF-AR`	Subscription Order Management	Remote Enabled Function Module	`CRMS4_HR_PER_VETO_CHECK` (Veto check for HR Master - S4CRM)

You have to activate all applications that should participate in the veto check for blocking data destruction

For this, if not already available, create an entry for `S4CRM` in Customizing under **Personnel Management** > **Personnel Administration** > **Tools** > **Data Privacy** > **Data Destruction** > **Block Data Destruction (Veto)** > **Activate Applications for Veto Check**. To activate the veto checks, set the entry with `S4CRM` to **Needs to be checked** (**Application Active** checkbox).

If needed, enter the logical system for `S4CRM` as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- HRPA_PERNR (Destruction Object for Personnel Number)

Veto Check Implementation

Subscription Order Management provides the following functionality for the veto check of the HR master data:

- The application searches for open transactions with relation to the personnel number. If open transactions exist, the application will not allow data related to the personnel number to be deleted.
- The end of business is reached when it reaches the end of the retention period.
- The application returns the following information, depending on the scenario:
  - For data destruction, the type of data (infotype/subtype) to be destroyed as well as a list of personnel numbers and the date on which the data for each personnel number is to be destroyed
  - For deletion of personnel numbers, the personnel number to be deleted.

  Based on this information, the function modules can return information on personnel number level about whether the data in the applications is still required. If the veto checkbox is selected, the system prevents the destruction of infotype and subtype data as well as the deletion of personnel numbers.

Analysis

Analysis Question	Answer
What is the purpose of the HR Master Data?	The fields from CRM tables always refer to the Business Partner ID, which is associated with (replicated from) an employee. In Service, it is mandatory to have an associated business partner.
Would a deletion in HR Master Data impact the integrity of your records?	Yes
Describe the impact.	The veto check against the blocking of the HR Master Data person is needed in case a 1Order is not completed as the data is retrieved from HR Employee and not stored in parallel in the BP.

Data Destruction Objects in Subscription Order Management

You use data destruction objects to delete data from the database that is no longer relevant to your business processes, and that you aren't legally required to keep.

SAP has defined a minimum residence time for some data objects. You can also define your own residence time. If more than one residence time is defined, the longer residence time applies.

You destroy data by using **Schedule Data Destruction Run**. Execute a data destruction run for each data destruction object.

Destroying Mass Runs Using `CRMS4_SOM_MR_DESTRUCTION`

You can use the data destruction object `CRMS4_SOM_MR_DESTRUCTION` for destroying Subscription Order Management mass runs.

For more information, see:

- Processing Audit Areas
- Editing ILM Policies
- Editing Retention Rules

ILM-Based Information for the Data Destruction Object

You create policies and rules for the related ILM object of this data destruction object.

The following fields for the ILM object are defined in the ILM policy and are visible when creating or processing the **ILM Policies**:

- Available **Time References**
  - `CREATION_DATE` - Creation date of the mass run
  - `LAST_CHANGE_DATE` - Last change date of the mass run
- Available **Condition Fields**
  - `SERVICEMASSRUNSTATUS` - Status of the mass run
  - `SERVICEMASSRUNTYPE` - Type of the mass run

Enablement for Information Retrieval

The ILM object `CRMS4_SOM_MR_DESTRUCTION` assigned to this data destruction object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation (`BADI_CRM_MRUN_ILM_OBJ_TABLES`) of BAdI: Tables Used by an ILM Object (`BADI_DTINF_ILM_OBJ_TABLES`) for `CRMS4_SOM_MR_DESTRUCTION`. For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, you must assign it to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction `DTINF_MAINTAIN_PURP`).

Once the profile has been created, it can be used for data collection using the transaction `DTINF_START_COLL` - Start Data Collection. You can view the collected data using the transaction `DTINF_PROC_COLL` - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Compatibility Scope

With this compatibility pack, SAP provides a limited use right to SAP S/4HANA customers to run certain classic SAP ERP solutions on their SAP S/4HANA installation. Use the compatibility scope to understand the compatibility packages offered.

For more information about the compatibility scope and the compatibility packages offered for SAP S/4HANA, see SAP Help Portal at http://help.sap.com/s4hana *<Choose an on-premise release>* **Discover** **Compatibility Packs**.

Note

- The features described in this entire compatibility scope are only available in SAP S/4HANA.
- The system behavior described in this entire compatibility scope is only valid in SAP S/4HANA.

Related Information

SAP Note 2269324